College as a Commitment Device: Parental Altruism and the Samaritan's Dilemma*

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Abstract

Why do parents invest so heavily in their children's college education? I argue that college is a commitment device against a Samaritan's dilemma: anticipating support, children under-save and over-consume, and college mitigates this by permanently raising the child's income, shrinking the states where parents optimally transfer. Consistent with this, in matched parent-child data parents of college children consume more and transfer less often, the consumption gain exceeding the fall in realized transfers—as if they release resources once they need no longer stand ready to help. I formalize the mechanism in a continuous-time dynastic model with endogenous college choice, estimated by the simulated method of moments. The lack of commitment lowers aggregate enrollment modestly on net, but the distortion's sign varies across ability and parental wealth, as in-college support subsidizes attendance for low-ability children while lifetime support insures those who forgo college. Yet the friction cannot be undone for free: the distorting transfers also insure children against income risk, so welfare rises only under full commitment—which removes the moral hazard while preserving insurance—not under implementable substitutes that sever it. The welfare effect of free college turns on the same distinction.

JEL: D15, D64, I22, I23

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1 Introduction

Parents in the United States transfer substantial resources to their adult children. Inter-vivos transfers—cash gifts, tuition, housing assistance—flow downward, from parents to children, and persist well into adulthood (McGarry, 1999, 2016); in any given year few families make one, but those they make are large. Nowhere are these transfers more consequential than in higher education: families remain the primary source of college financing, and the support a child anticipates is itself a key determinant of enrollment (Belley and Lochner, 2007; Brown et al., 2012; Abbott et al., 2019). This link matters for policy, since government aid crowds out private contributions precisely because family transfers respond to the enrollment decision (Abbott et al., 2019). Understanding why parents transfer, and how realized and anticipated transfers shape children's choices, is therefore fundamental both to explaining college attendance and to designing education policy.

This paper studies these questions through the Samaritan's dilemma (Lindbeck and Weibull, 1988). Altruistic parents who cannot credibly commit to withholding future support create an incentive problem: children who anticipate transfers under-save and over-consume, knowing parents will step in when resources are low. This induces larger and more frequent transfers than under commitment—parents support their children *because* they care, but the anticipation of support distorts the very decisions that determine how much will be needed. Because transfers are tied to the college decision, the distortion extends to human capital investment.

College plays a distinctive role in this dynamic. From the parent's perspective, a dollar of tuition differs fundamentally from a dollar transferred: tuition permanently shifts the child's income level, moving the child away from the states where support would be triggered, whereas a direct transfer is temporary and may itself reward under-saving. College thus serves as a commitment device—investing in human capital shrinks the *transfer region*, the set of states where the parent would optimally provide support, reducing the lifetime cost of the dilemma. The net effect on attendance is theoretically ambiguous: anticipated in-college transfers subsidize enrollment, while anticipated lifetime transfers insure children who forgo college. The structural model quantifies these opposing forces and shows that the sign of the moral-hazard effect on college varies across the ability–wealth distribution.

I formalize the mechanism in a continuous-time, non-cooperative dynastic model building on Barczyk and Kredler (2014a,b). Parents choose consumption and a non-negative transfer flow; children choose consumption and, at the start of the dynasty, whether to attend college. Each agent optimizes taking the other's policy as given, while the parent internalizes the child's welfare through an altruism parameter. In equilibrium the state space partitions endogenously into a transfer region, where the parent supports the child, and a no-transfer region, where the child is self-sufficient; the boundary—and hence the insurance value of support—is shaped by the child's education, income, and wealth. College contracts the transfer region by permanently raising expected earnings, so the college decision depends not only on the human capital return but on how education reshapes the future financial relationship with the parent.

I document two stylized facts from the Panel Study of Income Dynamics (PSID) and the Health and Retirement Study (HRS), comparing parents of college and non-college children while conditioning on parent income. *First*, parent consumption is markedly higher when the child attends college. *Second*, college lowers the probability of any within-family transfer, while conditional amounts are weakly larger. Both line up with the central mechanism: the consumption gain far exceeds the implied fall in realized transfers, consistent with parents releasing resources because they no longer need to *stand ready* to support the child rather than because they pay fewer transfers. The estimated model quantifies this option-value channel directly.

I estimate the model by the simulated method of moments, targeting college attainment by ability and parent wealth, the college premium, transfer patterns, and intergenerational education transmission. I then decompose the enrollment effect of the lack of commitment relative to an efficient benchmark in which the parent commits to education-contingent transfers at the start of the dynasty, separating the enrollment subsidy of anticipated in-college support from the insurance anticipated lifetime support extends to children who forgo college. On net the lack of commitment lowers aggregate enrollment modestly, but the sign of the distortion varies across the joint distribution of ability and parental wealth: the in-college subsidy raises attendance for low-ability children, while the lifetime insurance lowers it for the medium- and high-ability. The friction cannot be undone for free, because the distorting transfers also insure children against income risk: dynastic welfare rises only under full commitment, which removes the moral hazard while preserving that insurance, not under the implementable substitutes that sever it. The welfare effects of a free-college policy turn on the same distinction.

This paper connects several literatures. The first studies how families insure members against income risk. Since Becker and Tomes (1979, 1986), this literature has debated what motivates such transfers, distinguishing altruism, in which parents give because they value the child's welfare, from exchange, in which transfers buy services or attention in return (Cox, 1987; Bernheim et al., 1985). The evidence is mixed, and no single motive dominates across all transfer types: tests of *pure* altruism are rejected—redistributing a dollar from child to parent raises the parent's transfer by far less than the dollar that altruism predicts (Altonji et al., 1997)—and some transfers rise with recipient income, as exchange implies (Cox, 1987). For the transfers central to this paper, however—inter vivos support flowing from parents to adult children—the altruistic margin is the operative one: such transfers are compensatory, going disproportionately to lower-income children and responding to their income shocks (McGarry, 1999, 2016). I therefore model altruism as the motive for parent-to-child support, while remaining agnostic about bequests and old-age assistance, where exchange and accidental components loom larger. How complete the resulting insurance is remains debated: Altonji et al. (1992) and Hayashi et al. (1996) reject perfect family insurance, yet families still insure substantially—Attanasio et al. (2018) finds considerable insurance potential between parents and adult children, and Kaplan (2012) shows the option to move home insures young workers. Closest, Boar (2021) identifies dynastic precautionary saving; I build on this, showing college attenuates the dynastic precautionary motive by raising the child's income and shrinking the transfer region. Transfers also fund large, lumpy investments and transmit wealth through gifts and bequests (Nishiyama, 2002): Barczyk et al. (2023) emphasize how illiquid housing delays transfers until death, and Brandsaas (2025) finds parental transfers account for a sizable share of young-adult homeownership by relaxing down-payment constraints. I study the same insight—that family transfers shape children's long-run investment—for human capital.

Because parents here hold wealth to stand ready to support adult children, the paper connects to the literature on slow wealth decumulation in retirement. De Nardi et al. (2010) trace much of it to longevity and medical-expense risk and Love et al. (2009) show annualized wealth is roughly flat over retirement; family links do much of the rest, as De Nardi (2004), De Nardi and Yang (2014), and Lockwood (2018) show voluntary bequests reproduce the slow drawdown and the concentration of wealth, and Ameriks et al. (2016) find parents hold wealth to assist children when need is greatest. This literature models the family motive largely as a terminal bequest. Building on the forward-looking inter-vivos motive of Barczyk and Kredler (2014a,b), I instead let parents retain wealth for the option value of an endogenous transfer region whose extent depends on the child's college, and I target the pace of late-life drawdown directly. A related literature asks whether financial constraints bind for college: Cameron and Heckman (1998) and Keane and Wolpin (2001) found a small role for family income conditional on ability, but Belley and Lochner (2007) document its growing importance and Lochner and Monge-Naranjo (2011) find binding constraints for some students, while Brown et al. (2012) stress that the expected family contribution is neither guaranteed nor universal and Hotz et al. (2018) show parental income and housing wealth affect attendance and graduation. Quantitatively, Restuccia and Urrutia (2004), Caucutt and Lochner (2020), and Heckman and Mosso (2014) study how parental investment, borrowing constraints, and family environments shape skills and persistence. Closest, Abbott et al. (2019) show government aid crowds out private contributions; I add a channel—moral-hazard reduction through the Samaritan's dilemma—through which parental wealth affects enrollment beyond direct cost subsidies.

Methodologically, the model builds on the quantitative literature on family dynamics without commitment (Laitner, 1988; Lindbeck and Weibull, 1988), in which Lee and Seshadri (2019) develop a dynastic model with parental human-capital investment. I adopt the continuous-time framework of Barczyk and Kredler (2014a,b), which yields a well-defined equilibrium with an endogenous transfer region and has been applied to long-term care and housing (Barczyk and Kredler, 2018; Barczyk et al., 2023); I extend it by embedding endogenous college decisions. The Samaritan's dilemma at the center of the mechanism (Lindbeck and Weibull, 1988; Bruce and Waldman, 1990) here takes the form of strategic under-saving by the child, which college mitigates by permanently shifting income and reducing entry into the transfer region.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 documents the stylized facts. Section 4 describes the estimation strategy. Section 5 discusses the model fit and the role of parental altruism. Section 6 introduces the full commitment benchmark, formalizes the two opposing forces of moral hazard on college, and presents the quantitative decomposition. Section 7 conducts the welfare analysis. Section 8 concludes.

2 Model

This section presents a continuous-time, non-cooperative dynastic model without commitment that captures the interactions between altruistic parents and their adult children. The model extends the framework of Barczyk and Kredler (2014a,b) by incorporating endogenous college decisions, allowing me to study how family transfers affect college financial support, graduation, and parent retirement consumption. I first describe the demographic structure and income processes, then present the coupled parent-child optimization problem, the parent's transfer decision, and the college decision.

2.1 Demographics and Timing

Time is continuous. Each dynasty consists of one parent and one child who overlap for a finite period, as illustrated in Figure 1. The child enters the model at age 18, at which point the parent is 42. Parent and child are coupled for the parent's entire remaining life: the overlap runs from child age 18 / parent age 42 to the parent's death at age 72, when the child is 48. The parent works until age 66 ($t = T^{\text{ret}}$), earning labor income $y_p(t)$, and then receives social security income $\text{SS}(e_p)$ until death at age 72 ($t = T$, where $T = T^{\text{ret}} + 6$). The retired parent remains coupled with the child, so transfers $\tau(t) \geq 0$ are available throughout the overlap $[0, T]$, including during the parent's retirement; the coupled problem is solved over the parent's full remaining life. Death is deterministic at age 72 (the model has no mortality risk). At the parent's death, a fraction α_{beq} of remaining parent wealth transfers to the child as a bequest; the remaining $(1 - \alpha_{\text{beq}})$ share is not transmitted.[1] The child, now 48, then

[1]The bequest fraction α_{beq} separates the two roles that terminal wealth plays in the model. The warm-glow motive ϕ_b requires parents to hold substantial wealth at death—this is what matches the absence of decumulation at ages 60–70—while the conditional child-wealth moments (child wealth conditional on parent

continues alone—working until its own retirement at 66 and consuming social security until death at 72. The child in turn becomes the parent of a child of its own, so that its terminal state—ability, education, and wealth—becomes the starting point for the next generation, and the dynasty continues.

At the beginning of the dynasty, the child makes a one-time, irreversible college decision. Children who attend college spend four years (ages 18–22) earning a fraction ℓ_C of full-time income—paid at the high-school wage, since the degree is not yet completed—while paying annual tuition ϕ. After college, the child enters the labor force with a permanently higher income process. Children who do not attend college enter the labor market immediately at age 18.

Both parent and child can save in a risk-free bond paying interest rate r. The child borrows through the student loan during and after college, paying the fixed student-loan rate $r + \iota_s$ on the balance, which is amortized to zero by age 32 (the standard ten-year repayment plan).[2] Beyond the student loan, both the child and the parent may hold unsecured debt as working-age adults: following Abbott et al. (2019), education-specific limits of \$85,000 for college graduates and \$25,000 otherwise apply, the outstanding balance carries the estimated wedge $r + \iota_b$, and the limit ramps linearly to zero over the years before retirement, so no one retires or dies in debt (a no-Ponzi constraint) and bequests remain non-negative. Markets are incomplete: no state-contingent insurance is available against the child's income risk. The parent and child are connected through four financial links: a one-time college-conditional gift paid at enrollment, a continuous transfer flow that supports the child's consumption during the overlap, discrete lump-sum transfers that arrive at exogenous Poisson opportunities and move wealth between generations, and a bequest at the parent's death, when a fraction α_{beq} of remaining parent wealth transfers to the child. The flow alone cannot move wealth across generations—capped at the child's desired consumption, it only supports spending—so the lump-sum transfers and the bequest are the channels through which parental wealth

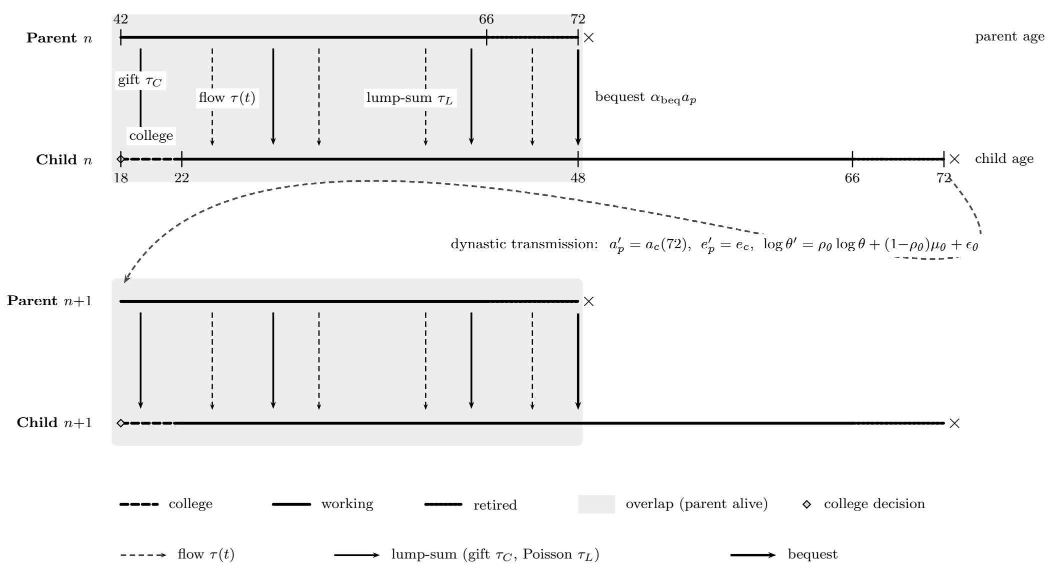
wealth) discipline how much of that wealth children actually receive. The untransmitted share $(1 - \alpha_{\text{beq}})$ is a reduced-form stand-in for end-of-life wealth absorption that the model's compressed lifespan omits: death is deterministic at 72, so the later years in which out-of-pocket medical and long-term-care spending absorb much of retirement wealth (De Nardi et al., 2010; Lockwood, 2018) are not modeled, and the wealth that would be consumed in them is treated as leaving the dynasty rather than reaching the child.

[2]The federal Standard Repayment Plan, the default plan for Direct Loans, fixes repayment at 120 equal monthly payments over ten years (U.S. Department of Education, Federal Student Aid, 2025).

actually reaches the child: the former match the discrete, life-event-driven gifts observed in the data, the latter captures transmission at death. Because the lump opportunities arrive exogenously rather than on a schedule the parent chooses, adding them leaves the no-commitment structure intact.

Two features deserve comment. First, altruism is one-sided: the parent cares about the child's welfare, but the child does not internalize the parent's utility. Second, transfers flow only downward—from parent to child—and the overlap is finite, ending at the parent's death (age 72).[3]

Figure 1. Dynasty Timeline



Notes: Two consecutive generations. The parent (upper line) and child (lower line) overlap for the parent's remaining life (parent ages 42–72, child 18–48; shaded), with the child's college phase dashed (18–22, decision $\diamond$). Four transfer channels run from parent to child: a college gift at enrollment (τ_C) and a continuous consumption-support flow ($\tau(t)$, dashed arrows); discrete lump-sum transfers at Poisson arrivals after college (τ_L, solid arrows); and a bequest at the parent's death ($\alpha_{\text{beq}}a_p$, bold arrow). The child's terminal state then initializes the next generation's parent (curved arrow), and the dynasty continues. Ages and labels are shown once, on generation n.

[3] McGarry (1999) documents that inter-vivos transfers from parents to children are substantial and go disproportionately to less well-off children, and McGarry (2016) shows that they respond to children's incomes over the life cycle; transfers from adult children to parents are comparatively rare, as old-age consumption is financed primarily through Social Security, pensions, and private savings.

2.2 State Variables and Income Processes

The state of a dynasty at any instant t during the overlap phase is $(a_p(t), a_c(t), z(t); \theta, z_p, e_p, e_c)$, where a_p and a_c are parent and child wealth, z is the child's stochastic productivity shock, θ is the child's cognitive ability, z_p is the parent's persistent productivity component (initialized at the dynastic hand-off by the state the parent held as the previous generation's child, and evolving over the parent's working life), and $e_p, e_c \in \{HS, C\}$ denote the education levels of parent and child.

Child income. The child's labor income at time t is:

$$y_c(t) = w_{e_c} \cdot \left(\frac{\theta}{\bar{\theta}}\right)^{\lambda_{e_c}} \cdot \exp\bigl(\gamma(t, e_c) + z(t)\bigr)$$

where $h(\theta, e_c) = (\theta/\bar{\theta})^{\lambda_{e_c}}$ maps ability into human capital with an education-specific ability gradient λ_{e_c}, following Abbott et al. (2019), and $\bar{\theta} = \exp(\mu_\theta)$ is the population mean ability. [4] The gradient $\lambda_C > \lambda_{HS}$ implies that cognitive ability and college education are complements: higher-ability individuals benefit more from college. The education-specific wage w_{e_c} captures the relative price of education-specific labor: $w_{HS} = w$ and $w_C = w \cdot \omega_C$. The deterministic age-income profile is quadratic in age, $\gamma(t, e_c) = \gamma_{1,e}\, t - \gamma_{2,e}\, t^2$, and $z(t)$ follows an Ornstein-Uhlenbeck process:

$$dz = -\kappa_{e_c}\, z\, dt + \sigma_{e_c}\, dW$$

with education-specific mean-reversion rate κ_{e_c} and diffusion σ_{e_c} (Appendix F). The channel through which college reshapes the child's income distribution, and hence the transfer region, is the income *level*: a higher ability gradient ($\lambda_C > \lambda_{HS}$), a steeper age-income profile ($\gamma_{1,C} > \gamma_{1,HS}$), and the relative wage ω_C jointly raise expected income and keep college-educated children far from the states where parental transfers would be triggered. During college ($t \in [0, 4]$ for $e_c = C$), the child works part-time, earning a fraction ℓ_C of full-time earnings. Because the degree has not yet been completed, these earnings are paid at the high-school level—the wage w_{HS}, ability gradient λ_{HS}, and age profile $\gamma(t, HS)$—and the

[4] The normalization by $\bar{\theta}$ ensures that the human capital component equals one at mean ability for both education groups, so that the average college premium comes entirely from the life-cycle profile and wage price ω_C, not from the level effect of $\lambda_C \neq \lambda_{HS}$ evaluated at a non-zero mean $\log \theta$.

college premium applies only after graduation.[5]

Parent income. Parent income depends on education, age, and a persistent productivity component $z_p(t)$ that evolves over the parent's working life: before retirement, $y_p(t) = w_{e_p} \cdot (\theta/\bar{\theta})^{\lambda_{e_p}} \cdot \exp\bigl(\gamma(t, e_p) + z_p(t)\bigr)$, where z_p follows the same education-specific Ornstein–Uhlenbeck process as any worker, $dz_p = -\kappa_{e_p} z_p \, dt + \sigma_{e_p} \, dW_p$, initialized at the dynastic hand-off (age 42) by the productivity state the parent held as the previous generation's child; after retirement at age 66, $y_p(t) = \mathrm{SS}(e_p)$. The parent therefore faces genuine, persistent labor-income risk over the overlap. This risk is a second source of within-dynasty uncertainty alongside the child's income shock z: it generates a precautionary saving motive and the cross-sectional dispersion of parent income and wealth, and—through the transmission channel below—a direct dependence of the child's earning capacity on the parent's realized income.[6]

Ability transmission. Cognitive ability is transmitted across generations through a log-normal AR(1) process: $\log \theta' = \rho_\theta \log \theta + \mu_\theta(1 - \rho_\theta) + \epsilon_\theta$, with $\epsilon_\theta \sim N(0, \sigma_\theta^2)$ and μ_θ the unconditional mean of log ability. The drift term $\mu_\theta(1 - \rho_\theta)$ ensures that the stationary distribution of ability has mean μ_θ, generating realistic intergenerational persistence while allowing for regression to the mean.

Intergenerational earnings transmission. Beyond ability and bequests, a third link operates through initial earning capacity. A child enters the labor market with an initial productivity draw $z_0 \sim N\bigl(\zeta_{pe} \, \mathbf{1}\{e_p = C\} + \rho_{zp} \, z_p^{\text{entry}}, \, \sigma^2_{z_0, e_c}\bigr)$, where $\zeta_{pe} \geq 0$ is a parent-education earnings premium, $\rho_{zp} \geq 0$ an intergenerational income-transmission coefficient (letting earning capacity depend on the parent's income *level*, not only education; Lee and Seshadri, 2019), and z_p^{entry} the parent's persistent productivity at the child's entry. Because the model begins at age 18, the childhood through which parental background shapes earning capacity (Restuccia and Urrutia, 2004; Lee and Seshadri, 2019) is pre-determined; ζ_{pe} and ρ_{zp} are its reduced form, the counterpart of the parental dependence of skill endowments in

[5]In-college earnings therefore rise only weakly with ability, so high-ability students cannot self-finance net tuition out of part-time work and parental resources remain relevant throughout the ability distribution.

[6]The process is initialized at age 42 by the persistent state the parent held as the previous generation's child and continues to evolve thereafter, generating the parent-income dispersion endogenously.

Abbott et al. (2019). Since the early-career advantage accumulates permanently into wealth even as the mean-reverting shock z decays, this channel routes part of the parent–child wealth correlation through inherited earning capacity rather than liquid transfers, leaving the child's age-18 college-financing constraint untouched; absent it, the wealth-transmission moments would load entirely on ability and the financial channels, overstating the role of family resources in college attendance.

Asset-return risk. Following Barczyk and Kredler (2014a), each agent's wealth carries a small multiplicative shock,

$$da_p = \dot{a}_p\,dt + \sigma_a\,a_p\,dB_p, \qquad da_c = \dot{a}_c\,dt + \sigma_a\,a_c\,dB_c,$$

where B_p and B_c are independent standard Brownian motions, the drifts $\dot{a}_p, \dot{a}_c$ are given in (8)–(9), and the volatility, $\sigma_a \max(a, 0)$, vanishes at zero wealth and is absent on debt. The shock represents idiosyncratic asset-return risk and, following Barczyk and Kredler (2014a), regularizes the equilibrium: a diffusion in the *wealth* state—which the income shock z cannot provide—renders the value functions C^2 and the transfer boundary smooth, restoring a pure-strategy Markov equilibrium and eliminating the "blast from the past" of Barczyk and Kredler (2021) (Appendix K). I set σ_a to about ten percent per year, a realistic diversified-portfolio return standard deviation, at which it both regularizes the equilibrium and, as genuine return risk, contributes to the cross-sectional dispersion of wealth.

Return heterogeneity. On positive balances the parent earns a return that rises with the level of wealth,

$$r_{\text{eff}}(a_p) = r + r_{a,\text{grad}}\,\frac{a_p}{a_p + a_{\text{ref}}},$$

a bounded, saturating premium that vanishes near the median ($a_p \ll a_{\text{ref}}$) and approaches $r + r_{a,\text{grad}}$ for the wealthy. This scale dependence in returns—documented in the wealth-inequality literature (Fagereng et al., 2020; Gabaix et al., 2016)—concentrates the right tail of the wealth distribution, lifting the p90/p50 ratio and aggregate wealth through the top without shifting the median or the pace of decumulation; and because the parent's productivity type is inherited, it also contributes to intergenerational wealth persistence. The premium applies only to positive assets, with the saturation scale a_{ref} fixed and the slope $r_{a,\text{grad}}$ estimated; debt instead carries the borrowing wedges defined below. Setting $r_{a,\text{grad}} = 0$ recovers a homogeneous return.

2.3 The Coupled Parent-Child Problem

During the overlap period, parent and child simultaneously choose consumption and savings, while the parent additionally chooses a non-negative transfer flow $\tau(t) \geq 0$. The interaction is non-cooperative: each agent optimizes taking the other's policy as given, but the parent internalizes the child's welfare through the altruism parameter $\eta > 0$.

The continuous-time formulation follows Barczyk and Kredler (2014a), who show that simultaneity of moves in continuous time yields a well-defined equilibrium—an advantage over discrete-time formulations where the timing protocol (who moves first within a period) can affect the equilibrium outcome.

2.3.1 Parent's Problem

The parent chooses consumption c_p and transfer flow $\tau \geq 0$ to maximize lifetime utility, weighting the child's instantaneous utility by η. The parent's value function $V^p(a_p, a_c, z, z_p; t)$—which depends on the parent's own persistent productivity state z_p as well as the child's—satisfies:

$$\begin{aligned}
\rho\, V^p = \max_{c_p,\, \tau \geq 0} \Big\{ & u(c_p) + \eta\, u(c_c^*) + V^p_{a_p}[r_{\text{eff}}(a_p)\, a_p + y_p - c_p - \tau] + V^p_{a_c}[\tilde{r}_c\, a_c + y_c - c_c^* + \tau] \\
& + V^p_z(-\kappa z) + \tfrac{1}{2}\sigma^2 V^p_{zz} + V^p_{z_p}(-\kappa_p z_p) + \tfrac{1}{2}\sigma_p^2 V^p_{z_p z_p} + \tfrac{1}{2}\sigma_a^2 a_p^2\, V^p_{a_p a_p} + \tfrac{1}{2}\sigma_a^2 a_c^2\, V^p_{a_c a_c} + V^p_t \\
& + \lambda_g\, [V^p(a_p - \tau_L^*,\, a_c + \tau_L^*,\, z, z_p;\, t) - V^p(a_p, a_c, z, z_p;\, t)] \Big\}
\end{aligned} \tag{1}$$

where ρ is the continuous-time discount rate, $u(c) = c^{1-\gamma}/(1-\gamma)$ is CRRA utility with $\gamma = 1.5$, c_c^* is the child's equilibrium consumption policy, taken as given by the parent, and $\tilde{r}_c$ is the child's effective interest rate, which accounts for student debt (defined in Section 2.3.5). Consumption is bounded below by a floor $\underline{c}$ = \$3,900 (in 2016 dollars), representing the effective consumption guaranteed by means-tested government transfers (SSI, SNAP, Medicaid), in the range estimated by De Nardi et al. (2010). The terms involving V^p_z and V^p_{zz} capture the effect of the child's income uncertainty on the parent's value function, while $V^p_{z_p}$ and $V^p_{z_p z_p}$ carry the parent's own persistent labor-income risk—the Ornstein–Uhlenbeck process for z_p of Section 2.2, with mean-reversion $\kappa_p \equiv \kappa_{e_p}$ and diffusion $\sigma_p \equiv \sigma_{e_p}$, and where $r_{\text{eff}}(a_p)$ is the wealth-dependent return defined there. The final term is the lump-sum transfer channel: at Poisson arrivals with intensity λ_g, the parent may execute a one-time wealth rebalancing τ^*_L toward the child.

2.3.2 Child's Problem

The child chooses consumption c_c to maximize own lifetime utility:

$$\begin{aligned}\rho V^c = \max_{c_c} \Big\{ & u(c_c) + V^c_{a_c}[\tilde{r}_c\, a_c + y_c - c_c + \tau^*] + V^c_{a_p}\left[r_{\text{eff}}(a_p)\, a_p + y_p - c^*_p - \tau^*\right] \\ & + V^c_z(-\kappa z) + \tfrac{1}{2}\sigma^2 V^c_{zz} + V^c_{z_p}(-\kappa_p z_p) + \tfrac{1}{2}\sigma_p^2 V^c_{z_p z_p} + \tfrac{1}{2}\sigma_a^2 a_p^2\, V^c_{a_p a_p} + \tfrac{1}{2}\sigma_a^2 a_c^2\, V^c_{a_c a_c} + V^c_t \\ & + \lambda_g\left[V^c(a_p - \tau^*_L,\, a_c + \tau^*_L,\, z, z_p;\, t) - V^c(a_p, a_c, z, z_p;\, t)\right] \Big\} \end{aligned} \tag{2}$$

where τ^* and c^*_p are the parent's equilibrium policies, taken as given, and the jump term reflects that the child's state moves with the *parent's* lump-sum policy τ^*_L at transfer opportunities—altruism is one-sided, so the child receives but does not choose. The child tracks parent wealth a_p because it affects expected future transfers: wealthier parents are more likely to provide support.

2.3.3 Transfer Decision

The parent's optimal transfer is characterized by a complementarity condition:

$$\tau^* \geq 0, \quad V^p_{a_p} \geq V^p_{a_c}, \quad \tau^*\left(V^p_{a_p} - V^p_{a_c}\right) = 0 \tag{3}$$

This partitions the state space into two regions. In the *no-transfer region* ($\mathcal{N}$), the parent's marginal value of own wealth exceeds the marginal value of child wealth ($V^p_{a_p} > V^p_{a_c}$), and $\tau^* = 0$. In the *transfer region* ($\mathcal{T}$), these marginal values are equalized ($V^p_{a_p} = V^p_{a_c}$) and $\tau^* > 0$. The boundary between $\mathcal{N}$ and $\mathcal{T}$ is a free boundary, endogenously determined as part of the equilibrium. The parent transfers when the child is relatively poor (low a_c), faces negative productivity shocks (low z), or when the parent is relatively wealthy (high a_p).

This complementary-slackness condition is the continuous-time analog of the transfer decision in discrete-time dynastic models, but with two key advantages. First, the flow transfer carries no within-period commitment: a discrete-time transfer is implicitly a promise held fixed for the length of the period, whereas the flow can be revised at every instant. Second, the boundary of the transfer region provides a clean characterization of the states where moral hazard is operative. Flow transfers are not the only instrument: at stochastically arriving opportunities the parent can also move wealth in discrete amounts, and the same free boundary governs when it chooses to do so.

Transfer size. Within the transfer region, the transfer is determined at the consumption margin. The static optimality condition $u'(c_p) = \eta\, u'(c_c)$ implies an altruistic consumption target $c_c = \eta^{1/\gamma} c_p$ for the child, and the parent funds the gap between this target and the child's own flow resources $R_c = \tilde{r}_c a_c + y_c$, capped at the consumption level c_c^{opt} the child would choose for itself from its first-order condition:

$$\tau^* = \max\Big\{0,\ \min\left(c_c^{\text{opt}} - R_c,\ \ \eta^{1/\gamma} c_p - R_c\right)\Big\}. \tag{4}$$

The child consumes $R_c + \tau^*$ inside the transfer region: the transfer is consumption support, not a wealth transfer. The cap at c_c^{opt} reflects this—any gift beyond the child's desired consumption would be saved by the child, converting parent wealth into child wealth.[7]

The flow transfer never moves the parent's wealth stock. Mechanically, moving a discrete amount in an instant would require an unbounded flow; economically, the absence of commitment makes pre-funding unattractive, since wealth placed in the child's account would be consumed too quickly. By retaining the wealth the parent preserves the option to support the child only when it is actually constrained, providing liquidity period by period rather than capital while continuing to optimize its own saving inside the transfer region ($\dot{a}_p = r_{\text{eff}}(a_p)a_p + y_p - c_p - \tau^*$). Appendix K derives these properties formally.

[7] Transfers are therefore not confined to children exactly at the borrowing constraint, as in the no-labor-income setting of Barczyk and Kredler (2014b) where gifts flow only to a recipient with zero wealth. With flow income, the gift (4) is positive whenever the child's flow resources fall short of both its desired consumption and the altruistic target, $R_c < \min(c_c^{\text{opt}}, \eta^{1/\gamma} c_p)$, so the transfer region concentrates at low child wealth and adverse income draws but its boundary lies at strictly positive a_c (Figure 2).

2.3.4 Lump-Sum Transfer Opportunities

The flow transfer is consumption support by construction—capped at the child's desired consumption, it props up the child's spending in bad states but never adds a dollar to the child's balance sheet. Observed inter-vivos giving is not so confined: parents make discrete, sizeable transfers—help with a house down payment, a wedding, support around the birth of a grandchild—that move *wealth* between generations well before any bequest. In the HRS, transfers of $500 or more in a two-year window occur for 5 to 25 percent of parent-child pairs depending on parent wealth, and their lumpiness is intrinsic to how they arrive: they are occasioned by life events, not paid as annuities. A model whose only wealth-moving instruments are a tuition gift at age 18 and a bequest at the parent's death cannot reproduce the lumpy inter-vivos transfers observed between enrollment and death—the very transfer incidence the extensive-margin moments measure—with an instrument set that, by construction, produces none of them in between.

I restore the lump without restoring the commitment.[8] At exogenous Poisson arrivals with intensity λ_g, the parent may execute a one-time transfer $\tau_L \in [0, \max(a_p, 0)]$, chosen at the moment of the opportunity to maximize its own value,

$$\tau_L^*(a_p, a_c, z, z_p;\, t) \;=\; \arg \max_{\tau_L \in [0,\, \max(a_p, 0)]} \; V^p(a_p - \tau_L,\; a_c + \tau_L,\; z, z_p;\, t)\,. \tag{5}$$

Because the arrival is exogenous, the parent chooses neither the timing nor a schedule: the exercise is an instantaneous action, not a promise, so no commitment is violated; and because opportunities arrive smoothly in time, the channel introduces none of the anticipation discontinuities that deterministic transfer dates would create (Barczyk and Kredler, 2021). Proposition 1 in Appendix M makes this precise: under the Poisson channel the value functions remain C^2 away from the transfer boundary, whereas deterministic dates would reintroduce the non-smoothness. The arrival intensity has a natural interpretation as the frequency of life events at which discrete transfers occur.

[8] There is also a methodological reason this channel is absent from continuous-time altruism models. Barczyk and Kredler (2014a) observe that discrete-time dynastic models implicitly assume the donor can commit to a lump-sum transfer for the length of a period—commitment smuggled in by the time aggregation itself—and the continuous-time reformulation removes it; but in removing the artificial commitment it also removes the lump, leaving only the flow.

The optimality condition for (5) ties the lump to the same free boundary that governs the flow. An interior $\tau_L^* > 0$ requires $V^p_{a_c} > V^p_{a_p}$ at the pre-transfer state—strictly inside the transfer region—and the optimal lump moves the state onto the boundary where the two marginal values are equalized. Lump-sum transfers are therefore infrequent, sizeable, and state-contingent: they fire when the child has fallen deep into the transfer region, and they rebalance the dynasty's wealth in one shot rather than drip-feeding consumption. This is precisely the empirical object the extensive-margin transfer moments measure—and it is what identifies the altruism parameter η, since the wealth gradient in transfer incidence pins down the size of the region in which the parent exercises the opportunity—giving altruism a channel through which it can *raise* child wealth in adverse states rather than only eroding the child's own saving through the Samaritan's dilemma. Routing general altruistic wealth rebalancing through these opportunities also keeps it off the college gift: the mean-support moment disciplines the college gift at enrollment, while the transfer-probability moments by parent wealth discipline the Poisson lumps over the rest of the overlap, so the two margins are separately identified rather than loaded onto a single college transfer. The remaining lump-sum transfers in the model are the college-conditional gift at enrollment and the bequest at death; lump opportunities begin at the child's college-exit age, so the college gift remains the only lump instrument during the schooling years.

2.3.5 Consumption Policies and Wealth Dynamics

The first-order conditions for consumption are:

$$u'(c_p^*) = V^p_{a_p}, \tag{6}$$

$$u'(c_c^*) = V^c_{a_c}. \tag{7}$$

Under CRRA utility, these invert to $c_p^* = (V^p_{a_p})^{-1/\gamma}$ and $c_c^* = (V^c_{a_c})^{-1/\gamma}$. The drifts of the two wealth processes (the deterministic part of the laws of motion $da_p = \dot{a}_p\,dt + \sigma_a a_p\,dB_p$ and

$da_c = \dot{a}_c\,dt + \sigma_a a_c\,dB_c)$ are:

$$\dot{a}_p = r_{\text{eff}}(a_p)\,a_p + y_p(t) - c_p - \tau, \tag{8}$$

$$\dot{a}_c = \tilde{r}_c\,a_c + y_c(t,z) - c_c + \tau, \tag{9}$$

where the parent's wealth is bounded below by the age- and education-dependent borrowing limit introduced above (and zero from retirement on). The child's wealth, by contrast, can be negative: a child who attends college may finance tuition and consumption with student loans, accumulating a debt balance ($a_c < 0$) that it carries into working life (Section 2.7). The child's effective interest rate therefore depends on the sign and source of its debt,

$$\tilde{r}_c = r + \iota_s\,\mathbf{1}\{\text{student debt}\} + \iota_b\,\mathbf{1}\{\text{unsecured debt}\},$$

so the child earns r on savings, pays the fixed student-loan rate $r + \iota_s$ on student debt, and pays the estimated wedge $r + \iota_b$ on other unsecured debt. During college the constraint is relaxed to $a_c \geq -\bar{b}$ at the student rate, and that balance is amortized so that $a_c \geq 0$ by age 32 (the standard ten-year repayment plan). As a working-age adult the child—like the parent—may also hold unsecured debt up to the education-specific limit (\$25,000 for high school, \$85,000 for college) at the rate $r+\iota_b$, with the limit ramping to zero before retirement; this borrowing generates the negative net worth observed among young adults.

2.4 Child-Alone Problem and the Dynastic Link

When the parent dies (parent age 72, child age 48), the child solves a consumption-savings problem with their own accumulated wealth—augmented by the bequest—for the remainder of their life:

$$\begin{aligned}\rho\,V^{c,\text{alone}} = \max_{c_c} \Big\{ & u(c_c) + V^{c,\text{alone}}_{a_c}[\tilde{r}_c\,a_c + y_c(t,z) - c_c] \\ & + V^{c,\text{alone}}_z(-\kappa z) + \tfrac{1}{2}\sigma^2 V^{c,\text{alone}}_{zz} + \tfrac{1}{2}\sigma_a^2 a_c^2\,V^{c,\text{alone}}_{a_c a_c} + V^{c,\text{alone}}_t \Big\}\end{aligned} \tag{10}$$

The child lives alone until its own death at age 72, working until 66 and then drawing social security. The dynasty is then closed as a stationary recursion: a new dynasty begins with a parent whose initial wealth equals the child's terminal wealth a_c at age 72, whose education is $e'_p = e_c$, and whose child has ability θ' drawn from the intergenerational transmission process.

2.5 Terminal Conditions

At the parent's death ($t = T$, parent age 72, child age 48), the coupled system terminates and the child transitions to the child-alone problem. The boundary conditions at $t = T$ are:

$$V^p(a_p, a_c, z, z_p; T) = \phi_b \frac{(a_p + \underline{a}_{\text{beq}})^{1-\gamma}}{1-\gamma} + \eta \cdot V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p,\, z;\, 0), \tag{11}$$

$$V^c(a_p, a_c, z, z_p; T) = V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p,\, z;\, 0). \tag{12}$$

The child's value function transitions to the child-alone problem (10), evaluated at $t = 0$ of the child-alone clock—the moment of the parent's death. Equation (12) states this continuity: the child carries forward its own wealth a_c plus the bequest $\alpha_{\text{beq}}\, a_p$.

The parent's terminal value (11) has two components. The first, $\phi_b\,(a_p + \underline{a}_{\text{beq}})^{1-\gamma}/(1-\gamma)$ with $\phi_b \geq 0$, is a warm-glow term following De Nardi (2004) that captures the parent's direct utility from holding wealth at the end of the overlap period.[9] The second component, $\eta \cdot V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p, z; 0)$, reflects the parent's altruistic concern for the child's continuation utility, with $\alpha_{\text{beq}} \in [0, 1]$ the fraction of parent terminal wealth bequeathed at death. The bequest channel creates two motives for wealth accumulation—warm glow (ϕ_b) and altruism through α_{beq}, which disciplines the intergenerational wealth correlation.

2.6 College Decision

At the beginning of the dynasty ($t = 0$), two decisions are taken in sequence: the parent makes a one-time inter-vivos gift, and the child then decides whether to attend college. Let $V^c_e(a_p, a_c)$ and $V^p_e(a_p, a_c)$ denote the child's and parent's equilibrium continuation values under education $e \in \{C, HS\}$, evaluated at $z = 0$.

Child's decision. Given the parent's college gift (described below), the child weighs its own value gain from attending—the college branch evaluated at the post-gift state, the high-

[9] The shifter $\underline{a}_{\text{beq}} \geq 0$ makes bequests a luxury good in the sense of De Nardi (2004): with $\underline{a}_{\text{beq}} > 0$ the marginal propensity to bequeath rises with wealth, so poor parents leave little while the rich bequeath a rising share—the mechanism that generates the fat right tail of the wealth distribution.

school branch at the no-gift state—

$$G \equiv V_C^c - V_{HS}^c,$$

against the psychic cost of college, which follows the Abbott et al. (2019) linear-in-log specification with an idiosyncratic shock:

$$\kappa(\theta, \varepsilon_\kappa, e_p) = \big(\kappa_0 + \kappa_\theta \log\theta + \kappa_w \,\mathbf{1}\{e_p = C\} + \sigma_\kappa\, \varepsilon_\kappa\big) \cdot \bar{V}, \qquad \varepsilon_\kappa \sim N(0,1),$$

where ε_κ is drawn once per individual at $t = 0$ and $\bar{V} = \text{mean}_{a_p,\theta}|V_C^c - V_{HS}^c|$ is a normalization constant that makes the dimensionless parameters κ_0, κ_θ, κ_w, and σ_κ comparable in scale to the college value gap. The gradient $\kappa_\theta < 0$ implies that higher-ability individuals face lower psychic costs, reflecting the non-pecuniary, "consumption value" view of schooling (Cunha et al., 2005; Heckman et al., 2006). The shifter $\kappa_w < 0$ lowers the psychic cost for children of college-educated parents, a non-pecuniary intergenerational channel of college taste, information, and preparation that operates through the human-capital preparedness parental education builds over childhood (Cunha and Heckman, 2007; Cunha et al., 2010).[10] It lets the socioeconomic gradient in attainment operate through the *non-pecuniary* cost of college, distinct from the credit-market frictions emphasized by Belley and Lochner (2007) and Caucutt and Lochner (2020), which the model captures separately through the borrowing channels of Section 2.7. This is the only channel through which parent education enters the college decision beyond altruism and resources; the parent derives no separate utility from the child's enrollment. The child attends whenever the value gain exceeds the realized cost, $G \geq \kappa(\theta, \varepsilon_\kappa, e_p)$. Integrating over the normally distributed cost shock yields a probit college-

[10] The socioeconomic gradient in attainment could reflect this child-side taste/preparation channel (κ_w) or a parent-side paternalistic motive (ξ, as in Abbott et al. 2019), in which the parent values the child's enrollment independently of the child's own preference. The two are not separately identified by the attendance gradient alone, but they differ in their transfer implications: paternalism operates only through the parent's college gift, so a paternalism-driven gradient implies counterfactually large inter-vivos transfers, which the transfer moments do not show. The baseline therefore omits the paternalistic motive and attributes the gradient to the child-side shifter κ_w.

attendance probability:

$$\Pr(e_c = C \mid a_p, a_c, \theta, e_p) = \begin{cases} \Phi\left(\dfrac{G/\bar{V} - \kappa_0 - \kappa_\theta \log\theta - \kappa_w \mathbf{1}\{e_p = C\}}{\sigma_\kappa}\right) & \text{if } G > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{13}$$

where Φ is the standard normal CDF and σ_κ governs the dispersion of college choices: a larger σ_κ flattens the mapping from the value gap to attendance and, through favorable cost draws, raises enrollment among low-ability types. The decision is the child's alone—consistent with one-sided altruism, the child does not weigh the parent's value in choosing its education.

Parent's college gift. Before the child's cost shock is realized, the parent chooses a one-time college-conditional gift τ_C, paid at enrollment, anticipating how it shifts the child's choice:

$$\begin{aligned} \tau_C^* = \arg\max_{\tau_C \in [0,\, a_p]} \; & \Pr(C \mid \tau_C)\, V_C^p(a_p - \tau_C,\, \tau_C) \\ & + \big(1 - \Pr(C \mid \tau_C)\big)\, V_{HS}^p(a_p,\, 0), \end{aligned} \tag{14}$$

where $\Pr(C \mid \tau_C)$ is the probit (13) with the college branch evaluated at the post-gift state $(a_p - \tau_C,\, a_c = \tau_C)$ and the high-school branch at $(a_p,\, 0)$. The conditional gift requires no commitment: payment and enrollment are a single transaction—the parent finances enrollment at the moment it occurs—so there is no promise to renege on, and a child who chooses high school receives no lump sum at $t = 0$. The gift is bounded only by the parent's wealth, $\tau_C \in [0,\, a_p]$: it is the conditional inter-vivos transfer the parent makes at enrollment, the broad earmarked support—tuition together with allowances and in-kind room and board—that the college-support data of Abbott et al. (2019) and the mean-support moment of Section 4 measure, rather than tuition narrowly. The size of the gift is therefore disciplined directly by that mean-support moment. The gift is purely altruistic: it raises the child's continuation value V_C^p that the parent already internalizes, and the parent derives no additional utility from enrollment per se.[11] Wealthy parents—whose marginal utility of consumption is low—make larger gifts, relaxing the child's college-financing constraint and raising the prob-

[11] An enrolling child enters the overlap with $a_c = \tau_C^*$ and the parent with $a_p - \tau_C^*$; a child who chooses high school receives no lump sum at $t = 0$.

ability of enrollment; together with the parent-education cost shifter κ_w, this is the force that generates college attendance among children of high-resource and college-educated families. During the schooling years the college gift is the only lump-sum instrument; after college exit, support moves through the state-contingent flow $\tau(t)$, the Poisson lump-sum opportunities of Section 2.3.4, and ultimately the bequest.

Parental altruism enters the college decision through the continuation values V_C and V_{HS}. Parents who anticipate supporting their child during and after college raise the value of attending, and because the continuation values incorporate the entire future trajectory of transfers—which depends on the child's income path and hence on education—parent wealth affects enrollment beyond the direct cost channel. Anticipated transfers, however, raise both V_C and V_{HS}: they subsidize college enrollment but also insure non-college children against income risk. The net effect of the Samaritan's dilemma on the college decision is therefore theoretically ambiguous; Section 6.1 formalizes this result.

2.7 College Financial System

I model the college financial system following Abbott et al. (2019); Appendix L gives the full specification. The annual net cost of college is $\phi = \$6{,}700$ (average net tuition from the NLSY97), and students work a fraction $\ell_C = 0.56$ of full-time hours, earning at the high-school rate until graduation. Financial aid follows a continuous expected-family-contribution schedule based on parental *income* at college entry: the maximum grant $\bar{g}$ phases out linearly in parent income, so the effective price of college rises with family resources. During college the child may borrow up to a cumulative limit

$$\bar{b}(a_p) = \bar{b}_0 + b_{\text{slope}} \cdot a_p$$

at the student-loan rate $r + \iota_s$, where the base limit $\bar{b}_0 = \$18{,}000$ is the federally available capacity and the collateral channel $b_{\text{slope}} \geq 0$ adds private/PLUS capacity requiring parental backing, so children of wealthier parents can borrow more; the balance amortizes to $a_c \geq 0$ by age 32. As Abbott et al. (2019) document, parental transfers supplement institutional aid and government grants can crowd out private contributions.

2.8 Equilibrium

A Markov-Perfect Equilibrium (MPE) consists of value functions V^p, V^c, consumption policies c_p^*, c_c^*, a transfer-flow policy τ^*, a lump-sum transfer policy τ_L^*, and a college enrollment rule e_c^*, for each (θ, e_p, e_c) pair, such that: (i) given c_c^*, τ^*, and τ_L^*, the parent's HJB (1), the complementary-slackness condition (3), and the lump-sum optimality condition (5) are satisfied; (ii) given c_p^*, τ^*, and τ_L^*, the child's HJB (2) is satisfied; (iii) after the parent's death ends the overlap ($t > T$), $V^{c,\text{alone}}$ satisfies (10); (iv) the college rule follows from (13); and (v) boundary conditions (11)–(12) and borrowing constraints hold at all times.

The equilibrium is solved backward: first the child-alone problem, then the coupled system during the overlap period, and finally the college choice at $t = 0$. Within the coupled system, the transfer is obtained in closed form at each state—the bounded gift (4)—so the transfer region is determined pointwise rather than by iterating on the complementary-slackness condition, and the value functions are updated by implicit time-stepping. The model is dynastic in that generations are *linked* rather than fully recursive in the value functions: altruism is embedded one generation forward—the parent values the child through $\eta V^{c,\text{alone}}$, while the child's post-overlap life is solved as a standalone life-cycle problem that does not itself carry the child's altruism toward a future child—and successive generations are connected through the simulation, where each child's terminal wealth, ability, and education initialize the next generation's parent and generate a stationary cross-generation distribution. I therefore solve the value functions once, by backward induction over a single parent–child cycle, rather than iterating them across generations to a recursive dynastic fixed point as in Barczyk et al. (2023); this is a deliberate simplification: the Samaritan's-dilemma mechanism operates within the parent–child relationship, and the child's own altruism toward the next generation is second order for the college decision at $t = 0$, which turns on the child's lifetime resources and the support it anticipates from its parent. Appendix K derives the equilibrium properties, including the Samaritan's dilemma distortion in continuous time, and Appendix M describes the numerical solution algorithm.

2.9 Empirical Implications

College reshapes the transfer region—the set of states in which the parent optimally supports the child—and the boundary shift maps into family outcomes observable in matched parent-child data.

Why college shrinks the transfer region. Recall that the parent transfers when the marginal value of own wealth equals the marginal value of child wealth ($V^p_{a_p} = V^p_{a_c}$). Because the parent weighs the child's utility by η in the objective (1), $V^p_{a_c}$ is increasing in η: more altruistic parents have a larger transfer region. This condition is more likely to bind when the child is relatively poor or faces negative income shocks. College raises the child's expected income through a higher ability gradient ($\lambda_C > \lambda_{HS}$) and a steeper life-cycle earnings profile; the resulting increase in child wealth pushes states away from the transfer boundary, so college-educated children spend less time receiving support and the region contracts relative to high-school children. Figure 2 illustrates this in the (a_c, a_p) state space: college shifts the boundary to the left, so that at any given level of parent wealth the child must be poorer before transfers become optimal. The gap between the two boundaries—the states in which a high-school child would receive transfers but a college child would not—is the contraction of the region; together with the overconsumption zone bordering it, where the child under-saves in anticipation of entering the region, this contraction captures the scope for moral hazard that college eliminates.

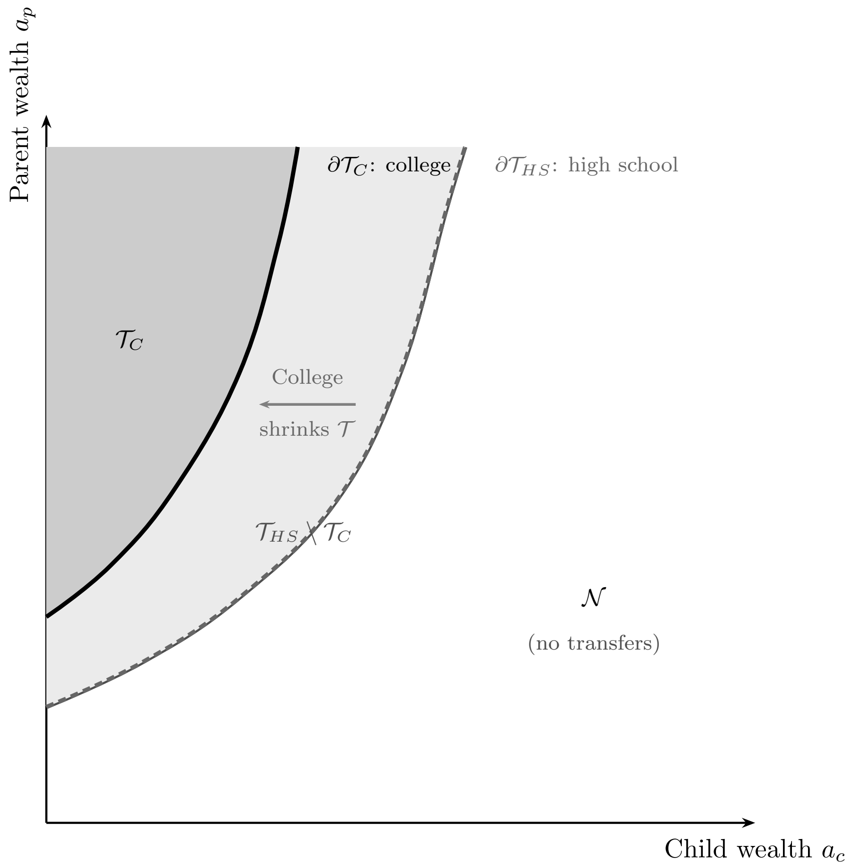


Figure 2. Transfer Region Boundary: College vs. High School

Notes: Schematic illustration of the transfer region in the (a_c, a_p) state space. In the transfer region (shaded), the parent optimally transfers to the child ($\tau > 0$). The solid boundary corresponds to a college-educated child; the dashed boundary to a high-school-educated child. College shrinks the region because higher expected income keeps child wealth further from the transfer boundary. The model-computed free boundary and transfer policy that this schematic illustrates are reported in Appendix K (Figures K.1 and K.2).

The contraction of the transfer region yields two predictions, both conditional on parent income, that I document in the descriptive evidence of Section 3. The first is a consumption release: a smaller region means parents spend less of their lifetime making transfers, so resources that would have flowed to the child under the high-school path are instead available for own consumption, and parent consumption is higher when the child attends college. The second concerns the two margins of transfers. Because a larger negative shock is required to push a college child into the region, the probability of transferring falls; conditional on transferring, the states reached lie further inside it, so the optimal transfer τ^* is weakly larger. The model thus predicts a negative extensive-margin effect and a zero-or-slightly-positive intensive-margin effect.

Parents hold wealth against the option of supporting a child who falls into the transfer region, so college releases resources even in states where no transfer is ever paid. The consumption release therefore reflects both fewer realized transfers and less precautionary saving, and the model predicts that the first exceeds the second: the consumption gain from college is larger than the fall in realized transfers. This is the continuous-time option-value channel of Barczyk and Kredler (2014b), in which the family inefficiency operates before transfers flow rather than through the transfers themselves. The estimated model quantifies this channel directly.

3 Empirical Evidence

This section documents how parent consumption and inter-vivos transfers vary with the child's college attainment in matched parent-child data (PSID, HRS, and NLSY97). Each fact compares outcomes for parents whose children attended college with those whose children did not, conditioning on parent income to absorb the parent's own resources. None of these moments is targeted in estimation, so Section 5.2 confronts the estimated model with each of them as an out-of-sample check.

3.1 Data

The empirical analysis draws on three data sources. The Panel Study of Income Dynamics (PSID) provides matched parent-child panels with consumption, income, wealth, and inter-vivos transfer data from 1999 onward. I link parents to adult children using the Family Identification Mapping System (FIMS). The summary-statistics sample (Appendix Tables D.1–D.2) comprises 53,326 parent-year observations from 8,362 households with heads aged 25–60; the parent-consumption analysis of Section 3.2 further restricts to parents aged 50+ with adult children aged 26+, collapsed to the parent-year level, yielding 15,830 observations. The Health and Retirement Study (HRS, 1992–2014 Family-Kids file) offers a larger sample of older parent-child pairs with detailed transfer information—548,239 parent-to-child wave observations, of which 78,541 involve a positive transfer—which I use to study the extensive and intensive margins of inter-vivos transfers. The National Longitudinal Survey of Youth 1997 (NLSY97) provides cognitive ability scores (ASVAB) and parental wealth for 5,400 individuals.

3.2 Parent Consumption and Child College Status

The model implies that, conditional on own income and wealth, parents of college children consume more (the consumption release of Section 2.9): college shifts children out of the transfer region by permanently raising their income, freeing parents from contingent obligations even when no transfers flow.

I examine this implication by estimating the following regression on the PSID sample at the parent-year level:

$$C_{p,t} = \beta_0 + \beta_1 \text{ShareCollege}_{p,t} + \beta_2 N_{p,t} + \delta_{Q_p^{inc}} + \beta_X \mathbf{X_{p,t}} + \varepsilon_t + \epsilon_{p,t}$$

where $C_{p,t}$ is household consumption (in 2016 dollars) of parent p at time t, $\text{ShareCollege}_{p,t}$ is the fraction of parent p's children with a bachelor's degree or higher, $N_{p,t}$ is the total number of children, $\delta_{Q_p^{inc}}$ are parent income quartile fixed effects (within-cohort ranking), and $\mathbf{X_{p,t}}$ is a comprehensive set of controls: parent total wealth, non-financial wealth, household income, wealth quartile, labor force participation, household size, head's birth year, education, U.S. state, a cubic in head's age, housing tenure, and race. The specification includes year fixed effects ε_t.

A higher share of college-educated children is associated with higher parent consumption: moving from none to all of a parent's children holding a BA raises consumption by about 8% (Table 1, Column 1). The association is robust to alternative definitions of college exposure—having any child with a BA is associated with 8% higher consumption (Column 2), and having all children hold a BA with 6% higher consumption (Column 3), both significant at the 1% level. In levels (Column 4), moving from no college children to all corresponds to roughly \$2,500 more in annual parent consumption.

Table 1. Parent Consumption and Child College Status (PSID)

	(1) Log C	(2) Log C	(3) Log C	(4) Level C (000s)
Share of Children with BA+	0.081***			2.470***
	(0.019)			(0.866)
Number of Children	0.014*	0.010	0.016*	0.170
	(0.008)	(0.008)	(0.008)	(0.287)
Any Child with BA+		0.080***		
		(0.017)		
All Children with BA+			0.056***	
			(0.017)	
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Parent income Q FE	Yes	Yes	Yes	Yes
Observations	15,830	15,830	15,830	15,831
R^2	0.558	0.558	0.557	0.357

Notes: OLS regressions of parent household consumption on child college indicators. "Share of Children with BA+" is the fraction of children with a bachelor's degree or higher. The dependent variable in Column 4 is total consumption in thousands of 2016 dollars. Controls include a cubic in parent age, household income, non-housing wealth, total wealth, wealth quartile, and fixed effects for year, education, parent income quartile, gender, housing tenure, state, labor force status, race, household size, and birth year. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

3.3 Inter-Vivos Transfers

If college frees parents from contingent obligations, do realized transfers fall correspondingly? Answering requires decomposing transfers into their extensive and intensive margins, which map directly onto the model's transfer-region structure (Section 2.9).

Most parent–child pairs report zero transfers in any given wave: only 14.4% of observations involve a positive transfer (Table 2). Conditional on one, amounts are substantial—a mean of $8,161, median $2,897, and 90th percentile $18,394. This mass at zero with a heavy right tail is what the model predicts: the transfer region is entered infrequently, but entry is a non-trivial insurance event.

Table 2. Distribution of Inter-Vivos Transfers from Parents to Children (HRS)

Pr(Transfer > 0)	0.144
E(Transfer | Transfer > 0)	\$8,161
Median(Transfer | Transfer > 0)	\$2,897
P75(Transfer | Transfer > 0)	\$7,968
P90(Transfer | Transfer > 0)	\$18,394
E(Transfer, unconditional)	\$1,172
Observations	548,239
Observations with Transfer > 0	78,541

Notes: Wave-level statistics for parent-to-child transfers from HRS 1992–2014 (Family-Kids file, waves 1–12). Sample restricted to parent age ≥ 50, child age ≥ 26. Transfer amounts in real 2016 US dollars. Unconditional mean includes zeros.

I decompose the within-family college effect along these two margins. Appendix Table E.1 reports linear probability models for Pr(Transfer > 0) with parent fixed effects:

$$\mathbf{1}[IVT_{ijt} > 0] = \beta_0 + \beta_1 \text{College}_j + \gamma_t + \beta_X \mathbf{X_{jt}} + \delta_{Q_i^p} + \varepsilon_i + \epsilon_{ijt}$$

College is associated with a 3.1 percentage-point lower probability of any transfer—a 21% reduction on the 14.4% mean (Appendix Table E.1, Column 1). Interacting college with parent income quartile (Column 2) shows a steep gradient: −1.0 pp for Q1, −2.8 for Q2, −3.4 for Q3, and −4.2 for Q4. Wealthier parents have larger transfer regions ex ante, so college contracts them by more.

Conditional on a positive transfer, college children receive \$557 more per event (Appendix Table E.2, Column 1, $p < 0.10$). The effect is heterogeneous (Column 2): −\$906 for Q1 parents but positive and large above—\$958 for Q2 and \$826 for Q4. This matches the boundary shift: college moves the transfer boundary left in (a_c, a_p) space, so the states that pull a college child into the region involve worse income draws; conditional on entry the child is further inside the region and the optimal transfer is larger, most so for wealthier parents whose boundary shift is more pronounced.

The 3.1-point fall in transfer probability corresponds to roughly \$253 per year in expected transfers ($0.031 \times \$8{,}161$), an order of magnitude below the \$2,500 consumption difference of Section 3.2. The comparison is suggestive, but it is the option-value pattern: college lowers the burden of standing ready to support the child, releasing resources beyond realized transfers. Altogether, college is associated with higher parent consumption and less frequent but conditionally larger transfers.

4 Estimation

I estimate the model in two stages. In the first stage, I set parameters that are well-identified from external data or the existing literature, including the income-process parameters, which I take from Abbott et al. (2019) and map from their discrete-time estimates to the continuous-time counterparts used in the model. In the second stage, I estimate the remaining thirteen structural parameters by the simulated method of moments (SMM), matching 30 targeted moments.

4.1 Externally Set Parameters

Table 3 reports the parameters set in the first stage. The coefficient of relative risk aversion is $\gamma = 1.5$, following Abbott et al. (2019). The annual real interest rate is $r = 0.03$, following Daruich and Kozlowski (2020). The continuous-time discount rate ρ is estimated internally by SMM to match the wealth-to-income ratio and wealth distribution moments. The college costs, borrowing limits, and financial-aid schedule parameters ($\bar{g}$, $\bar{y}_{\text{efc}}$) are set externally to the Abbott et al. (2019) values, as described in Section 2.7. The income-process parameters—the education-specific ability gradients λ_e, age profiles, OU mean-reversion rates κ_e and diffusions σ_e, and entry dispersions—and the intergenerational ability parameters (ρ_θ, σ_θ, μ_θ) are likewise taken from Abbott et al. (2019). The relative price of college labor ω_C is calibrated externally to reproduce the high-school-to-college income ratio (Section 4), and the student-loan rate wedge ι_s is fixed; the wedge on working-age unsecured borrowing ι_b is likewise fixed externally at the Abbott et al. (2019) value ($\iota_b = 0.064$). The multiplicative wealth-diffusion volatility is fixed at $\sigma_a = 0.10$ per year, following Barczyk and Kredler (2014a). The lump-sum transfer arrival intensity is set externally at $\lambda_g = 0.25$ per year, so a parent has on average one opportunity to make a lump-sum transfer every four years. These opportunities arrive at random, independently of the family's wealth, so λ_g governs only how often a transfer *can* occur, not how transfer behavior varies with wealth. The wealth gradient in transfers instead comes from whether the parent chooses to exercise an opportunity—the size of the transfer region, which altruism η governs. Fixing λ_g therefore keeps the identification of η from the transfer-probability moments clean. Retirement income is estimated directly from PSID data on households where the respondent is retired and over age 67, computing the average sum of Social Security, SSI, disability, and employer pension income by education group. Finally, the intergenerational income-transmission parameter ρ_{zp}, which governs the persistence of the parent's residual income type into the child's initial productivity net of the transmitted-ability and parental-college channels, is not separately identified from those channels by the targeted moments; I therefore fix it at 0.23, a conservative value for the component of intergenerational earnings persistence—a rank–rank slope of 0.34 in Chetty et al. (2014) and an intergenerational elasticity of roughly 0.4 in Solon (1992)—not already carried by transmitted ability (ρ_θ) and the parental-college earnings premium (ζ_{pe}). The bequest fraction is fixed externally on the same logic at $\alpha_{\text{beq}} = 0.34$, the bequeathed share of the estate, with the complement absorbed by end-of-life medical and long-term-care expenses (De Nardi et al., 2010; Lockwood, 2018). A robustness check confirms that perturbing α_{beq} and ρ_{zp} leaves the targeted moments and the counterfactuals essentially unchanged.

Table 3. Externally Set Parameters

Parameter	Description	Value	Source
Preferences			
r	Real interest rate	0.03	Daruich and Kozlowski (2020)
γ	CRRA risk aversion	1.5	Abbott et al. (2019)
$\underline{c}$	Consumption floor ($000s/yr)	3.9	De Nardi (2004)
σ_a	Wealth-return volatility (/yr)	0.10	Barczyk and Kredler (2014a)
a_{ref}	Return-premium saturation scale ($000s)	100	Normalization
λ_g	Lump-transfer arrival rate (/yr)	0.25	Normalization
Demographics (years)			
T^{ret}	Parent working, overlap (child 18–42)	24	—
$T - T^{\text{ret}}$	Parent retired, overlap (child 42–48)	6	—
T_{alone}	Child alone, working (48–66)	18	—
$T_{\text{retire},c}$	Child retirement (66–72)	6	—
Ability gradients			
λ_C	Returns to ability, college	0.797	Abbott et al. (2019)
λ_{HS}	Returns to ability, HS	0.517	Abbott et al. (2019)
Earnings process (annual)			
ρ_C^{ann}	Persistence, college	0.966	Abbott et al. (2019)
ρ_{HS}^{ann}	Persistence, HS	0.952	Abbott et al. (2019)
$(\sigma_\eta^{\text{ann}})^2$	Innovation variance (both groups)	0.017	Abbott et al. (2019)
σ_C	OU diffusion, college	0.133	Abbott et al. (2019)
σ_{HS}	OU diffusion, HS	0.134	Abbott et al. (2019)
$\bar{w}$	Avg. annual earnings	$70k	Census
College costs			
ϕ_C	Annual tuition (before aid)	$6.7k	Abbott et al. (2019)
ℓ_C	Time worked in college	0.56	Census
Intergenerational ability			
ρ_θ	Persistence of log ability	0.36	Abbott et al. (2019)
σ_θ	Std. dev. of ability innovation	0.40	Abbott et al. (2019)
μ_θ	Mean of log ability	1.30	Abbott et al. (2019)
Bequests & intergenerational transmission (fixed)			
α_{beq}	Bequest fraction (share of estate to children)	0.34	De Nardi et al. (2010); Lockwood (2018)
ρ_{zp}	Intergen. income transmission ($z_p \to z_0$)	0.23	Calibrated
Financial aid & borrowing			
$\bar{g}$	Max. annual grant ($000s)	2.82	Abbott et al. (2019)
$\bar{y}_{\text{efc}}$	EFC income phase-out ($000s)	72.0	Abbott et al. (2019)
$\bar{b}_0$	Student loan limit (base)	$18.0k	Federal
ι_s	Student-loan rate wedge (fixed)	0.031	Abbott et al. (2019)
ι_b	Unsecured borrowing rate wedge (fixed)	0.064	Abbott et al. (2019)
$\underline{a}_C$	Unsecured borrowing limit, college	$85k	Abbott et al. (2019)
$\underline{a}_{HS}$	Unsecured borrowing limit, HS	$25k	Abbott et al. (2019)
Retirement income (annual)			
SS_C	Soc. Security + pension, college	$14.3k	PSID
SS_{HS}	Soc. Security + pension, HS	$13.0k	PSID

Notes: All dollar amounts in 2016 dollars. The continuous-time income process is set externally from Abbott et al. (2019) (male estimates): persistence maps to the mean-reversion rate $\kappa_e = -\ln \rho_e^{\text{ann}}$, and the OU diffusion σ_e is chosen so the stationary variance matches AGMV (Appendix F). The transitory-income standard deviations $\sigma_{\varepsilon,e}$, the loan-limit slope b_{slope}, and the return-on-wealth gradient $r_{a,\text{grad}}$ are instead estimated by SMM (Table 5); σ_a regularizes the transfer boundary (Section 2). The financial-aid schedule $g(y_p^0) = \bar{g}\max(0, 1 - y_p^0/\bar{y}_{\text{efc}})$ keys on parent income at entry; it, the loan limit $\bar{b}_0$, and the rate wedges ι_s, ι_b follow Abbott et al. (2019). Retirement income is from the PSID (respondent retired, age > 67). The lump-transfer arrival rate λ_g is a normalization (Section 2.3.4): the transfer-region geometry, not the arrival rate, governs the wealth gradient in transfer incidence.

4.2 Income Process Estimation

The income process in continuous time is an Ornstein-Uhlenbeck process $dz = -\kappa_e z\, dt + \sigma_e\, dW$ with education-specific parameters, set externally from the male estimates of Abbott et al. (2019), hereafter AGMV.[12] I map their discrete-time estimates to the continuous-time process and hold it fixed rather than estimating the diffusion coefficients within the structural routine.

In discrete time, log labor income decomposes as $\log \epsilon_j = \lambda_e \log \theta + \gamma_{e,j} + z_j$, with education-specific ability gradient λ_e, deterministic age profile $\gamma_{e,j}$, and persistent shock z_j. AGMV estimate the gradient from the NLSY79 (log hourly wages on log AFQT, by education group), the age profile from a PSID age quadratic, and the shock persistence by minimum distance on the autocovariance structure of wage residuals:

$$z^e_{iat} = \log y_{it} - \widehat{f^e(a_{it})} - \hat{\lambda}_e \log(\text{AFQT}_i)$$

$$z^e_{iat} = \rho^{\text{ann}}_e\, z^e_{i,a-1,t-1} + \eta^e_{iat}$$

$$\eta^e_{iat} \sim N(0, (\sigma^e_\eta)^2), \;\; z^e_{i0t} \sim N(0, (\sigma^e_{z_0})^2)$$

I map the discrete-time persistence to its continuous-time mean-reversion counterpart κ_e and the innovation variance to the OU diffusion σ_e by matching the autocovariance structure; Appendix F derives the mapping formulas. Table 4 reports the discrete-time estimates and their continuous-time images. The mean-reversion rates are similar across education groups ($\kappa_{HS} = 0.049$, $\kappa_C = 0.035$), and the diffusions are nearly identical ($\sigma_{HS} = 0.134$, $\sigma_C = 0.133$). Because college earnings are more persistent, the implied stationary standard deviation of the persistent shock is in fact slightly *higher* for college graduates (0.51 versus 0.43 for high school), so the model's transfer-region contraction is driven by the income *level*, not by a reduction in income risk (Section 2).[13]

[12] AGMV estimate the income process separately by gender; I use their male estimates throughout, as the model abstracts from gender differences.

[13] Figure F.1 in Appendix F illustrates the implied life-cycle income profiles by education, including the deterministic age-income component, expected income for a median-ability worker, the role of OU shock volatility, and the complementarity between ability and education.

Table 4. Income Process and Age Profile

	High-School	College
Ability gradient		
λ_e	0.517	0.797
Deterministic age profile		
$\gamma_{1,e}$	0.0673	0.1197
$\gamma_{2,e} \times 1000$	-0.670	-1.191
Persistent shock — discrete-time AR(1)		
ρ_z^{ann}	0.952	0.966
$\sigma^2_{\eta,\text{ann}}$	0.017	0.017
$\sigma^2_{z_0}$ (entry var.)	0.059	0.094
Persistent shock — continuous-time OU (mapped, fixed)		
κ_e	0.049	0.035
σ_e	0.134	0.133
σ_{z_0} (entry s.d.)	0.243	0.307
stationary s.d. $\sigma_e/\sqrt{2\kappa_e}$	0.43	0.51
Relative price of college labor (calibrated)		
ω_C	—	0.41

Notes: Unless stated otherwise, all parameters are taken directly from Abbott et al. (2019) ("AGMV", male estimates): the ability gradient λ_e, the deterministic age profile $(\gamma_{1,e}, \gamma_{2,e})$, and the discrete-time persistent-shock parameters (ρ_z^{ann}, σ_η^{ann}, σ_{z_0}). The continuous-time mean-reversion rates κ_e and OU diffusions σ_e are mapped from the AGMV persistence ρ_z^{ann} and innovation variance σ_η^{ann} as described in the text and Appendix F; the same OU also governs the parent's persistent productivity z_p over the working life. Departing from AGMV, the two education-specific transitory-income standard deviations $\sigma_{\varepsilon,e}$ are estimated by SMM rather than set to zero, so that the model can match the cross-sectional dispersion of earnings—particularly for high-school workers—that the persistent process alone understates. The relative price of college labor ω_C (ω is normalized to one for high school) is calibrated to a single target: it is set to the value (0.41) at which the simulated high-school-to-college income ratio reproduces its data counterpart, and then held fixed during the SMM. It is anchored this way rather than estimated jointly because, although it maps essentially one-to-one onto the income ratio, it also shifts college attendance; left free, the many attendance moments would pull it off the income-ratio target. The implied stationary standard deviation of the persistent shock, $\sigma_e/\sqrt{2\kappa_e}$, is 0.43 for high school and 0.51 for college.

4.3 Ability Gradient

The ability gradient λ_e in the human capital function $h(\theta, e) = (\theta/\bar{\theta})^{\lambda_e}$ governs the returns to cognitive ability by education level. I take the estimates of λ_e directly from Abbott et al.

(2019), who estimate them from the NLSY79 by regressing log hourly wages (net of the PSID age profile) on log AFQT scores, separately by education group. The key estimates—reported in Table 3—are $\hat{\lambda}_{HS} = 0.517$ and $\hat{\lambda}_C = 0.797$, implying that a one-log-point increase in ability raises college earnings by 80% but high-school earnings by only 52%. This complementarity between ability and education is central to the mechanism: the college premium is strongly increasing in ability, so removing parental transfers has heterogeneous effects across the ability distribution.

4.4 SMM Estimation

The remaining thirteen parameters—parent altruism η, psychic cost parameters κ_0, κ_θ, and σ_κ, the parent-education psychic-cost shifter κ_w, the discount rate ρ, the warm-glow weight ϕ_b, the luxury-bequest shifter $\underline{a}_{\text{beq}}$, the collateral channel slope b_{slope}, the return-on-wealth gradient $r_{a,\text{grad}}$, the intergenerational earnings premium ζ_{pe}, and the two education-specific transitory-income standard deviations $\sigma_{\varepsilon,HS}$ and $\sigma_{\varepsilon,C}$—are estimated by the simulated method of moments (SMM; McFadden, 1989; Pakes and Pollard, 1989). SMM is well suited to this setting because the model does not admit closed-form expressions for the moments of interest: the coupled HJB system and the endogenous transfer region must be solved numerically. Following Duffie and Singleton (1993) and Lee and Ingram (1991), I simulate the model at each candidate parameter vector and match the simulated moments to their data counterparts.

Let $\mathbf{x} \in \mathbb{R}^{13}$ denote the parameter vector. For each $\mathbf{x}$, I solve the model and simulate $M = 4{,}000$ dynasties, computing the model counterparts of 30 targeted moments $\mathbf{m}^{\text{sim}}(\mathbf{x})$. The estimator minimizes a weighted percentage-deviation loss with fixed diagonal weights:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \sum_m w_m \left[\frac{m_m^{\text{sim}}(\mathbf{x}) - m_m^{\text{data}}}{m_m^{\text{data}}} \right]^2, \tag{15}$$

where the weights $\{w_m\}$ are diagonal and fixed prior to estimation. Following Lee and Seshadri (2019), I do not weight all moments equally: I combine a group-size normalization with deliberate upweighting of the moments most informative about the model's mechanisms, above all the sixteen college-attendance cells, the core target. Appendix M.6 details the full weighting scheme.

4.4.1 Target Moments

The moments are organized into the following blocks, each chosen to identify specific parameters. The model is overidentified—thirteen parameters from 30 targeted moments—and although all moments contribute jointly, I flag the dominant source of identification for each parameter as the corresponding block is introduced.

College graduation rates (16 moments). College graduation rates by ability quartile $\times$ parent wealth quartile, from the NLSY97. These moments form the core of the estimation: they trace out how college attainment varies across the joint distribution of ability and parental resources, which is precisely the variation the model is designed to explain. The *ability* gradient, holding wealth fixed, identifies the psychic-cost parameters: the constant κ_0 governs the overall college rate, the log-ability gradient κ_θ how steeply attendance rises with ability, and the shock scale σ_κ the dispersion of college choices among low-ability types (a larger σ_κ raises attendance at the bottom by giving some individuals favorable cost draws). The *wealth* gradient identifies the collateral slope b_{slope}—a positive value relaxes borrowing constraints for children of wealthy parents—and, jointly with the transfer block below, the altruism parameter η; the four high-wealth, low-ability cells, where the altruistic college gift among wealthy parents shows up most clearly, are the most informative about η.

Income (3 moments). The high-school-to-college mean income ratio and the two within-education cross-sectional variances of log earnings, all measured from individual gross earnings in the NLSY97. The two dispersions are matched by the education-specific transitory-shock standard deviations $\sigma_{\varepsilon,HS}$ and $\sigma_{\varepsilon,C}$, which the SMM estimates. The income ratio is instead matched by the relative price of college labor ω_C, which—being exactly pinned by this single moment—is calibrated to it outside the SMM rather than estimated jointly (Table 4).[14]

[14] Because each agent in the model represents a single individual—not a household—the income targets use individual gross earnings (wages and salary) from the NLSY97, not household income. This is consistent with the approach in Boar (2021) and Lee and Seshadri (2019), who also target individual labor income in single-agent dynastic models. The income ratio (0.69) is computed from the cross-section of individuals aged 25–37 to maximize sample size, while the two within-education variances of log earnings (0.45 for high school and 0.42 for college) are computed from ages 30–37. Earnings are individual gross earnings (wages and salary) deflated to 2016 dollars by the CPI-U; the sample conditions on positive earnings above $5,000 and defines college graduates as those with a bachelor's degree or higher.

Transfer moments (5 moments). The college-contingent inter-vivos transfer received *conditional on attendance* and the probability of a post-college inter-vivos transfer by parent wealth quartile. The support moment is constructed from Abbott et al. (2019) as the *in-college premium*: the difference between the average yearly inter-vivos transfer a male college graduate receives while enrolled and while not enrolled (\$8,203 against \$5,433 in year-2000 dollars), cumulated over the four college years and deflated to \$15.4k in 2016 dollars, conditional on attendance. Differencing the two enrollment states nets out the imputed co-residence rent—which Abbott et al. note makes up a large fraction of the raw transfer and which depends on the child's age rather than enrollment—together with the baseline transfer a child receives regardless of college, isolating the cash support *triggered by enrolling*: precisely the object the model's enrollment gift represents. Because in-college students co-reside less than their non-enrolled peers, the premium is a conservative lower bound on the college-contingent cash transfer, and its magnitude is corroborated by the inter-vivos transfers observed in the PSID (Boar, 2021). The model counterpart is the analogous difference: the in-college transfer—the enrollment gift τ_C plus the consumption-support flows over the college years—less the transfer received over an equal-length post-college window, averaged across college attendees.[15] The four transfer-probability moments are the HRS extensive margin of Section 3.3: the probability that a parent aged 50 or older gives \$500 or more to a non-coresident child aged 26 or older within a two-year survey wave, by parent-wealth quartile. The model counterpart splits the post-college overlap into two-year windows and counts a window as positive if cumulated flow transfers plus any lump-sum transfer reach the same threshold; these moments discipline the dynamic-game transfers—and through them the lump-sum channel and the free boundary—separately from the college-support level.[16] Because the arrival intensity λ_g is fixed and wealth-blind, the steep wealth gradient in the HRS transfer probabilities (5.5 percent in the bottom parent-wealth quartile to 24.8 percent

[15] Two definitional choices matter for identification. First, conditioning on attendance: an unconditional mean would conflate the attendance *rate*, already targeted by the sixteen college-rate moments, with the transfer *size*, taxing the transfer parameters twice for any attendance misfit. Second, differencing rather than matching the in-college *level*: the level bundles in-kind co-residence that the model's cash transfers do not represent, so the premium compares like with like and avoids over-targeting transfer magnitudes; the parent consumption and wealth paths, the college block, and the extensive margin below provide complementary discipline.

[16] The HRS quartiles are computed within survey-wave cohorts; Appendix A details the sample construction for both sources.

in the top) can be matched only by the free boundary itself, which is what identifies the altruism parameter η: a higher η enlarges the transfer region, converting more Poisson arrivals into realized transfers. The college-attendance wealth gradient above provides complementary identification, since a higher η also raises the lifetime transfer burden parents face when children do not attend college.

Wealth (4 targeted moments). From the PSID linked sample: the aggregate wealth-to-income ratio, which disciplines the overall level of wealth accumulation; the median parent wealth; the p90/p50 ratio of parent wealth (7.36), which disciplines the right tail and the luxury-bequest motive; and the fraction of parents with net worth below $25,000 (0.28), which disciplines the left mass of the parent wealth distribution. The wealth-to-income ratio and median parent wealth jointly identify the discount rate ρ—a lower ρ (more patient agents) generates more accumulation, raising both—while the p90/p50 ratio identifies the luxury-bequest shifter $\underline{a}_{\text{beq}}$ and the return-on-wealth gradient $r_{a,\text{grad}}$ through their distinct incidence on the upper tail: a positive $\underline{a}_{\text{beq}}$ makes bequests a luxury good (following De Nardi, 2004), so the rich bequeath a rising share of wealth and generate the fat right tail, while $r_{a,\text{grad}}$ raises top-end concentration without shifting the median. Following the standard of the continuous-time altruism literature (Barczyk and Kredler, 2014a,b) and the dynastic models of Boar (2021) and Abbott et al. (2019)—which discipline the parent (donor) wealth level and let the parent–child wealth relationship, and indeed the entire child wealth distribution, emerge as an equilibrium object—I target the parent wealth distribution only; the child wealth distribution and the parent–child wealth-transmission moments are left untargeted, and the child wealth distribution is reported as an out-of-sample check (Section 5.2).

Education transmission (1 targeted moment). From the NLSY97: the intergenerational education gap, the difference between the college-attainment rate of children whose parents attended college and of children whose parents did not. This is the persistence statistic the mechanism speaks to—human capital rather than liquid wealth. Net of family resources, the gap identifies the parent-education taste shifter κ_w: the part of the gap that the gift and borrowing channels cannot themselves reproduce. The parent-education earnings premium ζ_{pe}, which raises a child's earning capacity when the parent attended college, is disciplined separately by the income block rather than by this moment.

Decumulation (1 moment). The parent wealth decumulation rate over ages 60–70, computed from the PSID linked sample. This moment identifies the warm-glow weight ϕ_b: a stronger motive slows drawdown in retirement, as parents derive utility from holding wealth at T. The rate of +2.5% per year—parents ages 60–70 are still accumulating—is consistent with evidence that pre-retirees keep saving for precautionary and warm-glow reasons (De Nardi, 2004; Hurd, 1990); this is the donor-wealth role the bequest motive plays in the continuous-time altruism literature, explaining why the elderly do not decumulate independent of how much reaches the child.

Table 5. SMM-Estimated Parameters

Parameter	**Description**	**Value**
Preferences		
ρ	Discount rate (annual); $\beta = e^{-\rho} = 0.90$	0.100
Parental altruism & bequests		
η	Altruism weight	0.09
ϕ_b	Warm-glow terminal wealth weight	7.02
$\underline{a}_{\text{beq}}$	Luxury-bequest shifter (De Nardi, \$000s)	72.8
College psychic cost		
κ_0	Psychic-cost constant	-0.3
κ_θ	Psychic-cost gradient in $\log(\theta)$	-0.0
σ_κ	Psychic-cost idiosyncratic shock S.D.	1.13
κ_w	Psychic-cost shifter, college parent ($e_p = C$)	-0.50
College returns & financing		
b_{slope}	Collateral channel: borrowing-limit slope in a_p	0.018
Asset returns & income risk		
$r_{a,\text{grad}}$	Return-on-wealth gradient (premium for the rich)	0.105
$\sigma_{\varepsilon,C}$	Transitory income S.D., college	0.124
$\sigma_{\varepsilon,HS}$	Transitory income S.D., high school	0.196
Intergenerational transmission		
ζ_{pe}	Parental-college premium in initial productivity z_0	1.36

Notes: The thirteen parameters estimated jointly by the simulated method of moments, minimizing the percentage-deviation distance between the 30 targeted simulated moments and their data counterparts. The bequest fraction α_{beq}, the intergenerational income-transmission coefficient ρ_{zp}, and the unsecured-borrowing wedge ι_b are fixed externally and reported in Table 3. Parameter definitions and the identification argument are given in Section 4. Externally set parameters, including the income process and the financial-aid schedule, are reported in Table 3.

5 Results

5.1 Model Fit

Table 6 reports fit to the targeted moments: college graduation by ability $\times$ parent-wealth quartile (Panel A), income (Panel B), transfers (Panel C), and the PSID wealth moments that discipline the warm-glow weight ϕ_b and the parent wealth distribution (Panel D). The model matches the overall attainment level—41% against 42%—and both gradients. Averaged over wealth, attainment rises from 24% in the lowest ability quartile to 55% in the highest, a 31-point gradient against 36 in the data. The wealth profile is monotone and close: 33% to 54% across parent-wealth quartiles against 32% to 54%, reaching 68% for a top-ability child of a top-wealth parent against 74%.

The transfer side fits in step (Panel C): mean college support is \$14.7k against \$15.4k, and the transfer-probability wealth gradient tracks the data across quartiles (5%, 9%, 14%, 23% against 5%, 10%, 16%, 25%). The high-school-to-college income ratio is somewhat under-predicted (0.49 against 0.69), so the college premium is a little too large. On the wealth moments (Panel D) the model matches the median level of parent wealth (\$73k against \$85k), the share of parents holding little wealth (28% below \$25k, as in the data), and the pace of late-life accumulation (+2.6% against +2.5% per year). It under-predicts the aggregate wealth-to-income ratio (0.46 against 1.87) and the dispersion of parent wealth (a p90/p50 ratio of 4.1 against 7.4). Both misses have the same origin: the single-agent model, abstracting from housing and business wealth, does not reproduce the wealthiest households, who hold a disproportionate share of aggregate wealth and stretch the upper tail. The features that govern when parental support is triggered—the median, the left mass, and the decumulation pace—are matched.

Intergenerational persistence in education. The persistence the mechanism speaks to is in human capital, and there the model performs well. I target the intergenerational *education* gap—the difference between the college-attainment rate of children of college-educated parents and of children whose parents did not attend—which the model matches almost exactly (0.258 against 0.252 in the data).

Table 6. Targeted Moments: Data and Model

Panel A: College Attainment by Parent Wealth and Child Ability (NLSY97)				
Wealth Quartile \ Ability Quartile	1	2	3	4
1	0.19	0.36	0.37	0.47
	(0.19)	(0.24)	(0.33)	(0.53)
2	0.24	0.35	0.42	0.48
	(0.24)	(0.30)	(0.42)	(0.53)
3	0.25	0.38	0.39	0.53
	(0.26)	(0.39)	(0.51)	(0.63)
4	0.33	0.49	0.57	0.68
	(0.33)	(0.46)	(0.62)	(0.74)

Panel B: Income Moments	Model	Data
High-School/College mean income ratio	0.49	0.69
High-School Var(log income)	0.35	0.45
College Var(log income)	0.49	0.42

Panel C: Transfer Moments	Model	Data
Mean transfer ($k/yr)	14.7	15.4
Pr(transfer > 0) — Wealth Q1	0.05	0.05
Pr(transfer > 0) — Wealth Q2	0.09	0.10
Pr(transfer > 0) — Wealth Q3	0.14	0.16
Pr(transfer > 0) — Wealth Q4	0.23	0.25

Panel D: Parent Wealth Moments	Model	Data
Wealth–income ratio	0.46	1.87
Median wealth, parents ($k)	72.7	85.4
Var(IHS wealth), parents	4.57	5.99
Parent wealth growth rate	0.026	0.025

Notes: Model moments without parentheses; data moments in parentheses. Panel A: college graduation rates by parent wealth quartile (rows) and child ability quartile (columns), NLSY97. Panel B: income moments, NLSY97. Panel C: transfer moments; Panel D: parent wealth moments, PSID linked sample; wealth in thousands of 2016 dollars. Moment definitions, sources, and the targeted/untargeted distinction are given in Section 4; sample construction in Appendix A.

5.2 Untargeted Validation

The descriptive patterns of Section 3—the consumption release, the lower probability of any transfer, and the weakly larger conditional transfer—are untargeted in the SMM objective, which disciplines the attendance gradients, the wealth distribution, decumulation, and the level of support (Section 4). They are therefore out-of-sample checks. Table 7 reports each against its model counterpart over the post-college overlap.

The model reproduces all three signs. The consumption release is +\$27.0k per year (+25%) against +\$2.5k (+8%) in the PSID; the transfer probability falls with college by 49% in the model against 21% in the HRS (Panel C of Table 9); and the conditional transfer is positive in both data and model, so the model reproduces the intensive margin. The model reproduces the qualitative pattern—including the option-value pattern that the consumption gain exceeds the fall in realized transfers, present in both the data and the model—but overstates the magnitude of the consumption release and the extensive-margin contraction.

Table 7. Untargeted Moments: Data and Model

Untargeted moment (effect of child's college)	Data	Model
Parent consumption release (college − HS, within parent-wealth quartile)		
Level (\$k/yr)	+2.5	+27.0
Percent	+8%	+25%
Transfer extensive margin: Pr(any transfer)		
College effect (%)	−21	−49
Transfer intensive margin: $\mathbb{E}[\tau \mid \tau > 0]$, *college − HS*		
Direction	+	+

Notes: Each row reports the effect of the child's college attainment on a parent-side outcome that is *not* targeted in the SMM objective (Section 4); the data column reproduces the estimates of Section 3 and the model column is computed from $M = 4{,}000$ simulated dynasties over the post-college overlap, within parent-wealth quartile. The consumption release is the college−HS difference in mean parent consumption (\$k/yr and percent). The extensive margin is the percent change in the probability of any transfer; the model counterpart is the annual transfer probability, the data counterpart the HRS wave probability, so the levels are not directly comparable and only the college effect is. The intensive margin is the college−HS difference in the transfer conditional on a positive transfer; the data moment is measured per event and the model moment as a flow, so only the sign is comparable. Source: simulated model and Section 3.

Wealth distributions. Figure 3 compares the full parent and child wealth distributions with the data; the SMM targets only summary moments of parent wealth (Panel D of Table 6) and leaves the rest untargeted. The model reproduces the parent distribution—its right skew and the concentration of wealth among higher-income households—and the broad shape and tails of the child distribution, while placing somewhat more mass at low child wealth, so children hold a little less by their late thirties than in the data.

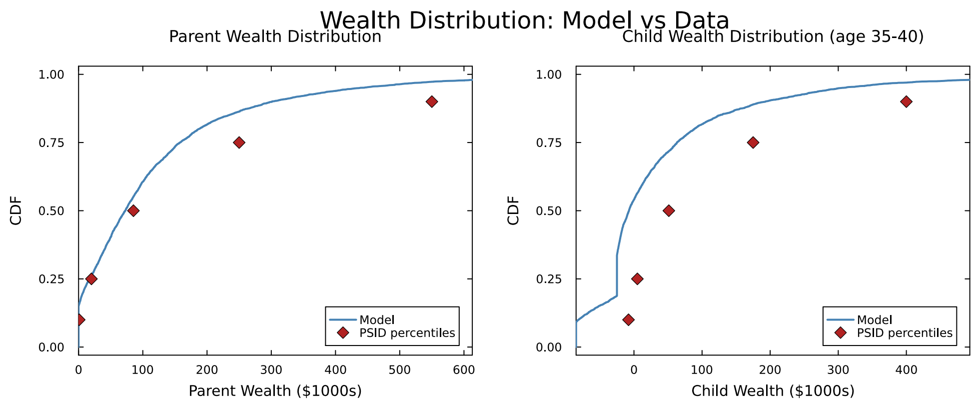


Figure 3. Parent and Child Wealth Distributions: Model vs. Data

Notes: Cumulative distribution functions of parent wealth (left panel) and child wealth at ages 35–40 (right panel) in the model (simulated) and the data (PSID). Wealth is measured in thousands of 2016 dollars.

5.3 The Role of Parental Altruism in College Attainment

To quantify altruism's role, I re-solve the model with $\eta = 0$: parents make no transfers, and children finance college through own earnings, loans, and grants. Removing altruism lowers attendance from 41% to 38% (Table 8), with the fall concentrated at the bottom of the ability distribution: the lowest-ability quartile drops from 24% to 17% (a 29% fall), while the middle and top quartiles move much less (2–6%). The effect is largest at the bottom because low-ability children are the marginal attendees—their private return to college is small, so parental support is pivotal to whether they enroll—whereas higher-ability children have a large enough return to attend on their own and are little affected. This isolates the *level* role of altruism: parental support finances enrollment, so removing it pushes marginal children out of college.

Table 8. College Attendance With and Without Parental Altruism

Ability quartile	Baseline Pr(C)	No altruism ($\eta = 0$) Pr(C)	Change (%)
Q1	0.24	0.17	-29
Q2	0.39	0.39	-2
Q3	0.44	0.43	-3
Q4	0.55	0.52	-6
Overall	0.41	0.38	-8

Notes: College attendance by child ability quartile in the baseline economy and in a counterfactual with parental altruism shut down ($\eta = 0$), so the parent makes no transfers and the child finances college through own earnings, loans, and grants. The final column reports the percentage change relative to the baseline. The model is re-solved at the estimated parameters with the altruism weight set to zero; statistics are computed from $M = 4{,}000$ simulated dynasties.

5.4 How College Reshapes the Transfer Region

College reshapes the transfer region by raising the child's income *level*—through a higher ability gradient and a steeper earnings profile—rather than by lowering income risk. The higher level holds college children further from the boundary, so a given shock is less likely to pull them into the region where the parent transfers. Table 9 and Figure 4 quantify the consequence.

Behavior inside and outside the transfer region. Panel A of Table 9 reports the child's own saving effort (which nets out the transfer itself), separating children who receive transfers from those who do not. Children who receive transfers are in the transfer region precisely because their resources are low, and both education groups run their own saving negative ($-\$2.5$k/yr for high-school and $-\$3.0$k for college children). Among children who receive no transfers the high-school child's own saving is close to balanced ($-\$0.4$k/yr), while the college child still runs it down ($-\$2.7$k/yr)—largely student-debt service, since interest on the outstanding loan balance depresses net resources early in the overlap. College children therefore receive transfers at the start just as high-school children do (Panel C); the college income premium pulls them out of the transfer region only as it accumulates.

Panel B isolates the behavioral distortion: the child's over-consumption relative to the full-commitment allocation evaluated at the same state, which holds resources fixed and so separates the Samaritan's dilemma from the level effect of the transfers themselves. Both education groups over-consume relative to the commitment benchmark, and the dollar amounts scale with the child's resources—larger outside the transfer region, where children hold more own wealth the planner would direct toward dynastic saving, and larger for college children, who earn more (inside the region, \$20.9k per year for college against \$14.0k for high school; outside, \$105.8k against \$50.1k). Because the distortion scales with resources, these dollar levels are not comparable across education groups: what makes college *reduce* the dilemma is not a smaller per-state distortion but reduced *exposure*—college children spend far less time in the transfer region (Panel C).

Exposure over the life cycle. Panel C reports exposure to the transfer region. College children are supported 48.2% of the college years—the in-college enrollment subsidy—and 10.5% of the post-college overlap, against 20.6% for high-school children. Panel (a) of Figure 4 traces this across parent age: the two profiles coincide early, while student debt holds the college child's wealth down, and diverge as the earnings premium accumulates.

Table 9. Transfer Region Exposure and Savings Behavior

	HS Child	College Child
Panel A: Child's own savings effort $s_c^{own} = ra_c + y_c - c_c$ *($k/yr)*		
When receiving transfers	-2.49	-2.96
When not receiving transfers	-0.38	-2.71
Difference (in − outside region)	-2.11	-0.26
Panel B: Over-consumption vs. full commitment $\Delta c_c = c_c^{MPE} - c_c^{FC}$ *($k/yr)*		
In transfer region	13.97	20.91
Outside transfer region	50.05	105.75
Panel C: Fraction of time receiving parental transfers		
During college (ages 18–22)	—	0.482
After college (ages 22–48)	0.206	0.105

Notes: Statistics from simulated dynasties ($M = 4{,}000$). Panel A reports the child's own saving effort $s_c^{\text{own}} = \tilde{r}_c\, a_c + y_c - c_c$, which excludes the mechanical effect of transfers on wealth accumulation, separately for child-states inside and outside the transfer region; the difference reflects the composition of resources across the two regions. Panel B isolates the behavioral distortion, reporting the child's over-consumption relative to the full-commitment allocation evaluated at the same state during the overlap: the planner pools total household resources $A = a_p + a_c$ and splits them by the optimal rule $c_c^{\text{FC}} = \beta\, c_{\text{tot}}^{\text{FC}}$ with $\beta = \eta^{1/\gamma}/(1 + \eta^{1/\gamma})$, so the comparison holds resources fixed and removes only the strategic anticipation of transfers. A positive value indicates over-consumption relative to the commitment benchmark. Panel C reports the fraction of time the child receives parental transfers, separately for the in-college years (ages 18–22, the enrollment-subsidy channel) and the post-college period (ages 22–48, the insurance channel); the in-college figure applies only to college children.

Transfer flows by parent wealth. Transfer levels follow the same logic, but only at the top of the wealth distribution (Panel (b) of Figure 4). In the top quartile a high-school child receives \$1.8k per year against \$0.8k for a college child, a 54% reduction; the reduction holds across the wealth distribution—72% in the bottom quartile and 61–73% in the middle two—so pooled across quartiles college children receive 44% less. In levels the gap is concentrated at the top (a \$1.0k difference in the fourth quartile against \$0.1–\$0.4k below), because the lump-sum channel carries most intergenerational rebalancing (Section 2.3.4) and college debt service keeps early-career resources, and hence transfer need, close to those of high-school children. The reduction reflects the income-insurance logic of college: a dollar of tuition permanently raises the child's expected income and lowers the probability of entering the transfer region, whereas a direct transfer smooths consumption without changing the child's income path.

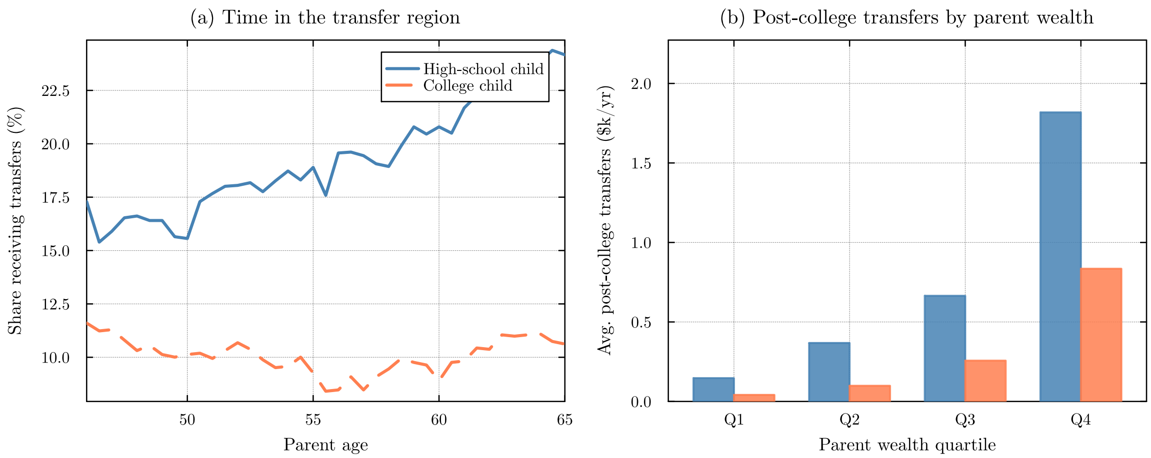


Figure 4. College, Transfer-Region Exposure, and Post-College Transfers

Notes: Simulated dynasties ($M = 4{,}000$). Panel (a) plots the fraction of children receiving positive transfers ($\tau > 0$) at each parent age over the post-college overlap window—child age 22–48, parent age 46–72—by child education. Panel (b) reports average post-college transfers received by the child (excluding the in-college years), by parent wealth quartile and child education.

6 Moral Hazard Decomposition

Parental altruism raises college attendance but generates the Samaritan's dilemma: anticipating transfers when their wealth is low, children under-save, imposing larger and more frequent transfer costs on parents. The effect on the college decision is theoretically ambiguous—anticipated in-college support subsidizes enrollment, while anticipated lifetime transfers insure non-college children against income risk. Which force dominates is quantitative. This section formalizes both forces, constructs counterfactual economies that isolate them, and reports the decomposition.

6.1 Full Commitment Benchmark and Two Opposing Forces

To isolate the effects of the Samaritan's dilemma, I first characterize the efficient allocation under full commitment. Suppose a benevolent planner commits at $t = 0$ to a state-contingent transfer schedule $\{\tau(a_p, a_c, z, t)\}$ for the entire overlap period. Because both agents pool resources under a single plan, the problem reduces to a single-agent optimization over total household wealth $A \equiv a_p + a_c$:

$$\begin{aligned}\rho V^{\mathrm{FC}}(A,z,t) = \max_{c_{\mathrm{tot}}} \Big\{ & u_{\mathrm{FC}}(c_{\mathrm{tot}}) + V_A^{\mathrm{FC}}\left[rA + y_p(t) + y_c(z,t) - c_{\mathrm{tot}}\right] \\ & + \tfrac{1}{2}\sigma_a^2 A^2 V_{AA}^{\mathrm{FC}} + \mathcal{L}_z V^{\mathrm{FC}} + V_t^{\mathrm{FC}} \Big\}\end{aligned} \tag{16}$$

Under CRRA preferences, the efficient allocation equates weighted marginal utilities, $u'(c_p) = \eta\, u'(c_c)$, so consumption shares are time-invariant: $c_c = \eta^{1/\gamma}\, c_p$. Total consumption $c_{\mathrm{tot}} = c_p(1+\eta^{1/\gamma})$ yields an aggregated flow utility $u_{\mathrm{FC}}(c_{\mathrm{tot}}) = \frac{c_{\mathrm{tot}}^{1-\gamma}}{1-\gamma}(1+\eta^{1/\gamma})^{\gamma}$, and the planner's HJB reduces to a single-state-variable problem. Full commitment differs from the baseline economy in three ways (Barczyk and Kredler, 2014a,b): it eliminates overconsumption by both agents, it makes the timing of transfers irrelevant—the planner pools the household's resources and fixes the consumption split, so it no longer matters when wealth moves from parent to child—and, most important for the college decision, it provides insurance without moral hazard. The child's college choice under full commitment follows the same probit structure as the baseline (equation 13), with the planner's continuation values replacing the baseline values.

Relative to full commitment, the Samaritan's dilemma creates two opposing forces on the child's college decision. From the parent's perspective, a college-educated child is cheaper to support: college permanently raises the income level, shifting the child away from the transfer boundary. Under full commitment, the parent could internalize this motive directly by conditioning transfers on e_c. In the baseline economy, however, the parent cannot commit: the transfer policy responds only to the current state (a_p, a_c, z), and the parent transfers whenever the child is sufficiently poor, regardless of the child's education choice.

The child chooses education to maximize own continuation utility, taking the parent's equilibrium transfer policy as given. The continuation values V_C and V_{HS} embed the entire expected path of future transfers. The child does not internalize the cost transfers impose on the parent, generating overconsumption relative to the efficient allocation whenever the child's state is near or inside the transfer region. Two forces emerge from this interaction:

Force 1: Anticipated transfers subsidize enrollment. The child starts college with zero assets and reduced income ($\ell_C \cdot y_c$ net of tuition), placing the initial state deep in the transfer region. The parent—bound by time consistency—optimally transfers resources that effectively subsidize enrollment costs. This implicit subsidy raises the child's valuation of the college path relative to full commitment, pushing attendance *above* the efficient level.

Force 2: Anticipated transfers insure non-college children. Children who forgo college face lower expected income, so they spend substantially more of their adult lives in the transfer region. The parent cannot withhold this support. The anticipation of this lifetime safety net raises the child's valuation of the high-school path—the child can under-save and over-consume, knowing that parental transfers will materialize when income falls. This insurance reduces the child's incentive to attend college, pushing attendance *below* the efficient level.

Whether Force 1 or Force 2 dominates depends on ability, parental wealth, and the income process parameters. The decomposition below quantifies these forces. In the estimated model the non-college insurance of Force 2 prevails in aggregate, lowering attendance modestly. The balance turns on the child's ability: the enrollment subsidy of Force 1 dominates only for low-ability children—moral hazard *raises* their attendance in every wealth quartile—while the non-college insurance of Force 2 dominates for medium- and high-ability children, *lowering* theirs, most sharply for medium-ability children of wealthy parents.

6.2 Decomposing the Enrollment Distortion: Pure Moral Hazard and Education Targeting

The goal of this section is to quantify how the lack of commitment distorts college enrollment and to split that distortion into two parts: the *pure moral-hazard effect*—the change in enrollment caused by the child's anticipation of future transfers—and the *education-targeting effect*—the further change available to a parent who can condition transfers on the child's education choice rather than support both states alike. I now construct two one-time-transfer counterfactuals that replace the dynamic transfer game with a single lump-sum transfer at $t = 0$. After the initial transfer the child solves the standard lifecycle problem alone, so these economies shut down the ongoing relationship entirely—both the Samaritan's dilemma and the insurance it provides—in contrast to the full-commitment benchmark, which removes the dilemma but keeps the insurance. The two counterfactuals differ in whether the parent gives the same transfer in both education states or an education-contingent menu (τ_C, τ_{HS}), which separates the pure moral hazard channel from the education-targeting channel.[17]

Counterfactual A: One-time college gift. The parent chooses a college-conditional gift τ_C at $t = 0$, paid at enrollment exactly as in the baseline:

$$\begin{aligned}\max_{\tau_C \in [0, a_p]} \quad & \Pr(C \mid \tau_C, \theta) \left[V_p^{\text{alone}}(a_p - \tau_C) + \eta \, V^{c,\text{alone}}(\tau_C \mid C) \right] \\ & + \big(1 - \Pr(C \mid \tau_C, \theta)\big) \left[V_p^{\text{alone}}(a_p) + \eta \, V^{c,\text{alone}}(0 \mid HS) \right] \end{aligned} \tag{17}$$

where $V_p^{\text{alone}}(a)$ is the parent's remaining lifetime utility with no further interaction with the child. The child's education choice follows the same probit rule as the baseline (equation 13), with the child-alone value gain replacing the coupled values:

$$\Pr(C \mid \tau_C, \theta) = \begin{cases} \Phi\left(\dfrac{G_0(\tau_C, \theta)/\bar{V} - \kappa_0 - \kappa_\theta \log \theta}{\sigma_\kappa} \right) & \text{if } G_0 > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{18}$$

where $G_0(\tau_C, \theta) \equiv V^{c,\text{alone}}(\tau_C \mid C) - V^{c,\text{alone}}(0 \mid HS)$ is the child-alone value gain from college given the gift τ_C. This economy retains the baseline's $t = 0$ decision structure—the college-conditional gift remains—and removes only the *ongoing* interaction: after the enrollment date, no further transfers flow. The education decision therefore reflects fundamentals—human capital returns net of psychic cost, given the gift—and not the anticipation of state-contingent support. Comparing baseline attendance with this economy isolates the *pure moral hazard effect*: the distortion to college enrollment caused solely by the child's anticipation of future transfers.

Counterfactual B: Education-contingent transfer menu. The parent commits at $t = 0$ to a transfer schedule that may differ by the child's education choice, choosing a transfer

[17] Appendix J provides a formal derivation of the full commitment benchmark and discusses its relationship to the one-time transfer counterfactuals. The key distinction is that the FC benchmark removes moral hazard while preserving insurance, whereas the one-time transfer counterfactuals remove both.

τ_C paid if the child attends college and a transfer τ_{HS} paid if the child chooses high school:

$$\max_{(\tau_C,\tau_{HS})\in[0,a_p]^2} \Pr(C \mid \tau_C, \tau_{HS}, \theta)\big[V_p^{\text{alone}}(a_p - \tau_C) + \eta\, V^{c,\text{alone}}(\tau_C \mid C)\big] + \big[1 - \Pr(C \mid \tau_C, \tau_{HS}, \theta)\big]\big[V_p^{\text{alone}}(a_p - \tau_{HS}) + \eta\, V^{c,\text{alone}}(\tau_{HS} \mid HS)\big] \quad (19)$$

$$\Pr(C \mid \tau_C, \tau_{HS}, \theta) = \begin{cases} \Phi\left(\dfrac{G_m(\tau_C, \tau_{HS}, \theta)/\bar{V} - \kappa_0 - \kappa_\theta \log\theta}{\sigma_\kappa}\right) & \text{if } G_m > 0, \\ 0 & \text{otherwise,} \end{cases} \quad (20)$$

where the value gain is the child's own, evaluated at the education-specific endowments:

$$G_m(\tau_C, \tau_{HS}, \theta) \equiv V^{c,\text{alone}}(\tau_C \mid C) - V^{c,\text{alone}}(\tau_{HS} \mid HS).$$

where $\kappa(\theta, \varepsilon) \geq 0$ is the psychic cost of college (zero for the HS alternative). This menu is the efficient one-time-transfer benchmark: it nests Counterfactual A (the restriction $\tau_{HS} = 0$) and an unconditional one-time transfer (the restriction $\tau_C = \tau_{HS}$), and because it relaxes both restrictions it weakly dominates them in parent welfare. Under the menu, the altruistic parent can support a child in the high-school state directly, so a low-return child is not pushed into college merely to capture parental resources. Comparing Counterfactual B with Counterfactual A isolates the *education-targeting effect*: the change in attendance when the parent can direct transfers by education rather than give the same endowment in both states.

Decomposition. Define:

$$\Delta_{\text{MH}}^{\text{pure}}(\theta, a_p) = \Pr(\text{College} \mid \text{baseline}) - \Pr(\text{College} \mid \text{college gift}) \quad (21)$$

$$\Delta_{\text{targ}}(\theta, a_p) = \Pr(\text{College} \mid \text{college gift}) - \Pr(\text{College} \mid \text{menu}) \quad (22)$$

$$\Delta_{\text{MH}}^{\text{total}}(\theta, a_p) = \Delta_{\text{MH}}^{\text{pure}} + \Delta_{\text{targ}} \quad (23)$$

A positive $\Delta_{\text{MH}}^{\text{pure}}$ indicates that the baseline raises college attendance relative to the no-moral-hazard economy—Counterfactual A, the one-time college gift, in which parent and child interact only at $t = 0$; a negative value indicates that the ongoing game depresses attendance. The education-targeting component Δ_{targ} isolates the change in enrollment when the parent can direct transfers by education: a positive sign indicates that targeting *lowers* attendance, because the parent diverts low-return children out of college and supports them in the high-school state instead.

Results. Table 10 reports the decomposition.

Table 10. Moral Hazard Decomposition of College Attendance

	Overall	Parent Wealth Quartile			
		Q1 (Low)	Q2	Q3	Q4 (High)
Panel A: College attendance rate (%), all ability types					
Baseline	40.7	32.9	36.9	38.9	54.2
College gift (no MH)	43.0	27.7	35.2	45.8	63.4
$\Delta_{\text{MH}}^{\text{pure}}$ (pp)	−2.3	+5.2	+1.7	−6.9	−9.2
Panel B: Pure moral-hazard effect Δ_{MH}^{pure} (pp), by ability					
Low ability		+16.5	+18.1	+12.8	+16.0
Medium ability		−0.9	−8.0	−17.4	−26.7
High ability		−5.1	−6.4	−14.6	−10.1
Panel C: Optimal college gift τ_C ($000s), by ability					
Low ability		0.0	3.2	9.2	23.6
Medium ability		0.0	11.0	30.5	75.2
High ability		0.0	6.4	15.1	45.9

Notes: Panel A reports average college attendance (in percent) under the baseline and under the no-moral-hazard economy with a one-time college gift, in which parent and child interact only at $t = 0$ and the child solves the lifecycle problem alone thereafter; the "Overall" column averages across wealth quartiles. $\Delta_{\text{MH}}^{\text{pure}} = \Pr(\text{College} \mid \text{baseline}) - \Pr(\text{College} \mid \text{college gift})$, in percentage points, isolates the overconsumption distortion. Panel B disaggregates $\Delta_{\text{MH}}^{\text{pure}}$ by child ability. Panel C reports the optimal college gift τ_C by ability. The education-contingent menu (τ_C, τ_{HS}) barely differs from the college gift—the optimal high-school transfer τ_{HS} never exceeds $4k—so the education-targeting effect is negligible and is omitted here. Source: simulated model ($M = 4{,}000$ dynasties).

Aggregate result. In the estimated model the ongoing game *lowers* aggregate attendance: $\Delta_{\text{MH}}^{\text{pure}} = -2.3$ percentage points (40.7% baseline against 43.0% under the one-time college gift), so on average the non-college insurance of Force 2 outweighs the in-college subsidy of Force 1. This aggregate masks sharp heterogeneity across wealth: moral hazard *raises* attendance in the bottom two quartiles (+5.2 and +1.7 pp), where the in-college subsidy finances enrollment for liquidity-constrained families, and *lowers* it in the top two (-6.9 pp in the third quartile and -9.2 pp in the fourth), where families are wealthy enough that the non-college insurance draws marginal attendees onto the high-school path. The education-targeting effect is negligible—the education-contingent menu barely differs from the college gift—so the total distortion essentially equals the pure effect ($\Delta_{\text{MH}}^{\text{total}} = -2.2$ pp). (Panel A is computed on the simulated population, so its baseline coincides with Table 6.)

Decomposition by ability. Panel B disaggregates the pure effect by ability and wealth, revealing large, offsetting effects beneath the aggregate. Moral hazard *raises* attendance for low-ability children in every wealth quartile—the pure effect runs from $+12.8$ to $+18.1$ pp, the in-college subsidy of Force 1 financing enrollment for children at the margin of affordability—and *lowers* it for medium- and high-ability children, for whom the non-college insurance of Force 2 dominates. The reduction is largest for medium-ability children and deepens with parental wealth, from -0.9 pp in the bottom quartile to -26.7 pp in the top; for high-ability children it ranges from -5.1 to -14.6 pp. The education-targeting effect is uniformly negligible: the optimal high-school transfer never exceeds \$4k, so conditioning support on enrollment moves almost no one. The distortion the lack of commitment creates therefore operates through the *pure* moral-hazard channel—the anticipation of ongoing state-contingent support—rather than through the parent's inability to target transfers by education.

Read with the no-altruism comparison of Section 5.3, the two exercises separate altruism's two roles. Shutting altruism down entirely (Table 8) lowers attendance by three points, concentrated among low-ability children—the *level* effect of altruism finances enrollment for those at the margin of affordability. Removing only the ongoing game while keeping the one-time college gift (Table 10) *raises* attendance by about two points on net, so the ongoing relationship *lowers* enrollment relative to the one-time benchmark. The lack of commitment is thus not a uniform force on enrollment: it raises attendance for low-ability children and lowers it for the medium- and high-ability, with a small negative net effect.

7 Welfare Analysis

This section quantifies the welfare consequences of the Samaritan's dilemma—weighing the cost of moral hazard against the insurance value of ongoing transfers—and evaluates a free-college policy with and without commitment.

7.1 Consumption-Equivalent Variation

I measure welfare using the consumption-equivalent variation (CEV): the permanent proportional change in consumption that makes an agent indifferent between the baseline (Samaritan's dilemma) economy and an alternative. Under CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, CRRA homogeneity implies that scaling all consumption flows by a factor $(1+\Delta)$ scales lifetime utility by $(1+\Delta)^{1-\gamma}$. Inverting gives the CEV

$$\Delta = \left(\frac{V^{\text{alt}}}{V^{\text{base}}}\right)^{\frac{1}{1-\gamma}} - 1, \tag{24}$$

where $\Delta > 0$ means the alternative is welfare-improving.

For the child, $V^c = \int_0^T u(c_c)\, e^{-\rho t}\, dt$ and equation (24) applies directly. For the parent, $V^p = \int_0^T [u(c_p) + \eta\, u(c_c)]\, e^{-\rho t}\, dt$ captures both own and altruistic consumption. Because u is homogeneous of degree $1-\gamma$, scaling both c_p and c_c by $(1+\Delta)$ scales V^p by $(1+\Delta)^{1-\gamma}$. The parent CEV therefore measures the permanent household-wide consumption change—own plus the altruistic component—that achieves indifference.

I compute CEV at each grid point (a_p, θ) by evaluating the value functions at the $t = 0$ decision state ($z = 0$), with the college branch at the post-gift state $(a_p - \tau_C^*, a_c = \tau_C^*)$ and the high-school branch at $(a_p, 0)$, integrating over the college-choice psychic-cost shock. I then average over the simulated population: each grid CEV is weighted by the model's stationary distribution of parent wealth and child ability, using the same quartile sort as the targeted moments, so the welfare averages are comparable to the model-fit and decomposition tables rather than to a uniform grid.

7.2 Welfare Across Commitment Structures

Table 11 reports CEV for three alternative economies relative to the baseline. The first two rows of Panel A correspond to the no-moral-hazard counterfactuals from Section 6: the one-time college gift eliminates ongoing transfers and pays the child only in the college state, while the education-contingent menu also lets the parent transfer in the high-school state (τ_C, τ_{HS}). Both allocations remove the Samaritan's dilemma by construction—the parent commits to a transfer schedule at $t = 0$ and the child solves an independent problem thereafter. In both, the parent's own continuation value is the single-agent parent HJB of Appendix M.7, solved with the same consumption floor, bequest motive, and wealth diffusion as the baseline coupled value, so the consumption-equivalent comparison differences like with like rather than against a permanent-income approximation.

Measured by the dynastic (η-weighted) CEV—the object the parent optimizes—both one-time-transfer economies are *worse* than the baseline (college gift -12.0%, menu -11.9%), while full commitment delivers $+15.4\%$. This contrast is the central result. The one-time transfers eliminate the Samaritan's dilemma but sever the ongoing relationship, removing the insurance repeated transfers provide against the child's income risk; the dynasty loses heavily, and the child—no longer exposed to moral hazard but also no longer insured against bad income draws—is itself worse off (-3.4% and -4.1%). The menu is barely distinguishable from the college gift on the dynastic measure—the high-school transfers it adds are nearly welfare-neutral for the parent even as they reshape enrollment (Section 6). Full commitment removes moral hazard while *preserving* insurance and is the only economy that raises dynastic welfare; the gain accrues to the parent while the child loses sharply (-47.6%), as commitment strips the child of the ability to extract resources by under-saving. Because $\eta = 0.09$, the dynastic measure $V^p_{\text{own}} + \eta V^c_{\text{own}}$ is dominated by the parent's own utility, so the positive figure is the value of commitment *to the parent*—the rents recovered by no longer being exploited—not a Pareto improvement.

Table 11. Welfare Analysis: CEV (%) Relative to Baseline

	Dynastic CEV (%)	Child CEV (%)
Panel A: Commitment Structure		
Baseline (Samaritan's dilemma)	0.00	0.00
College gift (no MH)	-11.95	-3.41
Education-contingent menu (no MH)	-11.90	-4.06
Full commitment (planner)	15.44	-47.60
Panel B: Free College Policy ($\phi = 0$, lump-sum tax)		
Free college, no commitment	-0.69	5.18
Free college, with commitment	-11.92	0.90
Panel C: Value of commitment for the free-college policy (CEV points)		
Value of commitment	-11.23	-4.29

Notes: Consumption-equivalent variation (CEV) measures the permanent proportional change in consumption that achieves indifference between each economy and the baseline. The *Dynastic CEV* column reports the η-weighted value $V_p^{\text{own}} + \eta V_c^{\text{own}}$ that the parent optimizes; the *Child* column uses the child's own value. Because the model contains no paternalistic motive, the dynastic CEV is precisely the consumption, insurance, and bequest value of the Samaritan's-dilemma mechanism. All averages are weighted by the model's stationary population distribution of (a_p, θ). Panel A compares commitment structures holding tuition at its estimated value. Panel B sets $\phi = 0$ and funds the policy with a budget-balancing lump-sum tax T^{LS} (equation 25) solved in general equilibrium for each regime. Panel C reports the value of commitment for the free-college policy—the difference in its CEV gain under commitment versus the time-consistent equilibrium, $\Delta^{\text{cmt}} - \Delta^{\text{no cmt}}$, in CEV points—reported as a signed level rather than a ratio so that it remains well defined when gains are small or negative. Source: model solution at estimated parameters.

The parent would like to implement full commitment but cannot, precisely because of altruism. Committing to give the child one transfer and then stand back is not time-consistent: ex ante the parent would promise to transfer τ_0 once and induce the child to self-insure, but ex post, whenever the child draws a bad shock and nears the transfer region, $\eta > 0$ makes the parent strictly prefer to transfer. The child anticipates this and under-saves, so the promise unravels—the only time-consistent (Markov-perfect) policy is the baseline. The commitment allocations are benchmarks, not policies the parent can choose unilaterally. This is why college matters: a tuition payment permanently raises the child's income and moves it away from the transfer region for life, achieving through human capital the commitment that cash cannot.

7.3 Free-College Policy

I consider a policy that eliminates tuition ($\phi = 0$) and finances the revenue shortfall through a lump-sum tax T^{LS} on initial parent wealth. No financial aid is needed when tuition is zero, so $\bar{g} = 0$. The tax is set in general equilibrium to balance the government budget: it must cover tuition for the children who enroll, while enrollment itself responds to the tax through the parent's post-tax wealth. The per-dynasty tax therefore solves the fixed point

$$T^{\mathrm{LS}} = 4\phi \cdot \mathbb{E}\big[\Pr\big(\mathrm{C} \mid a_p - T^{\mathrm{LS}},\, \theta\big)\big]\,, \tag{25}$$

where ϕ is the annual tuition defined above, 4ϕ is the four-year cost of tuition the government covers per enrolled child, and the expectation is over the (a_p, θ) distribution. Enrollment, transfers, and value functions are all evaluated at the post-tax wealth $a_p - T^{\mathrm{LS}}$. Because enrollment differs across the two behavioral regimes, the budget-balancing tax is solved separately for each. Parents with $a_p < T^{\mathrm{LS}}$ have post-tax wealth clamped at zero; they still benefit because their child pays no tuition.

I solve the free-college economy under two regimes. Under *no commitment*, the model retains its Markov-perfect equilibrium structure—parents and children interact strategically with $\phi = 0$—and the Samaritan's dilemma persists. Under *commitment*, the parent commits to a one-time education-contingent menu (τ_C, τ_{HS}) at $t = 0$ and the child solves independently with $\phi = 0$.

Panel B reports the free-college results. Under no commitment, free college raises enrollment modestly (40.7% to 46.8%)—the parent already finances much of college through the gift and ongoing support—so the policy acts largely as a transfer to the family: the child gains +5.2% while the dynasty is roughly neutral (−0.7%, despite a lump-sum tax of present value \$12.5k), with the dynastic effect positive only in the bottom quartile (Table 12). Under commitment, enrollment rises to 56.0%—mostly from removing the non-college insurance, not the tuition subsidy—but the dynasty bears a heavy burden (−11.9%) and the child is essentially unchanged (+0.9%), for a sharply negative dynastic change. That burden is not the policy but the cost of severing the ongoing relationship, the same loss the one-time menu carries in Panel A. The lesson reverses the usual commitment story: free college delivers more to the family precisely *when* the parent cannot commit and the ongoing relationship is preserved.

Panel C reports the *value of commitment* for the policy—the dynastic CEV gap $\Delta^{\text{cmt}} - \Delta^{\text{no cmt}}$, in consumption-equivalent points, signed rather than a ratio so it stays interpretable when gains are small or negative. It is *negative* (about -11.2 points): committing to the one-time menu is worse than letting the policy play out under the ongoing relationship, because the implementable form of commitment forfeits the insurance value of repeated transfers. The permanent income increase from college, not a transfer schedule, is the commitment technology that matters; the first-best full-commitment planner of Panel A is not implementable by the altruistic parent. The shortfall widens steeply with wealth—from -2.7 points in the bottom parent-wealth quartile to -25.3 in the top (Table 12)—because wealthier families hold more of their insurance in the ongoing relationship and lose more by replacing it with a fixed schedule.

Table 12. Free-College Welfare and the Value of Commitment by Parent Wealth Quartile

	Dynastic CEV (%)		Value of
Wealth Q	No commit	Commit	commitment (pp)
Q1	0.55	-2.11	-2.66
Q2	-1.08	-6.46	-5.38
Q3	-1.43	-13.03	-11.60
Q4	-0.80	-26.08	-25.28

Notes: Consumption-equivalent variation (CEV, %) of the free-college policy relative to the baseline, by parent wealth quartile, all on the η-weighted dynastic value the parent optimizes. The first two columns report the dynastic CEV under the no-commitment (Markov) equilibrium and under the education-contingent menu commitment. The final column is the value of commitment for the policy—the dynastic CEV gap $\Delta^{\text{cmt}} - \Delta^{\text{no cmt}}$, in CEV points. Because the model contains no paternalistic motive, the dynastic CEV is the consumption, insurance, and bequest value of the mechanism. All averages are weighted by the model's stationary population distribution of (a_p, θ) within each parent-wealth quartile. Source: model solution at estimated parameters.

8 Conclusion

This paper argues that college serves as a commitment device against the Samaritan's dilemma. In matched parent-child data, parent consumption is higher when children attend college and transfers are less frequent though conditionally larger, with the consumption gain far exceeding the fall in realized transfers—the hallmark of the mechanism: college shrinks the transfer region, releasing the precautionary resources parents hold to stand ready to support their children.

The net effect of the Samaritan's dilemma on college attendance is theoretically ambiguous—the lack of commitment creates an enrollment subsidy (Force 1) that raises attendance and an insurance value (Force 2) that depresses it—and which force dominates, and for whom, is a quantitative question. The structural decomposition makes that question answerable: it separates the pure moral-hazard effect from the education-targeting effect and traces both across the joint distribution of ability and parental wealth, where the balance of the two forces differs systematically across families. Whether eliminating the friction improves welfare depends on what replaces it: the ongoing transfer relationship that distorts the college margin also insures children against income risk, so severing it with a one-time transfer can leave the dynasty worse off, whereas a commitment that preserves insurance raises welfare. The welfare effects of a free-college policy likewise hinge on whether parents can commit. The same altruism parameter η that generates the Samaritan's dilemma also drives parents to invest in college as a way to mitigate it.

The framework carries implications for the design of college subsidies and financial aid. Because tuition policy operates on a margin embedded in an ongoing family relationship, its incidence is not confined to students: lowering the price of college reallocates resources within the dynasty, and how the gains divide between parent and child depends on the transfer relationship the policy leaves in place. The welfare effects of financial aid therefore depend on how policies interact with private transfers: grants that crowd out parental contributions may be less effective at raising enrollment than those that complement family support, while policies that expand parents' capacity to transfer without encouraging education may deepen the Samaritan's dilemma.

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A PSID Sample Construction

This appendix describes the construction of the analysis sample from the Panel Study of Income Dynamics (PSID). The sample combines four PSID data products: the Family Interview File, the Individual File, the Transition into Adulthood Supplement, and the Family Identification Mapping System. I use the thirteen biennial waves covering 1999 through 2023. The exact survey variable codes underlying each measure are documented in the replication package.

A.1 Data Sources

Family Interview File. The core source is the PSID family-level interview file, which provides household-level information on demographics, income, wealth, employment, and housing tenure for each survey wave from 1999 to 2023. A small number of supplementary measures that are not part of the main interview module—principally liquid savings and detailed industry classifications—are drawn from a companion family-level release and merged by household and wave.

Individual File. The PSID individual file provides person-level information for every sample member, including age, relationship to the household head, position within the family unit, and—beginning in 2013—years of completed education.

Transition into Adulthood Supplement. This supplement covers young adults aged roughly 18 to 28 in the 2015 and 2017 waves. I use it to confirm college enrollment directly, complementing the education-based measures available in the main files.

Family Identification Mapping System. This product provides the intergenerational links between children and their parents. Each child record carries identifiers for up to four parent types—biological father, biological mother, adoptive father, and adoptive mother—together with a generation marker that distinguishes original 1968 sample members from their children, grandchildren, and later descendants. The target sample consists of third-generation and later children (born roughly 1975–2005, reaching college age between 1993 and 2023) matched to their parents.

A.2 Variables Used in the Analysis

Table A.1 lists the measures drawn from the PSID, grouped by concept, together with the survey years over which each is available and the data product from which it is obtained. Variable names in the table are descriptive; they correspond to underlying PSID concepts rather than to any particular survey code.

Table A.1. Measures Drawn from the PSID

Measure	Waves	Source
Panel A: Demographics		
Age of household head	1999–2023	Family File
Age of spouse or partner	1999–2023	Family File
Number of children in the family unit	1999–2023	Family File
Age of youngest child	1999–2023	Family File
State of residence	1999–2023	Family File
Race of household head	1999–2023	Family File
Education of household head	1999–2023	Family File
Education of spouse or partner	1999–2023	Family File
Panel B: Income		
Total family income	1999–2023	Family File
Labor income of household head	1999–2023	Family File
Labor income of spouse or partner	1999–2023	Family File
Panel C: Wealth		
Total net worth (including home equity)	1999–2023	Wealth Module
Net worth excluding home equity	1999–2023	Wealth Module
Home equity	1999–2023	Wealth Module
Retirement accounts (IRAs and annuities)	1999–2023	Wealth Module
Liquid savings	2003–2023	Family File
Stocks and mutual funds	1999–2023	Wealth Module
Vehicles	1999–2023	Wealth Module
Other assets	1999–2023	Wealth Module
Other real estate	1999–2023	Family File
Student loan debt	2011–2023	Family File
Other debts	2013–2023	Family File
Panel D: Employment		
Employment status of head	1999–2023	Family File
Currently working (indicator)	1999–2023	Family File
Reason for leaving last job	1999–2023	Family File
Industry of employment	2003–2023	Family File
Occupation of employment	2003–2023	Family File
Panel E: Consumption		
Food expenditure	1999–2023	Family File
Housing expenditure	1999–2023	Family File
Transportation expenditure	1999–2023	Family File
Health-care expenditure	1999–2023	Family File
Clothing expenditure	2005–2023	Family File
Recreation expenditure	2003–2023	Family File

continued on next page

Table A.1 (continued)

Measure	Waves	Source
Panel F: Individual-Level		
Relationship to household head	1999–2023	Individual File
Position within the family unit	1999–2023	Individual File
Age of individual	1999–2023	Individual File
Years of completed education	2013–2023	Individual File

Notes: All monetary measures are deflated to 2016 dollars using the CPI-U. Wealth components for 1999–2007 use the PSID imputed wealth supplement; from 2009 onward they are recorded directly in the family interview file. Liquid savings is unavailable before 2003. Industry and occupation classifications change granularity in 2017; I harmonize them to a consistent classification (see below).

A.3 Sample Construction

The analysis sample is built in six sequential steps.

Step 1: Assemble the family-year panel. For each of the thirteen waves I extract the wave-specific family-level measures, place them on a common set of definitions, and stack the waves into a single panel with one observation per family per wave. Supplementary measures available only in the companion family release—liquid savings and detailed industry codes for 2003–2015—are merged by household and wave. I harmonize the industry and occupation classifications across the 2017 change in granularity, retaining a coarser classification that is consistent across all waves, and I define an involuntary job separation as a separation due to a plant closing or a layoff.

Step 2: Assemble the individual-year panel. I reshape the individual file into a person-by-year panel containing each member's position in the family unit, relationship to the household head, age, and (from 2013 onward) years of completed education. A time-invariant person identifier, constructed from each member's 1968 family lineage number and within-lineage person number, links individuals across waves and underlies the parent–child match.

Step 3: Construct the household panel. The family-year and individual-year panels are merged by household and wave. I retain current family-unit members who are the household head, the legal spouse, or a cohabiting partner. Including spouses ensures that a parent who is not the household head—typically the mother in a two-parent household—can still be matched to her children. Rare duplicate person-year records are resolved by giving preference to head status.

Step 4: Attach intergenerational links. Using the Family Identification Mapping System, each child is linked to the household of one parent. When more than one parent is available, the link follows a fixed priority: biological father, then adoptive father, then biological mother, then adoptive mother. Children with no matchable parent are dropped.

Step 5: Identify college attendance. I combine three approaches to maximize coverage across all thirteen waves. For 2013 onward, where years of completed education are recorded at the individual level, a child is coded as attending college if completed education lies strictly between high-school completion and a bachelor's degree at ages 18–24, or if completed education rises between consecutive waves above the high-school level before age 26. For 1999–2011, where individual education is not yet recorded, children aged 18–22 whose highest education observed in later waves exceeds high school are coded as having attended college. For the 2015 and 2017 waves, the direct enrollment indicators from the Transition into Adulthood Supplement take precedence and override the education-based codes.

Step 6: Form the parent-household sample. The child–parent records are merged with the household panel, collapsed to one observation per parent household and wave, and augmented with the number of children currently enrolled in college. I then merge state-level college costs—in-state public tuition and private tuition by state and year, both from the Integrated Postsecondary Education Data System (IPEDS)—and deflate all monetary measures to 2016 dollars using the CPI-U. Finally, I assign each parent to a permanent-income group based on average real household income observed while the head is aged 26–59 (with overall average income used as a fallback when the head is never observed in that range). The groups are: \$30,000 to \$60,000; \$60,000 to \$100,000; \$100,000 to \$160,000; and above \$160,000. Households with average income below \$30,000 or otherwise unclassifiable are dropped.

A.4 Sample Restrictions

I drop observations meeting any of the following conditions:

1. Missing or invalid demographic information (age of head or spouse not reported).
2. Missing or invalid education for the head or spouse.
3. Top-coded or missing wealth or income (any wealth or income component recorded with a PSID missing or refused code).
4. Negative total net worth.
5. Internally inconsistent portfolios (total net worth below home equity, or below other real estate).
6. Implausible portfolio shares (home-equity share below -40%, other-real-estate share below -40%, or vehicle share above 50% of total net worth).
7. Average real income below $30,000 or otherwise outside the classification bounds.

A.5 Variable Definitions

Household consumption is the sum of expenditures on food at home, food away from home, housing (rent or imputed rent for homeowners), utilities, transportation, health care, education, and child care; from 2005 onward, clothing, recreation, and vacation are also included. Income is total household income before taxes. Wealth is total net worth: the sum of home equity, other real estate, vehicles, farm and business assets, stocks, checking and savings accounts, bonds, retirement accounts, and other assets, minus all debts. Inter-vivos transfers are recorded separately for transfers from parents to children and from children to parents, and include cash gifts, help with expenses, and regular financial support. The key explanatory variable throughout the empirical analysis is child college attainment, an indicator for whether the child holds a bachelor's degree or higher. To condition on parental resources, I construct within-cohort parent income quartiles, ranking each parent against others born in the same year.

A.6 Key Constructed Variables

Portfolio shares. Each asset class—retirement accounts, home equity, other real estate, and vehicles—is expressed as a share of total net worth and used to screen out internally inconsistent portfolios (see the sample restrictions above).

College-cost instrument. The instrument used in the empirical analysis interacts the (log) in-state tuition in the parent's state of residence with the number of the parent's children currently enrolled in college, so that exposure to tuition costs is increasing in the number of children of college age.

Involuntary job separation. An indicator equal to one when the head left the previous job because of a plant closing or a layoff, and zero otherwise.

Real values. All monetary values are deflated to 2016 dollars using the CPI-U for all urban consumers. Log transformations are applied where indicated in the regression specifications.

A.7 Consumption Data

PSID consumption expenditure data are available for every wave from 1999 onward. The consumption measure aggregates spending across six categories: food (at home and away), housing (rent, utilities, property taxes, and insurance), transportation (vehicle costs, fuel, parking, and public transit), health care (insurance premiums and out-of-pocket spending), clothing (available from 2005), and recreation (available from 2003). Clothing and recreation are added to the aggregate only in the years in which they are collected.

A.8 Merge Attrition

Table A.2 documents the sample size at each stage of construction: from the full PSID individual file, through the intergenerational link to parents, to the merge with household-level consumption, income, and wealth. Panel B breaks the parent–child links down by relationship type, and Panel C quantifies the observations lost at each merge stage. The main source of attrition is the requirement that the matched parent appears as a household head or spouse in the family interview file during the sample period.

Table A.2. PSID Sample Construction and Merge Attrition

Stage	Observations	Unique IDs	% Retained
Panel A: Building the Analysis Sample			
Individual file (all persons × years)	441 158	45 386	—
After college identification	441 152	45 386	100.0
Matched to FIMS (child–parent link)	358 874	34 916	81.3
Matched to parent household (hh_data)	148 541	10 492	41.4
Collapsed to parent-HH × year	71 896	10 492	—
After sample restrictions	62 320	10 145	86.7
Panel B: Parent Identification Source			
Linked via biological father	297 284		
Linked via adoptive father	4114		
Linked via biological mother	56 856		
No identifiable parent (dropped)	0		
Panel C: Merge Losses			
Individuals not in FIMS	82 278		
Child–years without parent in hh_data	210 333		
Parent–years without state tuition data	16 488		
Final analysis sample	62 320	10 145	
of which: HH–years with child in college	6566		

Notes: Children are linked to parents via the Family Identification Mapping System. The parent–child link follows a hierarchical priority: biological father, adoptive father, biological mother, adoptive mother. "Unique IDs" refers to unique persons (first two rows) or unique parent households (remaining rows). "% Retained" is relative to the previous stage. All monetary measures are winsorized at the 1st and 99th percentiles and deflated to 2016 dollars.

B NLSY97 Sample Construction

This appendix describes the construction of the NLSY97 analysis sample. The National Longitudinal Survey of Youth 1997 (NLSY97) tracks a nationally representative cohort of 8,984 Americans born between 1980 and 1984, interviewed annually from 1997 through 2011 and biennially thereafter. I use the NLSY97 for two purposes: constructing college graduation rates by ability and parent-wealth quartile (the sixteen cell-level moments used in estimation), and estimating the education-specific income process. The exact survey variable codes underlying each measure are documented in the replication package.

B.1 Data Sources

Baseline and demographics. The baseline survey provides each respondent's gender, race and ethnicity, birth year, and census region, together with parental education for the biological mother and father and family income during the late 1990s and early 2000s.

Cognitive ability and academic performance. Cognitive ability is measured by the math-verbal percentile from the computer-adaptive aptitude battery administered in 1999. High-school transcripts provide SAT and ACT scores and overall high-school grade point average.

Schooling module. The schooling module records detailed college enrollment and financing information at the college-by-term-by-year level: tuition charged, room and board, grants and scholarships, total loans, out-of-pocket expenses, credit hours attempted, and withdrawal from college.

Income and transfer module. The income module records annual labor income and annual hours worked (used to estimate the income process) and, for the college-age years, total parental transfers and parental allowances.

B.2 Variables Used in the Analysis

Table B.1 lists the measures drawn from the NLSY97, grouped by concept, together with their coverage and source. Variable names are descriptive and correspond to underlying survey concepts rather than to any particular survey code.

Table B.1. Measures Drawn from the NLSY97

Measure	**Years**	**Source**
Panel A: Demographics and Ability		
Gender	1997	Baseline
Race and ethnicity	1997	Baseline
Birth year	1997	Baseline
Math-verbal aptitude percentile	1999	Aptitude battery
SAT verbal and math scores	Cumulative	Transcript
ACT composite score	Cumulative	Transcript
Overall high-school GPA	Cumulative	Transcript
Panel B: Family Background		
Biological mother's highest grade	1997	Baseline
Biological father's highest grade	1997	Baseline
Gross family income	1997–2003	Annual
Census region at baseline	1997	Baseline
Panel C: College Enrollment and Costs		
Tuition charged per term	1997–2017	Schooling
Grants and scholarships	1997–2017	Schooling
Total student loans	1997–2011	Schooling
Out-of-pocket college expenses	1998–2001	Schooling
Room and board	1998–2001	Schooling
Credit hours attempted	1997–2017	Schooling
Panel D: Education Attainment		
Highest grade attended	1998–2003	Annual
Highest degree ever received	Cumulative	Cumulative
Withdrawal from college	Annual	Schooling
Panel E: Income Process		
Annual labor income	1997–2011	Income
Annual hours worked (all jobs)	1997–2007	Employment
Panel F: Parental Transfers		
Total parental transfers	1997–2003	Income
Parental allowance	1997–2003	Income

Notes: Cumulative measures are computed by NLS staff from the respondent's full survey history. The aptitude percentile is the math-verbal score from the computer-adaptive battery administered in 1999. Schooling-module measures are recorded by college enrollment, term, and survey year. Family income refers to the gross household income of the respondent's family of origin.

B.3 Sample Construction

Step 1: Assemble the respondent panel. I retain core demographics (gender, race, birth year, and census region), parental education for the biological mother and father, cognitive ability (the 1999 math-verbal aptitude percentile), academic performance (SAT and ACT scores and high-school GPA from transcripts), and family income from 1997 to 2003.

Step 2: Build the college-spell panel. The detailed schooling measures are organized into panels indexed by individual, college enrollment, term, and year—tuition, grants and scholarships, total loans, out-of-pocket expenses, room and board, credit hours, and withdrawal—and merged into a unified college-spell panel.

Step 3: Identify education attainment. College attendance is identified from the highest grade attended (available 1998–2003): a respondent is coded as having attended college if the maximum grade attended exceeds high school. College graduation is defined as attaining a bachelor's degree by age 25, using the cumulative highest-degree measure.

Step 4: Estimate the income process. Using annual labor income and annual hours worked, I construct hourly wages by education group. Following Abbott et al. (2019), I estimate the high-school-to-college income ratio and the education-specific income standard deviations, which provide three of the moments used in estimation.

Step 5: Assemble parental transfers. Parental financial transfers and allowances are organized into a person-year panel for 1997–2003. These measure financial support from parents during the college-age years and discipline the transfer structure in the model.

B.4 Key Constructed Variables

Ability quartiles. I rank respondents by their 1999 math-verbal aptitude percentile and partition them into quartiles, restricting the sample to respondents with a non-missing aptitude score.

Parent-wealth quartiles. Total household net worth at age 17 is used to construct within-cohort parent-wealth quartiles. Together with the ability quartiles, these define the 4×4 grid of college-graduation rates used as the core estimation moments.

College-graduation rates. The sixteen cell-level graduation rates (ability quartile by parent-wealth quartile) are the primary target moments in estimation; they trace how college attainment varies across the joint distribution of ability and parental resources.

Average tuition. The gross annual cost of college (\$12,200) is computed from average tuition reported in the NLSY97 schooling module, following Abbott et al. (2019). The model uses a net annual tuition (after institutional discounts) of $\phi = \$6{,}700$, with the remaining gap covered by the endogenous financial-aid schedule $g(a_p)$ described in Section 2.7.

C HRS Sample Construction

This appendix describes the construction of the HRS analysis sample. The Health and Retirement Study (HRS) is a biennial longitudinal survey of Americans over age 50, conducted by the University of Michigan since 1992. I use the RAND HRS Longitudinal File, which harmonizes variables across all waves (1992–2022), together with four supplements: the RAND Family-Kids file, the RAND Family-Resident file, the Consumption and Activities Mail Survey (CAMS), and the supplemental survey on college support. Bequest information is taken from the HRS exit interview. The exact survey variable codes underlying each measure are documented in the replication package.

C.1 Data Sources

RAND HRS Longitudinal File. The core source is the RAND HRS longitudinal file, in which RAND staff harmonize variable definitions, handle skip patterns, and construct consistent wealth, income, and health measures across the sixteen waves spanning 1992–2022. From this file I draw respondent and spouse demographics, household wealth and its components, household income and its sources, and respondents' subjective probabilities of leaving a bequest.

RAND Family-Kids file. This supplement records information about each respondent's children—demographics (age, gender, education, marital status), geographic proximity, financial transfers in both directions, care provision, and bequest intentions—for waves spanning 1992–2014. I organize it into a child-by-wave panel.

RAND Family-Resident file. This supplement provides household composition by wave—the number of living children, sons, and daughters—which I organize into a respondent-by-wave panel.

Consumption and Activities Mail Survey (CAMS). CAMS is a supplemental mail survey that collects detailed consumption and spending data for a subset of HRS households, biennially from 2001 to 2017. It provides total household spending, durable and non-durable spending, housing expenditure, and mortgage payments.

Supplemental survey on college support. This supplement collects retrospective information on parents' financial contributions to their children's college education. For each child who attended college it records tuition, room and board, years of college, and the share of each cost paid by the parent, which I use to construct the total parental contribution to college in 2016 dollars.

Exit interview. The HRS exit interview, conducted with a proxy informant after a respondent's death, records bequest and inheritance information. I use it to construct child-level bequests (dollar transfers and shares of the estate) and housing inheritance (whether each child inherited the main residence and its imputed value).

College tuition data. Regional in-state college tuition by census division and year is merged with HRS respondents using their census division of residence, providing the regional college-cost variation used in the analysis.

C.2 Variables Used in the Analysis

Table C.1 lists the measures drawn from the HRS, grouped by concept, together with their coverage and source. Variable names are descriptive and correspond to underlying survey concepts rather than to any particular survey code.

Table C.1. Measures Drawn from the HRS

Measure	**Coverage**	**Source**
Panel A: Respondent Demographics		
Age of respondent	1992–2022	RAND HRS
Age of spouse	1992–2022	RAND HRS
Gender	Time-invariant	RAND HRS
Race	Time-invariant	RAND HRS
Years of education	Time-invariant	RAND HRS
Highest degree	Time-invariant	RAND HRS
Birth year	Time-invariant	RAND HRS
HRS sample cohort	Time-invariant	RAND HRS
Census division of residence	1992–2022	RAND HRS
Panel B: Household Wealth		
Total net worth	1992–2022	RAND HRS
Net home value	1992–2022	RAND HRS
Retirement accounts (IRA/Keogh)	1992–2022	RAND HRS
Checking and savings	1992–2022	RAND HRS

continued on next page

Table C.1 (continued)

Measure	Coverage	Source
Vehicles	1992–2022	RAND HRS
Business and farm assets	1992–2022	RAND HRS
Other real estate	1992–2022	RAND HRS
Panel C: Income		
Total household income	1992–2022	RAND HRS
Social Security retirement income	1992–2022	RAND HRS
Social Security disability / SSI	1992–2022	RAND HRS
Pension income	1992–2022	RAND HRS
Panel D: Bequest Expectations		
Probability of bequest $\geq$ \$10K	1992–2022	RAND HRS
Probability of bequest $\geq$ \$100K	1992–2022	RAND HRS
Probability of bequest $\geq$ \$500K	1992–2022	RAND HRS
Probability of any bequest	1992–2022	RAND HRS
Panel E: Child Information		
Child's years of education	1992–2014	Family-Kids
Child's age	1992–2014	Family-Kids
Child's gender	Time-invariant	Family-Kids
Child's marital status	1992–2014	Family-Kids
Child's income bracket	1992–2014	Family-Kids
Child lives within ten miles	1992–2014	Family-Kids
Panel F: Inter-Vivos Transfers		
Parent gave a transfer (indicator)	1992–2014	Family-Kids
Amount the parent gave	1992–2014	Family-Kids
Child gave a transfer (indicator)	1992–2014	Family-Kids
Amount the child gave	1992–2014	Family-Kids
Panel G: College Support		
Tuition and board paid by the parent	Retrospective	College support
Number of children who attended college	Retrospective	College support
Parent contributed a positive amount	Retrospective	College support
Panel H: Consumption		
Total household consumption	2001–2017	CAMS
Non-durable spending	2001–2017	CAMS
Housing expenditure	2001–2017	CAMS
Panel I: Bequests		
Dollar bequest to child	Exit interview	Exit interview
Share of estate to child	Exit interview	Exit interview
Housing inheritance value	Exit interview	Exit interview

Notes: The Family-Kids and Family-Resident supplements cover waves spanning 1992–2014. Monetary measures from the RAND HRS file are nominal at the source and are deflated to 2016 dollars in the analysis. The college-support measures are retrospective, covering each child's entire college period. Child income brackets are reported by the parent respondent; I use the bracket midpoint.

C.3 Sample Construction

Step 1: Assemble the respondent panel. I reshape the RAND HRS longitudinal file into a person-by-wave panel containing time-invariant identifiers and demographics (birth year, gender, race, education, religion) together with wave-varying measures of wealth, income, health, bequest expectations, family structure, employment, and spouse characteristics.

Step 2: Organize the child file. The Family-Kids supplement is reshaped from one record per child to a child-by-wave panel, with a child identifier that links each child to the parent respondent. Measures are grouped into child demographics, child-to-parent transfers, parent-to-child transfers, and bequest intentions.

Step 3: Add household composition. The Family-Resident supplement, which records the number of living children, sons, and daughters by wave, is merged into the respondent panel to provide the child-count breakdown.

Step 4: Add consumption. CAMS is organized into a respondent-by-wave panel of total household spending, durable and non-durable spending, housing expenditure, mortgage payments, and total household consumption. CAMS covers a subsample of HRS households biennially from 2001 to 2017.

Step 5: Construct college support. For each child who attended college, the total parental contribution is the sum of the real tuition paid by the parent (annual tuition, deflated to 2016 dollars, multiplied by years of college and the share of tuition the parent paid) and the real room-and-board paid by the parent (annual room and board, deflated, multiplied by years and the share the parent paid). Records with missing cost or share information are dropped; an explicit "did not contribute" response is recoded to zero. The total family contribution sums across all children in the household.

Step 6: Identify widowhood. I extract each respondent's year of death and attach it to the spouse's record, which yields, for each respondent, the year of the partner's death and supports the analysis of widowhood.

Step 7: Construct bequests from the exit interview. The exit interview provides two kinds of child-level bequest information. Financial bequests are the dollar amounts and shares of the estate transferred to each child; survey missing codes are removed, and when the estate share is missing but dollar amounts are available, the share is imputed as each child's portion of total transfers. Housing inheritance is constructed for respondents whose main residence passed to children rather than to a surviving spouse: I identify which children received the home and, when the home was left to all children jointly, divide its imputed value equally among them.

Step 8: Add regional college costs. Average in-state college tuition by census division and year is organized into a region-by-year panel and merged with HRS respondents using their census division of residence.

C.4 Sample Restrictions

I impose the following restrictions, analogous to those applied to the PSID sample:

1. **Age restrictions:** parents over age 50 and children over age 26.
2. **Interview status:** only respondents with completed interviews in a given wave.
3. **Couple households:** for analyses requiring spouse information, I restrict to coupled households.
4. **Missing data:** survey "don't know," "refused," and "missing" responses are carried through the construction stage and handled within each specific analysis, so that wealth and income are not discarded prematurely.

The final HRS analysis sample contains 19,179 parent–child pairs and 98,861 observations.

D Sample Summary Statistics

Tables D.1–D.2 report summary statistics for the PSID sample, stratified by average household income group. Parents in higher income groups are older, more likely to hold a college degree, more likely to own their home, and substantially wealthier: total wealth ranges from \$103,000 in the lowest income group to \$966,000 in the highest, and total consumption rises monotonically from \$36,000 to \$73,000. Among children, the college attendance rate among 18–24 year olds rises steeply with parental income. A wealth-portfolio breakdown is in Appendix Table G.1 and a consumption decomposition by expenditure category in Appendix Table H.1.

Table D.1. Parent Demographics and Financial Characteristics by Income Group (PSID)

	By Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
Panel A: Demographics					
Age of head	45.6	45.2	46.8	49.1	46.3
	(16.2)	(14.4)	(13.4)	(13.2)	(14.6)
Children in household	1.3	1.2	1.1	1.0	1.1
	(1.4)	(1.3)	(1.2)	(1.1)	(1.3)
Total children	2.1	2.0	2.1	2.2	2.1
	(1.3)	(1.0)	(1.0)	(1.0)	(1.1)
Head has college degree	0.07	0.19	0.42	0.69	0.29
	(0.26)	(0.39)	(0.49)	(0.46)	(0.45)
White	0.44	0.63	0.77	0.85	0.64
	(0.50)	(0.48)	(0.42)	(0.36)	(0.48)
Black	0.45	0.29	0.18	0.09	0.28
	(0.50)	(0.45)	(0.38)	(0.28)	(0.45)
Homeowner	0.50	0.71	0.84	0.85	0.70
	(0.50)	(0.45)	(0.36)	(0.36)	(0.46)
Panel B: Income, Wealth, and Consumption					
Household income	44,223	76,829	120,169	242,277	102,569
	(21,535)	(37,964)	(51,447)	(253,591)	(121,904)
Head labor income	23,523	40,976	64,728	123,504	53,930
	(19,369)	(28,162)	(42,902)	(123,970)	(64,202)
Total wealth	102,555	200,522	367,076	965,941	326,288
	(230,260)	(377,059)	(536,534)	(1,052,497)	(612,283)
Total consumption	36,426	46,131	57,243	73,130	50,076
	(18,374)	(20,389)	(24,489)	(33,277)	(26,252)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. Income groups defined by parent average real household income (2016 dollars) observed between ages 25 and 60. Panel A reports demographics; Panel B reports income, wealth, and consumption. "Children in household" is the number of children under age 18 in the parent's PSID family unit; "Total children" is the total number of children linked to the parent household via the Family Identification Mapping System (FIMS). All monetary variables in 2016 dollars, winsorized at the 1st and 99th percentiles. A detailed wealth portfolio breakdown is in Appendix Table G.1.

Table D.2. Children Descriptive Statistics by Parent Income Group (PSID)

	By Parent's Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
Panel A: Child Demographics					
Age	17.9	15.9	16.7	17.4	16.9
	(19.0)	(15.7)	(13.6)	(13.4)	(16.0)
Number of siblings in sample	1.9	1.6	1.5	1.6	1.6
	(1.7)	(1.2)	(1.2)	(1.1)	(1.4)
College-age (18–24)	0.14	0.14	0.16	0.17	0.15
	(0.34)	(0.35)	(0.36)	(0.38)	(0.36)
Panel B: Education					
Years of education (age 18+)	13.7	13.7	13.8	14.4	13.9
	(11.8)	(9.9)	(7.8)	(8.1)	(9.7)
Max years of education (age 25+)	16.1	15.7	15.9	16.5	16.0
	(14.8)	(11.5)	(9.9)	(8.9)	(11.9)
Currently in college	0.05	0.06	0.08	0.10	0.07
	(0.21)	(0.23)	(0.27)	(0.29)	(0.25)
Ever attains post-secondary ed.	0.33	0.37	0.47	0.56	0.41
	(0.47)	(0.48)	(0.50)	(0.50)	(0.49)
Panel C: College Rate Among 18–24 Year Olds					
College attendance rate	0.32	0.40	0.47	0.54	0.42
	(0.47)	(0.49)	(0.50)	(0.50)	(0.49)
College-age observations	4308	4984	4270	2907	16 469
Total child–year observations	31 731	35 484	27 026	16 962	111 203
Unique children	5680	5560	3784	2186	17 210

Notes: Means with standard deviations in parentheses. Each observation is a child-year. Children linked to parents via FIMS. Income groups refer to parent average real household income (2016 dollars). "Years of education" reported conditional on age ≥ 18; "Max years of education" conditional on age ≥ 25. "College attendance rate" computed conditional on the child being aged 18–24. "Ever attains post-secondary ed." indicates maximum observed years of education exceeds 12. Years of education are directly observed only from 2013 onward; for earlier waves, college attendance is inferred from a proxy that combines the typical college-age window with the individual's eventual attainment (maximum years of education observed in 2013 or later exceeding 12). Individuals not observed in any post-2012 wave cannot be assigned by this proxy and are excluded from the pre-2013 attendance measure.

E Inter-Vivos Transfer Margin Regressions (HRS)

This appendix reports the extensive- and intensive-margin regressions discussed in Section 3.3: the effect of child college attainment on the probability of a positive inter-vivos transfer (Table E.1) and on the transfer amount conditional on a positive transfer (Table E.2), both estimated with parent fixed effects.

Table E.1. Extensive Margin: Effect of College on Transfer Probability (HRS)

	Extensive Margin (Linear Probability)	
	(1)	(2)
	Pr(Parent $\rightarrow$ Child)	Pr(Parent $\rightarrow$ Child)
College (BA+)	-0.031***	-0.010***
	(0.002)	(0.003)
College $\times$ Q2		-0.018***
		(0.004)
College $\times$ Q3		-0.024***
		(0.004)
College $\times$ Q4		-0.032***
		(0.005)
Parent FE	Yes	Yes
Wave FE	Yes	Yes
Dep. Var. Mean	0.144	0.144
Observations	548,164	548,164
Within R^2	0.012	0.012

Notes: Linear probability models. Dependent variable is an indicator for any positive transfer in the HRS wave. All specifications include parent fixed effects, wave dummies, child cohort controls, and parent income quartile dummies. Standard errors clustered at parent level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Table E.2. Intensive Margin: Transfer Amount Conditional on Positive Transfer (HRS)

	Intensive Margin (Amount | Transfer > 0)	
	(1)	(2)
	Parent → Child	Parent → Child
College (BA+)	557*	-906*
	(288)	(486)
College × Q2		1864***
		(607)
College × Q3		965*
		(542)
College × Q4		1732**
		(684)
Parent FE	Yes	Yes
Wave FE	Yes	Yes
Dep. Var. Mean	8161	8161
Observations	78,541	78,541
Within R^2	0.002	0.002

Notes: OLS regressions on the subsample with positive transfers. Dependent variable is the real transfer amount (2016 US$). All specifications include parent fixed effects, wave dummies, child cohort controls, and parent income quartile dummies. Standard errors clustered at parent level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

F Income Process: Discrete-to-Continuous-Time Mapping and Life-Cycle Profiles

This appendix derives the mapping from the discrete-time AR(1) income process estimated by Abbott et al. (2019) to the continuous-time Ornstein-Uhlenbeck (OU) specification used in the model, and presents the implied life-cycle income profiles.

F.1 Mapping Derivation

The discrete-time income shock process is an AR(1) at annual frequency:

$$z^e_{t+1} = \rho^{\text{ann}}_e \, z^e_t + \eta^e_{t+1}, \qquad \eta^e_{t+1} \sim N\big(0,\, (\sigma^e_\eta)^2\big). \tag{26}$$

The continuous-time counterpart is the OU process:

$$dz^e = -\kappa_e \, z^e \, dt + \sigma_e \, dW, \tag{27}$$

where $\kappa_e > 0$ is the mean-reversion rate and σ_e is the diffusion coefficient.

Mean-reversion rate. The exact discretization of (27) at unit time intervals ($\Delta = 1$ year) gives the conditional mean $\mathbb{E}[z^e_{t+1} \mid z^e_t] = e^{-\kappa_e} z^e_t$. Matching this with the AR(1) coefficient ρ^{ann}_e yields:

$$e^{-\kappa_e} = \rho^{\text{ann}}_e \quad \Longrightarrow \quad \kappa_e = -\ln\big(\rho^{\text{ann}}_e\big). \tag{28}$$

Diffusion coefficient. The conditional variance of the discretized OU process is:

$$\text{Var}(z^e_{t+1} \mid z^e_t) = \frac{\sigma_e^2}{2\kappa_e}\big(1 - e^{-2\kappa_e}\big). \tag{29}$$

The conditional variance of the AR(1) is simply $(\sigma_\eta^e)^2$. Equating the two and using $e^{-\kappa_e} = \rho_e^{\text{ann}}$:

$$(\sigma_\eta^e)^2 = \frac{\sigma_e^2}{2\kappa_e}\big(1 - (\rho_e^{\text{ann}})^2\big)$$
$$\implies \quad \sigma_e = \sigma_\eta^e \cdot \sqrt{\frac{2\kappa_e}{1 - (\rho_e^{\text{ann}})^2}}. \tag{30}$$

This mapping preserves both the autocorrelation and the stationary variance $\text{Var}(z^e) = \sigma_e^2/(2\kappa_e)$ of the process. Note that the stationary variance of the OU process equals the unconditional variance of the AR(1), $(\sigma_\eta^e)^2/(1 - (\rho_e^{\text{ann}})^2)$, confirming internal consistency.

Numerical values. Applying (28)–(30) to the male estimates from Abbott et al. (2019) (Table 3.2): for high-school graduates the annual persistence is $\rho_{HS}^{\text{ann}} = 0.952$ with innovation variance $(\sigma_\eta^{HS})^2 = 0.017$, giving $\kappa_{HS} = -\ln(0.952) = 0.049$ and $\sigma_{HS} = \sqrt{2\kappa_{HS}(\sigma_\eta^{HS})^2/(1 - (\rho_{HS}^{\text{ann}})^2)} = 0.134$; for college graduates $\rho_C^{\text{ann}} = 0.966$ with $(\sigma_\eta^C)^2 = 0.017$, giving $\kappa_C = 0.035$ and $\sigma_C = 0.133$. Both the mean-reversion rates and the diffusions are set externally to these AGMV-implied values rather than estimated. Because college earnings are more persistent ($\rho_C > \rho_{HS}$) while the innovation variance is common, the implied stationary variance of the persistent shock, $\sigma_e^2/(2\kappa_e) = (\sigma_\eta^e)^2/(1 - (\rho_e^{\text{ann}})^2)$, is *higher* for college graduates: 0.254 (standard deviation 0.51) versus 0.182 for high school (standard deviation 0.43). The model therefore does not generate the transfer-region contraction through a reduction in income risk; the operative channel is the income level (Section 2).

F.2 Life-Cycle Profiles

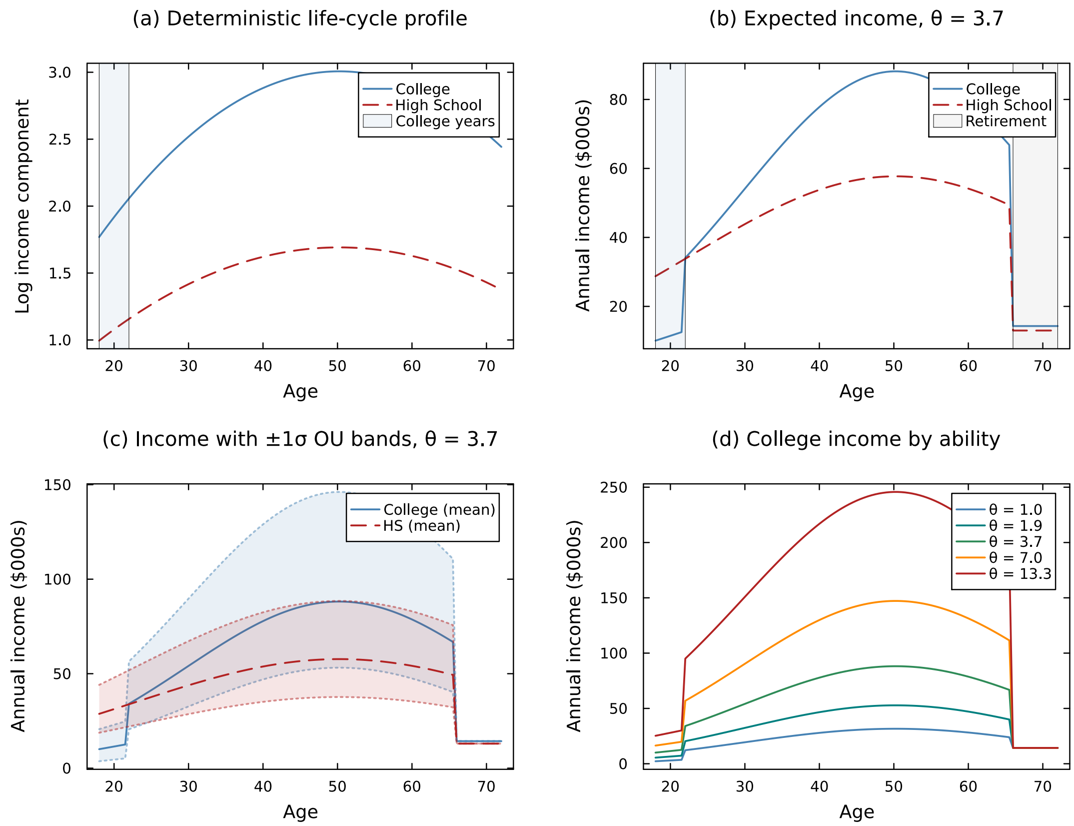

Figure F.1. Income Process: Life-Cycle Profiles by Education

Notes: Panel (a): deterministic age-income component $\gamma(t, e_c) = \gamma_{1,e}\, t - \gamma_{2,e}\, t^2$ from Abbott et al. (2019). Panel (b): expected annual income $y_c(t) = w \cdot \theta^{\lambda_e} \cdot \exp(\gamma(t, e_c))$ for median ability ($\theta = 3.9$); college workers earn $\ell_C = 56\%$ of full-time income during ages 18–22 and pay net tuition. Panel (c): ± 1 stationary standard deviation of the OU income shock ($\sigma_e/\sqrt{2\kappa_e}$) around expected income. Panel (d): college income profiles for low ($\theta = 1.5$), medium ($\theta = 3.9$), and high ($\theta = 10.2$) ability workers, illustrating the complementarity between ability and education ($\lambda_C > \lambda_{HS}$).

G Parent Wealth Portfolio

Table G.1 provides a detailed breakdown of parent income and wealth by income group. The table complements the summary variables reported in Table D.1 in the main text.

Table G.1. Parent Income and Wealth by Income Group (PSID, 2016 Dollars)

	By Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
Household income	44,223	76,829	120,169	242,277	102,569
	(21,535)	(37,964)	(51,447)	(253,591)	(121,904)
Head labor income	23,523	40,976	64,728	123,504	53,930
	(19,369)	(28,162)	(42,902)	(123,970)	(64,202)
Total wealth	102,555	200,522	367,076	965,941	326,288
	(230,260)	(377,059)	(536,534)	(1,052,497)	(612,283)
Home equity	48,321	76,931	127,798	223,253	102,056
	(86,276)	(105,763)	(136,410)	(195,989)	(138,176)
IRA/annuity	6,437	20,026	49,226	107,888	35,668
	(38,692)	(68,967)	(108,878)	(163,964)	(98,914)
Non-housing fin. wealth	53,469	121,496	234,059	698,124	214,727
	(182,888)	(318,143)	(453,318)	(895,431)	(505,467)
Savings	7,950	16,013	36,321	68,743	25,676
	(31,182)	(39,546)	(61,142)	(87,368)	(56,070)
Vehicles	12,549	19,741	24,701	32,096	20,737
	(16,601)	(19,952)	(22,565)	(28,691)	(22,256)
Other real estate	7,938	13,934	22,799	55,426	20,513
	(42,810)	(57,200)	(75,043)	(122,965)	(73,634)
Student loans	1,736	4,134	7,932	7,755	4,881
	(7,821)	(12,628)	(18,488)	(20,246)	(14,759)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. All monetary variables deflated to 2016 dollars using the CPI-U and winsorized at the 1st and 99th percentiles. Income groups defined by parent average real household income observed between ages 25 and 60.

H Parent Consumption Detail

Table H.1 decomposes total household consumption into major expenditure categories. Housing constitutes the largest category across all income groups, followed by transportation and food. Discretionary categories—clothing, recreation, and vacation—are only available from the 2005 wave onward and increase sharply with income.

Table H.1. Parent Household Consumption by Income Group (PSID, 2016 Dollars)

	By Average HH Income Group				
	\$30–60	\$60–100	\$100–160	\$160+	All
Total consumption	36,426	46,131	57,243	73,130	50,076
	(18,374)	(20,389)	(24,489)	(33,277)	(26,252)
Food	7,915	9,211	10,582	12,872	9,718
	(4,854)	(4,944)	(5,296)	(6,269)	(5,470)
Housing	15,030	18,677	24,487	33,215	21,206
	(9,071)	(10,757)	(13,588)	(18,356)	(13,873)
Transportation	9,310	11,880	13,811	15,562	12,166
	(8,213)	(8,709)	(9,844)	(11,999)	(9,661)
Health care	2,108	3,402	4,195	5,027	3,469
	(3,027)	(3,649)	(4,157)	(4,832)	(3,950)
Clothing (2005+)	1,356	1,577	1,912	2,808	1,776
	(1,704)	(1,762)	(1,907)	(2,626)	(1,988)
Recreation (2005+)	503	807	1,178	1,805	957
	(920)	(1,211)	(1,464)	(1,897)	(1,396)
Vacation (2005+)	835	1,482	2,376	3,886	1,870
	(1,505)	(2,020)	(2,583)	(3,604)	(2,554)
Observations	15 096	17 356	13 011	7863	53 326
Unique parent households	2718	2770	1833	1041	8362

Notes: Means with standard deviations in parentheses. Consumption data available from the PSID beginning in 1999. All expenditure categories deflated to 2016 dollars and winsorized at the 1st and 99th percentiles. Clothing, recreation, and vacation are available from 2005 onward only. Total consumption is the sum of all listed categories.

I Additional Robustness: Parent Fixed Effects

As additional robustness for the consumption stylized fact (Section 3.2), I exploit parent fixed effects to identify the college effect from within-family (between-sibling) variation. For consumption, the specification uses child-pair-year observations: within a given family and year, siblings who differ in college attainment generate variation in the college indicator while holding all parent-level characteristics constant.

Table I.1 reports parent fixed effects regressions of log parent consumption on the child's college indicator, at the child-pair-year level. The positive and significant coefficient confirms that the consumption difference documented in Section 3.2 is not driven by unobserved parental characteristics: even within the same family, parent consumption is higher in years when the college-educated child's characteristics are more salient.

Table I.1. Parent Consumption and Child College Status: Parent Fixed Effects (PSID)

	(1) OLS + Income Q FE	(2) Parent FE
College (BA+)	0.068***	0.027**
	(0.018)	(0.013)
Controls	Yes	Yes
Parent FE	No	Yes
Parent income Q FE	Yes	–
Observations	13,841	13,841
R^2 / Within R^2	0.469	0.840

Notes: Regressions of log parent household consumption on child college indicator (BA+), at the child-pair-year level. Column 1: OLS with parent income quartile fixed effects and comprehensive controls. Column 2: parent fixed effects with child age polynomial and education controls. Standard errors clustered at the parent household level in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

J Full Commitment Benchmark

This appendix provides additional details on the full commitment benchmark introduced in Section 6.1 of the main text.

Under full commitment, a benevolent planner commits at $t = 0$ to a state-contingent transfer schedule $\{\tau(a_p, a_c, z, t)\}$ for the entire overlap (parent ages 42–72), solving a single-agent problem over total household wealth $A \equiv a_p + a_c$, whose law of motion carries the same multiplicative wealth diffusion $\sigma_a A\, dB$ as the decentralized economy (equation 16 in the main text). With CRRA preferences, the efficient allocation equates weighted marginal utilities:

$$u'(c_p) = \eta\, u'(c_c), \qquad \text{so that} \qquad c_c = \eta^{1/\gamma}\, c_p \tag{31}$$

Consumption shares are time-invariant. Substituting yields:

$$u_{\mathrm{FC}}(c_{\mathrm{tot}}) = \frac{c_{\mathrm{tot}}^{1-\gamma}}{1-\gamma} \left(1 + \eta^{1/\gamma}\right)^{\gamma} \tag{32}$$

Relationship to the one-time transfer counterfactuals. The full commitment benchmark and the one-time transfer counterfactuals (Section 6.2) remove the Samaritan's dilemma through different mechanisms. Full commitment eliminates moral hazard while preserving state-contingent insurance: the planner adjusts transfers in response to income shocks throughout the overlap period. The one-time transfer counterfactuals eliminate both moral hazard and ongoing insurance: the child receives a lump sum at $t = 0$ and is subsequently on their own. The difference between FC and the one-time transfer economies therefore isolates the *insurance-removal effect*—the change in college attendance and welfare attributable to shutting down ongoing parental support, holding the absence of moral hazard fixed.

Comparing FC with the baseline economy yields the pure moral hazard effect holding insurance constant. As discussed in Section 6.1, this comparison reveals two opposing forces: anticipated transfers subsidize enrollment (raising college attendance above the efficient level) but also insure non-college children (lowering attendance below the efficient level). The net sign depends on the child's position in the ability–wealth distribution.

K Equilibrium Properties

This appendix derives the key properties of the Markov-Perfect Equilibrium in the continuous-time coupled problem.

K.1 Optimality Conditions

Following Barczyk and Kredler (2014a), the equilibrium partitions the state space into three regions: a *self-sufficient* region, where the possibility of transfers is remote; an *overconsumption* region, where no transfer currently flows ($\tau^* = 0$) but the state is drifting toward the transfer region; and the *transfer* region, where $\tau^* > 0$. The child's first-order condition $u'(c_c^*) = V_{a_c}^c$ holds throughout. In the self-sufficient region, the child's continuous-time Euler equation takes the standard form $\dot{c}_c/c_c = (r-\rho)/\gamma$. In the overconsumption region, the child anticipates that accumulating wealth will reduce future parental support, so the effective return on saving is $r + \partial\tau^*/\partial a_c$ and the Euler equation becomes:

$$\frac{\dot{c}_c}{c_c} = \frac{1}{\gamma}\left(r - \rho + \frac{\partial \tau^*}{\partial a_c}\right), \qquad \frac{\partial \tau^*}{\partial a_c} < 0. \tag{33}$$

This anticipation wedge is the Samaritan's dilemma in continuous time: it lowers the child's return on saving below r and induces over-consumption *before* any transfer flows. Inside the transfer region, by contrast, the parent equalizes the marginal values of own and child wealth ($V_{a_p}^p = V_{a_c}^p$) and acts as a family dictator, setting the child's consumption directly; the child's own Euler equation is therefore not the operative condition there (Barczyk and Kredler, 2014a).

The parent's first-order condition gives $u'(c_p^*) = V_{a_p}^p$. In the no-transfer region:

$$\frac{\dot{c}_p}{c_p} = \frac{1}{\gamma}\left(r_{\text{eff}}(a_p) - \rho\right) \tag{34}$$

In the transfer region, $V_{a_p}^p = V_{a_c}^p$: the parent equalizes the marginal values of own and child wealth and makes a positive gift $\tau > 0$. The parent funds the gift out of its own resources but continues to optimize its wealth, so its drift $\dot{a}_p = r_{\text{eff}}(a_p)a_p + y_p - c_p - \tau$ need not be zero—the parent makes a *bounded* transfer (it lifts the child only toward its own optimum) and keeps saving.

K.2 Transfer Region Characterization

The complementary-slackness condition (3) partitions the state space into:

1. **No-transfer region** $\mathcal{N} = \{(a_p, a_c, z, t) : V^p_{a_p} > V^p_{a_c}\}$: Parent and child accumulate wealth independently.

2. **Transfer region** $\mathcal{T} = \{(a_p, a_c, z, t) : V^p_{a_p} = V^p_{a_c}\}$: The parent makes a positive gift that brings the child toward the altruistic target $c_c = \eta^{1/\gamma} c_p$, bounded by the child's own optimum, $\tau^* = \max\{0,\ \min(c_c^{\text{opt}} - R_c,\ \eta^{1/\gamma} c_p - R_c)\}$ with $R_c = \tilde{r}_c a_c + y_c$. The parent funds the gift from its resources but continues to save ($\dot{a}_p = r_{\text{eff}}(a_p) a_p + y_p - c_p - \tau^*$).

The no-transfer region $\mathcal{N}$ subdivides into a *self-sufficient* subregion, where the wealth distribution is balanced and future transfers are remote, and an *overconsumption* subregion bordering $\mathcal{T}$, where no transfer currently flows but the state is drifting toward $\mathcal{T}$ and the anticipation wedge of (33) is active. The boundary of $\mathcal{T}$ is a free boundary determined endogenously. Parents transfer when the child is relatively poor (low a_c), faces negative income shocks (low z), or when the parent is wealthy (high a_p). The region shrinks as the parent approaches death ($t \to T$).

Lump-sum exercise rule. The same partition governs the lump-sum channel of Section 2.3.4. At a Poisson arrival, the parent solves (5); the first-order condition for an interior lump is

$$V^p_{a_c}(a_p - \tau^*_L,\, a_c + \tau^*_L,\, z, z_p;\, t) \;=\; V^p_{a_p}(a_p - \tau^*_L,\, a_c + \tau^*_L,\, z, z_p;\, t)\,,$$

so $\tau^*_L > 0$ if and only if $V^p_{a_c} > V^p_{a_p}$ at the pre-transfer state—the state lies strictly inside $\mathcal{T}$—and the optimal lump moves the dynasty onto the free boundary $\partial\mathcal{T}$ along the constant-total-wealth line $a_p + a_c$. In $\mathcal{N}$ and on $\partial\mathcal{T}$ the opportunity expires unused ($\tau^*_L = 0$). With bounded flow transfers, income and wealth shocks can carry the state strictly into $\mathcal{T}$ between arrivals, which is what gives the lump policy positive mass: the flow props up the child's consumption while the dynasty waits for an opportunity to rebalance the stocks. Anticipated lumps also feed the child's saving incentive: the jump term in (2) adds an expected-rebalancing component to the child's continuation value that is decreasing in own wealth, reinforcing the anticipation wedge of (33)—the Samaritan's dilemma extends to the lump margin.

K.3 The Samaritan's Dilemma

Both agents over-consume relative to the commitment solution. The child's distortion arises in the overconsumption region: anticipating that accumulating wealth would move the state out of $\mathcal{T}$ and eliminate support, the child faces a reduced effective return on saving. This marginal disincentive to save is captured by $\partial\tau^*/\partial a_c$ in (33), and it operates *before* transfers flow, not inside $\mathcal{T}$ itself, where the parent dictates the allocation.

The parent's behavior is also shaped by the lack of commitment: unable to commit to withholding support, the parent transfers whenever the child is sufficiently constrained, and the child anticipates this. Unlike a reflecting-barrier formulation, the parent does not exhaust its wealth in $\mathcal{T}$; it makes a bounded gift—lifting the child only toward its own optimum—and continues to save. The distortion therefore operates through the size and likelihood of transfers, and through the child's reduced self-insurance, rather than through a degenerate parent saving rule.

The continuous-time formulation offers three advantages over discrete-time alternatives. First, no transfer carries within-period commitment: the flow is revisable at every instant, and the lump-sum channel operates only at exogenous arrivals, so a discrete transfer is an action taken when an opportunity strikes rather than a promise held for the length of a period. Second, the free boundary provides a sharp characterization of where moral hazard operates—for both the flow and the lump margins. Third, the multiplicative wealth diffusion $\sigma_a a\, dB$ (Section 2) renders the value functions C^2 almost everywhere and the free boundary smooth, so the Markov-perfect equilibrium exists in pure strategies and is computed without the non-smoothness—the "blast from the past"—that afflicts deterministic and discrete-time altruism models (Barczyk and Kredler, 2014a, 2021); the stochastic timing of lump opportunities preserves this smoothness, where deterministic transfer dates would reintroduce anticipation kinks (Proposition 1).

K.4 College and Parental Influence

The college decision at $t = 0$ is forward-looking: the child compares lifetime values under each education path, incorporating the entire future trajectory of transfers. Altruistic parents affect the decision through two channels. First, they provide generous transfers during the college years (when child income is low), effectively subsidizing attendance—the transfer region $\mathcal{T}$ is larger when $e_c = C$ during the early years. Second, the continuation value under $e_c = C$ incorporates higher future income, which feeds back into the transfer policy: wealthier children require fewer transfers, freeing parent resources. The net effect generates a positive gradient of college enrollment in parent wealth.

K.5 Computed Transfer Region

The schematic in the main text (Figure 2) illustrates how college contracts the transfer region. Figures K.1 and K.2 report the model-computed objects at the estimated parameters: the free boundary $\partial\mathcal{T}$ for a college versus a high-school child, and the full transfer-flow policy τ^* across the parent–child wealth state space.

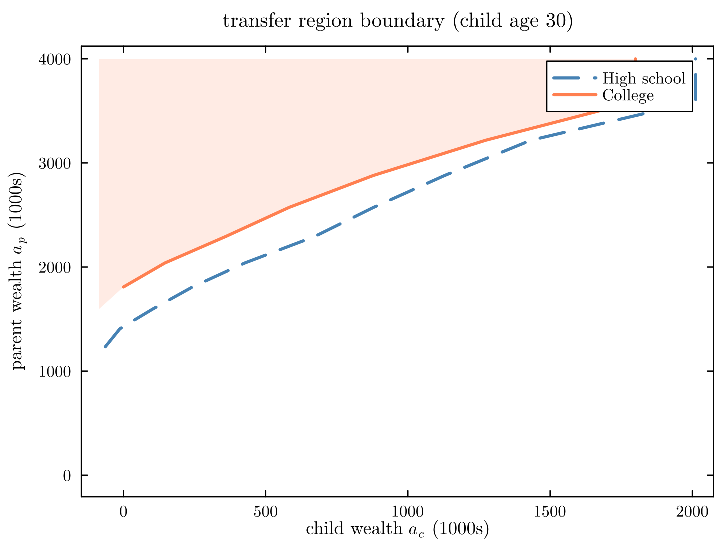

Figure K.1. Computed Transfer-Region Free Boundary: College vs. High School

Notes: Free boundary $\partial\mathcal{T}$ in the (a_c, a_p) state space at a representative post-college age, for a median-ability child of a high-school parent in the estimated model. To the left of each boundary the parent transfers ($\tau^* > 0$); to the right the child is self-sufficient. The shaded area is the college child's transfer region $\mathcal{T}_C$; the dashed line is the high-school boundary, the solid line the college boundary. College shifts the boundary left—the child must be poorer before transfers are optimal—because higher expected income keeps child wealth further from the boundary.

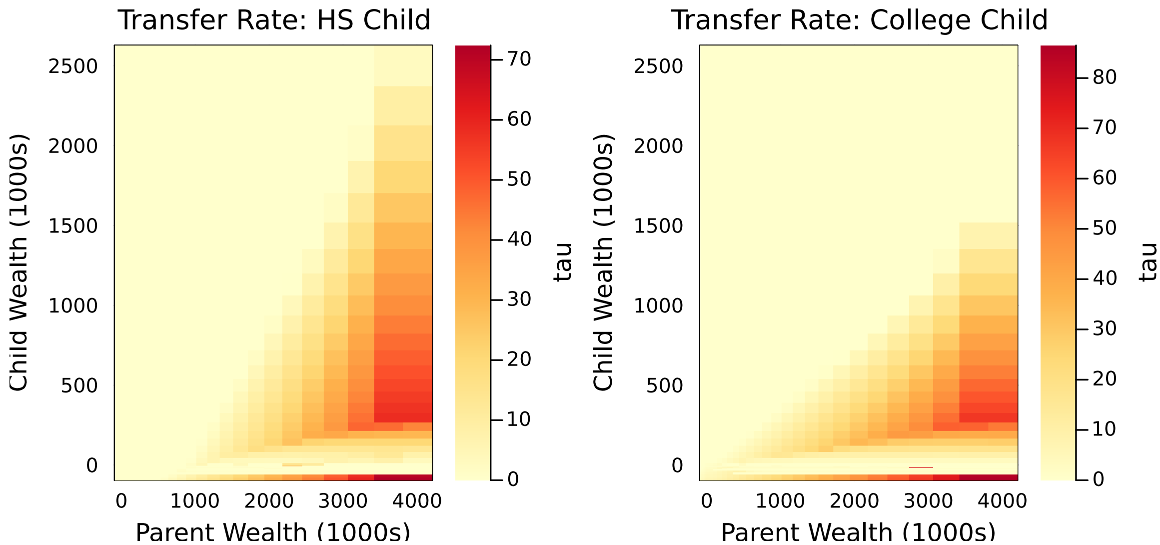

Figure K.2. Computed Transfer Policy τ^* by Child Education

Notes: Optimal transfer flow τ^* (thousands of 2016 dollars per year) over the parent-wealth (a_p) and child-wealth (a_c) state space, for a high-school child (left) and a college child (right), at the estimated parameters. The region where $\tau^* > 0$ is the transfer region $\mathcal{T}$; the white region is the no-transfer region $\mathcal{N}$. The two panels use independent color scales.

L College Financing Detail

This appendix details the college financial system summarized in Section 2.7, following Abbott et al. (2019). Let $y_p^0(\theta, e_p)$ denote the parent's income when the child enters college (parent age 42). The annual grant is

$$g(y_p^0) = \bar{g} \cdot \max\left(0,\ 1 - \frac{y_p^0}{\bar{y}_{\text{efc}}}\right),$$

where $\bar{g}$ is the maximum grant and $\bar{y}_{\text{efc}}$ the parental-income level at which aid phases out completely. Borrowing up to $\bar{b}(a_p) = \bar{b}_0 + b_{\text{slope}}\, a_p$ is allowed at the student rate $r + \iota_s$, implemented as $a_c \geq -\bar{b}(a_p)$ during $t \in [0, 4]$; the base $\bar{b}_0 =$ \$18,000 is the federal capacity available without parental collateral, and the accumulated balance amortizes so that $a_c \geq 0$ by age 32 (the standard ten-year repayment plan). The child's wealth dynamics during college are

$$\dot{a}_c = (r + \iota_s)\, a_c + \ell_C \cdot y_c(t, z) - c_c - \phi_{\text{net}}(y_p^0) + \tau, \qquad a_c \geq -\bar{b}(a_p),$$

where $\phi_{\text{net}}(y_p^0) = \phi - g(y_p^0)$ is net tuition. The schedule generates heterogeneity in the effective cost of college across the parental income distribution, and—as Abbott et al. (2019) document—government grants can crowd out the private contributions that supplement them.

M Solution Algorithm

The model is solved numerically using an upwind finite difference scheme with implicit time-stepping, following Achdou et al. (2022) and the transfer-region methods of Barczyk and Kredler (2014a,b). The algorithm proceeds in six stages: grid construction, solving the child-alone HJB, solving the coupled parent-child HJB system, computing college choice probabilities, forward simulation, and moment computation for SMM estimation.

M.1 State Space and Grid Construction

The model features three continuous state variables during the overlap period: parent wealth $a_p \in [0, \bar{a}_p]$, child wealth $a_c \in [\underline{a}_c, \bar{a}_c]$, and the persistent income shock $z \in [-\bar{z}, \bar{z}]$. In addition, child ability θ and education $e \in \{\mathrm{HS}, \mathrm{C}\}$ are discrete states that index separate HJB problems.

Asset grids. Both wealth grids use exponential spacing to concentrate points near the origin, where policy functions exhibit greater curvature. For N grid points from $a_{\min}$ to $a_{\max}$ with growth factor $g > 0$:

$$a_i = a_{\min} + a_{\max} \cdot \frac{(1+g)^{i-1} - 1}{(1+g)^{N-1} - 1}, \qquad i = 1, \ldots, N. \tag{35}$$

The parent grid has $N_{a_p} = 30$ points over $[0, 4{,}000]$ (thousands of dollars). The child grid has $N_{a_c} = 34$ points: 10 points in the negative (student-debt) region $[\underline{a}_c, 0)$ and 24 exponentially spaced points over $[0, \bar{a}_c]$. The negative segment mirrors the exponential spacing of the positive segment, so that nodes are dense near zero—where amortizing balances spend most of their time—and sparse toward the debt floor. This spacing matters for convergence: with a uniform debt segment, college-attendance cells fail to converge as the segment is refined, whereas the mirrored-exponential segment is converged at 10 points. We set the child-grid floor to $\underline{a}_c = -25$, below the cumulative student borrowing limit $\bar{b}_0 = 18$ (Section 2.7) so the constraint is bracketed by interior grid points, and $\bar{a}_c = 2{,}500$.

Income shock grid. The Ornstein-Uhlenbeck process $dz = -\kappa_e z\,dt + \sigma_e\,dW$ is discretized on a uniform grid of $N_z = 9$ points spanning ± 3 stationary standard deviations:

$$z_k \in \left\{-3\,\frac{\sigma_e}{\sqrt{2\kappa_e}}, \ldots, +3\,\frac{\sigma_e}{\sqrt{2\kappa_e}}\right\}, \qquad k = 1, \ldots, N_z.$$

The grid width is set to the maximum stationary standard deviation across education types to ensure both HS and college processes are well-covered.

Ability grid. Child ability follows an AR(1) in logs across generations: $\log\theta' = \rho_\theta \log\theta + \mu_\theta(1-\rho_\theta) + \sigma_\theta\varepsilon$. We discretize this using the Tauchen (1986) method with $N_\theta = 5$ grid points spanning ± 3 stationary standard deviations of $\log\theta$.

Time grid. The backward induction uses $N_t = 60$ equally spaced time steps over the parent's full remaining life ($T = 30$ years, parent ages 42–72, child ages 18–48), yielding $\Delta t = 0.5$ years per step. The implicit scheme is unconditionally stable, so this step size is accurate at half the cost of a finer grid. The coupled parent-child system is solved over the entire horizon $[0, T]$: the parent retires at $T^{\text{ret}} = 24$ (age 66) and switches to social security income, but remains coupled to the child and can continue to transfer until death at T (age 72).

M.2 Stage 1: Child-Alone HJB

After the parent dies (child age 48), the child solves a standard consumption-savings problem independently for the remainder of life ($T_{\text{child alone}} + T_{\text{child retire}} = 24$ years, child ages 48–72, working until 66 and then on social security). The HJB equation, with the same multiplicative wealth diffusion as the coupled problem, is:

$$\begin{aligned}\rho V^{c,\text{alone}}(a,z,t) = \max_{c\geq 0}\Big\{\, & u(c) + V_a^{c,\text{alone}}\,[r\,a + y(t,z) - c] + V_z^{c,\text{alone}}(-\kappa z) \\ & + \tfrac{1}{2}\sigma^2 V_{zz}^{c,\text{alone}} + \tfrac{1}{2}\sigma_a^2 a^2 V_{aa}^{c,\text{alone}} + V_t^{c,\text{alone}}\Big\}.\end{aligned} \tag{36}$$

We solve this problem separately for each ability type θ and education level $e_c \in \{\text{HS}, \text{C}\}$, giving $2 \times N_\theta = 10$ independent HJB problems that are solved in parallel.

Terminal condition. At the child's death (age 72, i.e. $t = T_{\text{child}}$ on the child-alone clock), the terminal value is $V^c(a, z, T_{\text{child}}) = u(c_T)/\rho$ where $c_T = \max\{r\,a + \text{SS}_{e_c} + 0.1\,a,\ \epsilon\}$ represents consuming all remaining resources.

Optimal consumption via the first-order condition. At each grid point and time step, the FOC $u'(c^*) = V_a^c$ yields the candidate consumption $c^* = (V_a^c)^{-1/\gamma}$, where γ is the CRRA coefficient. We compute V_a^c using upwind finite differences: the *forward* difference $\partial_a^+ V$ and *backward* difference $\partial_a^- V$ yield two candidate consumption levels, c^{fwd} and c^{bwd}, with associated savings drifts $s^{\text{fwd}} = ra + y - c^{\text{fwd}}$ and $s^{\text{bwd}} = ra + y - c^{\text{bwd}}$. The upwind scheme selects:

$$c^* = \begin{cases} c^{\text{fwd}} & \text{if } s^{\text{fwd}} > 0 \quad \text{(saving: use forward difference)}, \\ c^{\text{bwd}} & \text{if } s^{\text{bwd}} < 0 \quad \text{(dissaving: use backward difference)}, \\ ra + y & \text{otherwise (drift} \approx 0\text{: consume resources).} \end{cases}$$

This upwind selection ensures numerical stability by using the derivative in the direction from which information flows.

Implicit time stepping. Given optimal policies at time step n, we construct the sparse infinitesimal generator $\mathbb{A}^n$ that encodes the upwind finite-difference operator in a and the discretized OU operator in z (combining upwind drift and central-difference diffusion terms). The implicit update solves:

$$\left[\left(\frac{1}{\Delta t} + \rho\right)\mathbf{I} - \mathbb{A}^n\right]\mathbf{V}^n = \mathbf{u}^n + \frac{1}{\Delta t}\,\mathbf{V}^{n+1}, \tag{37}$$

where $\mathbf{V}^n$ and $\mathbf{u}^n$ are the vectorized value function and flow utilities at step n, $\mathbf{I}$ is the identity matrix of dimension $N_{a_c} \times N_z$, and $\mathbf{V}^{n+1}$ is the value at the next (future) time step. This system is solved via sparse LU factorization.

OU generator. The infinitesimal generator for the OU process on the uniform z-grid combines a central-difference approximation for diffusion with upwind differencing for the drift:

$$
\begin{aligned}
(\mathbb{A}_z)_{k,k+1} &= \frac{\sigma^2/2}{\Delta z^2} + \frac{\max(-\kappa z_k, 0)}{\Delta z}, \\
(\mathbb{A}_z)_{k,k-1} &= \frac{\sigma^2/2}{\Delta z^2} + \frac{\max(\kappa z_k, 0)}{\Delta z}, \\
(\mathbb{A}_z)_{k,k} &= -(\mathbb{A}_z)_{k,k+1} - (\mathbb{A}_z)_{k,k-1},
\end{aligned} \tag{38}
$$

with reflecting boundary conditions at $k = 1$ and $k = N_z$.

Wealth-diffusion generator. The multiplicative wealth shock $\sigma_a a\, dB$ contributes a second-order term $\frac{1}{2}\sigma_a^2 a^2 V_{aa}$, discretized by a central second difference on the (non-uniform) wealth grid. With diffusion coefficient $D_i = \frac{1}{2}\sigma_a^2 a_i^2$, forward spacing $\Delta_i = a_{i+1} - a_i$ and backward spacing $\Delta_{i-1} = a_i - a_{i-1}$, the generator entries are

$$
\begin{aligned}
(\mathbb{A}_a)_{i,i+1} &= D_i \, \frac{2}{\Delta_i(\Delta_i + \Delta_{i-1})}, \\
(\mathbb{A}_a)_{i,i-1} &= D_i \, \frac{2}{\Delta_{i-1}(\Delta_i + \Delta_{i-1})}, \\
(\mathbb{A}_a)_{i,i} &= -(\mathbb{A}_a)_{i,i+1} - (\mathbb{A}_a)_{i,i-1}.
\end{aligned} \tag{39}
$$

The coefficient D_i vanishes at $a_i \leq 0$ (no volatility at the borrowing constraint, following Barczyk and Kredler, 2014a) and at the grid boundaries, so no extra boundary condition is required. Because D_i depends only on the grid, this block is assembled once and reused across all time steps; it preserves the generator's M-matrix structure, so the implicit step remains stable.

M.3 Stage 2: Coupled Parent-Child HJB

During the overlap period, the parent and child interact strategically. The parent's HJB equation is given by (1), with the transfer decision characterized by the complementarity condition (3). That is, the parent transfers whenever the marginal value of an additional dollar in the child's account exceeds the marginal value of retaining it.

Terminal condition. At $t = T$ (parent age 72, the parent's death), the coupled system terminates. The parent's terminal value combines a warm-glow term over remaining wealth and altruistic concern for the child's continuation value (equation 11). The child's terminal value is $V^c(a_p, a_c, z, T) = V^{c,\text{alone}}(a_c + \alpha_{\text{beq}}\, a_p, z)$, the child-alone value function from Stage 1 evaluated at $t = 0$ with effective wealth including the bequest.

Transfer allocation in closed form. We solve the coupled system backward in time. At each time step n and grid point (i, j, k) the transfer is obtained in *closed form*: because altruism is one-sided (only the parent gives) and the recipient's marginal value enters through a static first-order condition, the transfer region $\mathcal{T}^n$ and the gift size are determined pointwise, without iterating on the region. The allocation is the continuous-time analog of the rule in Barczyk and Kredler (2014a) and the `GetBKalloc` routine of Barczyk et al. (2023), specialized to one-sided altruism.

1. **Own-consumption first-order conditions.** Each agent's unconstrained optimal consumption is recovered from its own marginal value of wealth using the upwind selection above:
$$c_p = (V^p_{a_p})^{-1/\gamma}, \qquad c_c^{\text{opt}} = (V^c_{a_c})^{-1/\gamma},$$
with implied parental saving $s_p = ra_p + y_p - c_p$.

2. **Bounded gift.** The static transfer FOC $u'(c_p) = \eta\, u'(c_c)$ implies a target child consumption $c_c = \eta^{1/\gamma} c_p$, and hence a first-best gift $g_{\text{fb}} = \eta^{1/\gamma} c_p - R_c$, where $R_c = \tilde{r}_c\, a_c + y_c$ is the child's own resources. The gift is capped at the level that lifts the child to its *own* optimum (the second-best ceiling) and floored at zero:
$$\tau = \max\Big\{ 0,\ \min\big(c_c^{\text{opt}} - R_c,\ \ \eta^{1/\gamma} c_p - R_c \big) \Big\}. \tag{40}$$
The point lies in the transfer region, $(i, j, k) \in \mathcal{T}^n$, if and only if $\tau > 0$. The lower bound is the one-sided no-negative-transfer constraint; the upper bound prevents a wealthy parent from transferring more than brings the child to its own optimum, which is what disciplines the gift to a finite, liquidity-insurance role.

3. **Wealth drifts.** The parent funds the gift out of its own saving and continues to

optimize its wealth—it is *not* frozen in the transfer region:

$$\dot{a}_p = s_p - \tau, \qquad c_c = \min\left(c_c^{\text{opt}},\ R_c + \tau\right), \qquad \dot{a}_c = R_c + \tau - c_c.$$

Since $\tau \leq c_c^{\text{opt}} - R_c$, the child consumes its resources plus the entire gift, so $\dot{a}_c = 0$ inside the transfer region: the transfer is pure liquidity insurance to a constrained child, consistent with the mechanism in Section 2.9.

4. **Construct the generator and solve.** Given the policies and drifts, assemble the sparse infinitesimal generator $\mathbb{A}^n$ over the full state space (a_p, a_c, z) of dimension $N_{a_p} \times N_{a_c} \times N_z$. The generator combines upwind drift operators in both wealth dimensions, the multiplicative wealth-diffusion blocks in a_p and a_c, and the OU operator in z.

   A key computational insight is that the parent and child value functions share the *same* state transitions—both wealth drifts and the OU process are identical for both agents given the equilibrium policies. Therefore, the system matrix

   $$\mathbf{B} = \left(\frac{1}{\Delta t} + \rho\right)\mathbf{I} - \mathbb{A}^n$$

   is factorized *once* (via sparse LU), and used for two triangular solves with different right-hand sides:

   $$\mathbf{B}\,\mathbf{V}^{p,n} = \mathbf{u}^p + \frac{1}{\Delta t}\,\mathbf{V}^{p,n+1} + \mathbf{J}^{p,n}, \tag{41}$$

   $$\mathbf{B}\,\mathbf{V}^{c,n} = \mathbf{u}^c + \frac{1}{\Delta t}\,\mathbf{V}^{c,n+1} + \mathbf{J}^{c,n}, \tag{42}$$

   where $\mathbf{u}^p = u(c_p) + \eta\, u(c_c)$ and $\mathbf{u}^c = u(c_c)$ are the flow utilities and $\mathbf{J}^{p,n}$, $\mathbf{J}^{c,n}$ carry the lump-sum jump terms defined in the next step.

5. **Lump-sum stage (operator splitting).** The Poisson lump-sum channel (Section 2.3.4) is handled *explicitly*, so the nonlocal jump operator never enters the matrix $\mathbf{B}$ and the symbolic factorization is reused unchanged. At each time step with the channel active (child age ≥ 22), the lump policy is computed on the already-known continuation

values: at each node,

$$\tau_L^*(i,j,k) = \arg \max_{\tau_L \in \mathcal{M}(a_{p,i})} V^{p,n+1}(a_{p,i} - \tau_L,\ a_{c,j} + \tau_L,\ z_k)\,,$$

where the candidate set $\mathcal{M}(a_p)$ combines a fixed menu of small lumps with fractions of the parent's available wealth, all bounded by $\max(a_p, 0)$, and off-node candidates are valued by bilinear interpolation in the two wealth dimensions. Because the value function is concave, interpolation *under*states off-node values while $\tau_L = 0$ is evaluated exactly, so discretization error biases against spurious lumps. The jump contributions to (41)–(42) are then

$$\mathbf{J}^{p,n} = \lambda_g\big[\mathcal{M}\mathbf{V}^{p,n+1} - \mathbf{V}^{p,n+1}\big]\,, \qquad \mathbf{J}^{c,n} = \lambda_g\big[\mathcal{M}\mathbf{V}^{c,n+1} - \mathbf{V}^{c,n+1}\big]\,,$$

where $\mathcal{M}V$ denotes evaluation at the rebalanced state $(a_p - \tau_L^*, a_c + \tau_L^*, z)$ under the *parent's* policy in both lines (one-sided altruism: the child's value jumps with the parent's choice). The explicit treatment is unconditionally well-behaved here: the per-step jump weight is $\lambda_g \Delta t = 0.125$ at the baseline calibration, far inside the stability region, and the splitting follows the standard treatment of nonlocal terms in Achdou et al. (2022).

Regularity and the role of stochastic timing. The exogenous, exponentially-timed arrival of lump opportunities is what keeps the lump channel from disturbing the smoothness that the wealth diffusion delivers. The following statement makes this precise.

Proposition 1 (Regularity under stochastic timing)**.** *Fix the child's equilibrium policy and suppose the wealth diffusion is non-degenerate ($\sigma_a > 0$) on the interior of the state space. Then the parent's value function solving the HJB (1)—with the bounded nonlocal jump term $\lambda_g\,[\mathcal{M}V^p - V^p]$ and the complementary-slackness condition that characterizes the transfer region $\mathcal{T}$—is the unique viscosity solution of the associated integro-differential variational inequality. It is continuously differentiable on the interior of the state space and twice continuously differentiable away from the free boundary $\partial\mathcal{T}$ (hence C^2 almost everywhere), and the optimal lump maps the state onto $\partial\mathcal{T}$. The Poisson lump channel therefore introduces no* *anticipation discontinuity. Restricting transfers instead to a deterministic set of dates* $\{t_k\}$ *would impose the interior relinking conditions* $V^p(\cdot, t_k^-) = \mathcal{M}V^p(\cdot, t_k^+)$ *at known instants, whose non-smoothness propagates backward in time.*

Argument. The nonlocal operator $\mathcal{M}V^p(x) = \max_{\tau_L \in [0,a_p]} V^p(a_p - \tau_L,\, a_c + \tau_L,\, z, z_p;\, t)$ is a bounded, zeroth-order term: it is Lipschitz in x whenever V^p is, and by the envelope theorem inherits the differentiability of V^p wherever the maximizer is interior. Adding such a bounded nonlocal term to the uniformly elliptic generator $\mathcal{L}$—uniform ellipticity following from $\sigma_a > 0$—keeps the problem within the class of *nondegenerate* controlled jump-diffusion problems, for which the value function is the unique viscosity solution of the integro-differential variational inequality and attains $C^{1,2}$ regularity on the continuation region (Pham, 1998); the accompanying verification theorem for controlled jump diffusions is standard (Øksendal and Sulem, 2007), and the explicit treatment of the nonlocal term follows Achdou et al. (2022). Because inter-arrival times are exponential (memoryless), no instant carries positive probability of a transfer, so V_t^p develops no kink; under deterministic dates the *certain* relinking at each t_k creates a kink that propagates backward along characteristics—the "blast from the past" of Barczyk and Kredler (2021). The statement is conditional on the opponent's policy: regularity of each agent's value function given the other's policy is what the diffusion-plus-Poisson structure secures, while existence of the coupled Markov-perfect equilibrium is verified numerically (smoothness of V^p, V^c, and the policies, and invariance of the solution to grid and time-step refinement).

This procedure is repeated for each combination of child ability θ, child education $e_c \in \{\mathrm{HS}, \mathrm{C}\}$, parent education $e_p \in \{\mathrm{HS}, \mathrm{C}\}$, and the parent's permanent productivity component z_p (on a grid of $N_{z_p} = 5$ nodes), yielding $N_\theta \times 2 \times 2 \times N_{z_p} = 100$ coupled HJB problems that are solved in parallel across threads.

College-period modifications. When the child is in college (age < 22), two modifications apply: (i) the child earns a fraction ℓ_C of full-time income and pays net tuition $\phi_{\text{net}}(y_p^0) = \phi - \bar{g} \cdot \max(0,\, 1 - y_p^0/\bar{y}_{\text{efc}})$, which varies continuously with parent *income* at college entry through the EFC system; and (ii) the child may borrow up to $\bar{b}(a_p)$ at the student loan rate $r + \iota_s$.

Borrowing constraints. The child's wealth drift is constrained to be non-negative at the borrowing limit: if $a_c \leq \underline{a}_c + \epsilon$ and $\dot{a}_c < 0$, we set $\dot{a}_c = 0$. During college, $\underline{a}_c = -\bar{b}$; after college, $\underline{a}_c = 0$. The parent faces a natural borrowing constraint $a_p \geq 0$.

M.4 Stage 3: College Choice

At $t = 0$, the decision is solved in two stages on each (a_p, θ, e_p, z_p) node. First, the parent's college gift: candidate gifts τ_C combine a fixed menu of small gifts, the non-negative child-grid nodes, and their midpoints, all bounded by the parent's wealth a_p (Section 2.6); off-node candidates are valued by bilinear interpolation in (a_p, a_c), while the high-school branch is fixed at $(a_p, 0)$; for each candidate, the child's probit (13) and the parent's objective (14) are evaluated at $z = 0$, $t = 0$, and the maximizing τ_C^* is selected. No HJB re-solve is required: the coupled solutions already cover the (a_p, a_c) plane. Second, the child's choice: the value gain $G = V_C^c(a_p - \tau_C^*, \tau_C^*) - V_{HS}^c(a_p, 0)$ is compared to the realized psychic cost $\kappa(\theta, \varepsilon_\kappa) = (\kappa_0 + \kappa_\theta \log\theta + \sigma_\kappa \varepsilon_\kappa) \cdot \bar{V}$ with $\varepsilon_\kappa \sim N(0,1)$, where $\bar{V} = \text{mean}_{a_p,\theta}|V_C^c - V_{HS}^c|$ (evaluated at $a_c = 0$) normalizes the cost to the scale of the value gap. Because the cost shock is Gaussian, the attendance probability $\Phi\big((G/\bar{V} - \kappa_0 - \kappa_\theta \log\theta)/\sigma_\kappa\big)$ is available in closed form, so no numerical integration is required. In the simulation, each dynasty re-solves the gift stage at its own (e_p, z_p) node with the ex-ante probit, and the child's choice is then deterministic given its drawn ε_κ; an enrolling child enters the overlap with $a_c = \tau_C^*$ and the parent with $a_p - \tau_C^*$, while a high-school child starts at $a_c = 0$ with the parent's wealth unchanged.

Boundary monotonicity correction. At very low parent wealth ($a_p \approx 0$), the coupled HJB can produce spurious college probabilities due to convergence difficulties at the corner of the state space. We enforce monotonicity by scanning from the bottom of the a_p grid upward: if the college probability at point i exceeds that at point $i + 1$ for the first few grid points, we cap it at the first interior-monotone value. This correction affects at most the bottom third of the grid.

M.5 Stage 4: Dynastic Simulation

We simulate $M = 4{,}000$ dynasties across $N_{\text{coh}} = 11$ overlapping generations by forward-simulating the model's discretized law of motion at the same time step used to solve the HJB system.

Initialization. The first cohort begins at median ability $\theta = \theta_{(N_\theta+1)/2}$. Subsequent generations draw ability from the AR(1) process: $\log\theta' = \rho_\theta \log\theta + \mu_\theta(1-\rho_\theta) + \sigma_\theta\varepsilon$. Each new generation's parent wealth equals the previous generation's terminal child wealth (clamped to $[0, \bar{a}_p]$). The initial income shock z_0 is drawn from an education-specific entry distribution $z_0 \sim \mathcal{N}\big(\zeta_{pe}\,\mathbf{1}\{e_p = C\},\ \sigma_e^2/2\kappa_e + \sigma^2_{z_0,e_c}\big)$: the variance combines the OU stationary variance with the Abbott et al. (2019) entry-dispersion component σ_{z_0,e_c}, and the mean carries the parent-college earnings premium ζ_{pe} (Section 2).

College decision. For each dynasty m in cohort g, we interpolate the college probability $\Pr(\mathrm{C} \mid a_p^{m,g}, \theta^{m,g})$ from the solved grid and draw education status from a Bernoulli distribution.

Forward simulation. Given education choices and initial conditions, we simulate the state dynamics forward in discrete time steps. At each time step $n = 1, \ldots, N_{\text{steps}}$:

1. **Lump-sum arrival**: with probability $1 - e^{-\lambda_g \Delta t}$ per step (active from child age 22), a transfer opportunity arrives; the lump policy τ_L^* is interpolated at the current state and executed as a discrete rebalancing, $a_p \mathrel{-}= \tau_L^*$, $a_c \mathrel{+}= \tau_L^*$, *before* the flow policies are applied—the discrete analog of jump-then-drift, consistent with the HJB timing in which the flow policies already price arrivals through the jump term.

2. **Interpolate policies**: trilinear interpolation of (c_p, c_c, τ) from the solved HJB policy grids at the (post-jump) state (a_p^n, a_c^n, z^n).

3. **Compute income**: parent income y_p (deterministic or Social Security if retired) and child income y_c (net of tuition during college), including the education-specific transitory shock with standard deviation $\sigma_{\varepsilon,e}$ (estimated by SMM; it enters realized income but not the state, with the Jensen correction $\mathbb{E}[e^{\varepsilon}] = 1$).

4. **Update wealth** (drift plus the multiplicative wealth-diffusion shock):

$$\begin{aligned} a_p^{n+1} &= a_p^n + \big(r_{\text{eff}}(a_p^n)\, a_p^n + y_p - c_p - \tau\big)\, \Delta t + \sigma_a\, a_p^n\, \sqrt{\Delta t}\, \xi_p^n, \\ a_c^{n+1} &= a_c^n + (\tilde{r}_c\, a_c^n + y_c - c_c + \tau)\, \Delta t + \sigma_a\, a_c^n\, \sqrt{\Delta t}\, \xi_c^n, \end{aligned} \tag{43}$$

with independent $\xi_p^n, \xi_c^n \sim \mathcal{N}(0,1)$; the diffusion term is set to zero whenever $a \leq 0$, matching the HJB.

5. **Update income shock**: $z^{n+1} = z^n - \kappa_e\, z^n\, \Delta t + \sigma_e \sqrt{\Delta t}\, \xi^n$, where $\xi^n \sim \mathcal{N}(0,1)$.

6. **Enforce constraints**: clamp a_p to $[0, \bar{a}_p]$ and a_c to $[\underline{a}_c, \bar{a}_c]$ (with $\underline{a}_c = -\bar{b}$ during college, $\underline{a}_c = 0$ after). Clamp z to the grid bounds.

Reproducibility. All random draws (initial z_0, college choice, OU innovations, transitory shocks, the parent and child wealth-diffusion innovations ξ_p, ξ_c, and the lump-sum arrival uniforms) are pre-generated sequentially before the threaded simulation loop, ensuring bit-identical results regardless of the number of parallel threads; the arrival uniforms are drawn last and only when $\lambda_g > 0$, so the $\lambda_g = 0$ economy reproduces the no-lump model draw-for-draw.

M.6 Stage 5: Moment Computation and SMM Estimation

The model is estimated by the simulated method of moments. It produces 43 moments—30 targeted—in the eight blocks defined in Section 4 (college attendance, 16; income, 3; transfers, 5; wealth, 4 targeted and 8 untargeted; education transmission, 1; decumulation, 1; college by parent income, 4 untargeted; negative net worth, 1 untargeted). All are computed from a settled cohort (the second-to-last generation) to avoid initial-condition transients. Two computations are specific to the simulation: the transfer *premium* differences the in-college transfer (the enrollment gift plus consumption-support flows) against an equal-length post-college window, and the transfer-*probability* moments count two-year windows from child age 26 in which cumulated flow plus lump-sum transfers reach the $500 HRS threshold.

Loss function. The objective is a weighted percentage-deviation loss with fixed per-moment diagonal weights,

$$\mathcal{L}(\boldsymbol{\vartheta}) = \sum_{m} w_m \left(\frac{m_m(\boldsymbol{\vartheta}) - \hat{m}_m}{\hat{m}_m} \right)^2, \tag{44}$$

with the weights $\{w_m\}$ fixed prior to estimation. Following Lee and Seshadri (2019), I do not weight all moments equally but combine a group-size normalization with deliberate adjustments toward the moments most informative about the model's mechanisms. The transfer and income blocks enter at their group-size-normalized weight. The sixteen college-attendance cells, the core target, carry elevated mass—downweighting them produces well-fitting wealth distributions but implausibly flat attendance gradients—and within that block the four high-wealth, low-ability cells, where the altruistic college gift shows up most clearly, are doubled. The two moments that discipline the parent wealth tail (the p90/p50 ratio and the share of parents with net worth below \$25,000) and the intergenerational education gap are upweighted, while the intergenerational-wealth-transmission moments (rank–rank slope, conditional child wealth, child-wealth dispersion) and the share of children with negative net worth are left untargeted and reported only as validation. Table M.1 lists the exact weight on every moment. Deliberately weighting some moment groups more heavily than others is standard in the structural life-cycle literature (De Nardi et al., 2010; Abbott et al., 2019; Lee and Seshadri, 2019; Boar, 2021).

Table M.1. Per-Moment Weights in the SMM Objective

Moment block	#	Weight (each)
College attendance (parent-wealth Q × ability Q)	16	0.075
of which high-wealth, low-ability cells	(4)	0.150
Income: HS/C income ratio, var. log income (HS, C)	3	0.333
Transfers: mean transfer, $\Pr(\tau > 0)$ by parent-wealth Q	5	0.200
Aggregate wealth-to-income ratio	1	0.091
Median parent wealth	1	0.091
Parent decumulation rate	1	1.000
p90/p50 parent wealth	1	0.150
Share of parents with net worth < $25k	1	0.500
Intergenerational education gap	1	0.500

Notes: Per-moment diagonal weights w_m in the SMM objective (44), fixed prior to estimation. The table lists the 30 targeted moments; the remaining 13 moments are computed but left untargeted (weight 0). Within the college-attendance block, the four high-wealth, low-ability cells (parent-wealth $Q3$–$Q4$ × ability $Q1$–$Q2$) carry twice the weight of the other twelve.

Optimization. We minimize $\mathcal{L}$ over the 13 free parameters using a parallel $(\mu/\mu_w, \lambda)$-CMA-ES optimizer restricted to the free (non-fixed) subspace. The relative price of college labor ω_C lies outside this subspace: it is held fixed during the optimization, pinned to the value that reproduces the high-school-to-college income ratio (Section 4, Table 4), so the income-ratio moment is matched by construction rather than by a free parameter. Each candidate vector is evaluated by solving the full model (Stages 1–4) and computing the loss, and multi-start initialization ensures coverage of the parameter space.

M.7 Stage 6: Counterfactual Economies

Full commitment. The parent and child commit to a transfer schedule $\{\tau_t\}_{t\geq 0}$ at $t = 0$. This is solved as a single planner's problem that jointly maximizes parent and child welfare, eliminating the Samaritan's dilemma. The resulting college rates isolate the moral hazard effect against the full-commitment benchmark: $\Delta_{\text{MH}}^{\text{FC}} = \Pr(\text{C} \mid \text{baseline}) - \Pr(\text{C} \mid \text{FC})$. This full-commitment comparison is distinct from the main-text decomposition, which uses the one-time-transfer economies ($\Delta_{\text{MH}}^{\text{pure}}$ and Δ_{targ} in equations 21–23).

No-moral-hazard (college gift). The parent makes a one-time college-conditional gift τ_C at $t = 0$, paid only if the child attends college; thereafter the child solves the child-alone problem for its entire remaining life. The parent's problem is:

$$\max_{\tau_C \in [0, a_p]} \Pr(\mathrm{C} \mid \tau_C, \theta) \left[V_p^{\mathrm{alone}}(a_p - \tau_C) + \eta \, V^{c,\mathrm{alone}}(\tau_C \mid \mathrm{C}) \right]$$
$$+ (1 - \Pr(\mathrm{C} \mid \tau_C, \theta)) \left[V_p^{\mathrm{alone}}(a_p) + \eta \, V^{c,\mathrm{alone}}(0 \mid \mathrm{HS}) \right],$$

so the child receives τ_C in the college state and nothing in the high-school state. Here $V_p^{\mathrm{alone}}(\cdot)$ is the value of a single-agent parent problem—a continuous-time consumption-savings HJB over the parent's remaining life (labor income to age 66, social security thereafter, the consumption floor, the warm-glow bequest, and the same wealth diffusion $\sigma_a a \, dB$ as the baseline). Solving it as a genuine dynamic program, rather than approximating it by a permanent-income annuity, places the parent's counterfactual value on the same footing as the baseline coupled value, so the consumption-equivalent comparisons in Section 7 difference like with like. The child-alone value $V^{c,\mathrm{alone}}(\tau_0 \mid e_c)$ is solved over the child's entire post-transfer life and likewise carries the wealth diffusion.

No-moral-hazard (menu). Same as above, but the parent commits to an education-contingent menu: a transfer τ_C paid if the child attends college and τ_{HS} paid if the child chooses high school, both optimized in $[0, a_p]$. This is the efficient one-time-transfer benchmark; it nests the unconditional transfer ($\tau_C = \tau_{HS}$) and measures the effect of removing moral hazard *plus* optimal education targeting.

Four-way decomposition. The difference in college rates across these allocations decomposes as:

$$\underbrace{\Pr(\mathrm{C} \mid \mathrm{BL}) - \Pr(\mathrm{C} \mid \mathrm{menu})}_{\Delta_{\mathrm{MH}}^{\mathrm{total}}}$$
$$= \underbrace{\Pr(\mathrm{C} \mid \mathrm{BL}) - \Pr(\mathrm{C} \mid \mathrm{uncond.})}_{\Delta_{\mathrm{MH}}^{\mathrm{pure}}} + \underbrace{\Pr(\mathrm{C} \mid \mathrm{uncond.}) - \Pr(\mathrm{C} \mid \mathrm{menu})}_{\Delta_{\mathrm{targ}}}. \quad (45)$$

The signs are consistent with the main text (Section 6): a positive $\Delta^{\text{pure}}_{\text{MH}}$ indicates that baseline moral hazard raises college attendance relative to the no-moral-hazard economy.